

CLINICAL STUDY REPORT

NEOCART®

STUDY 8-01

A RANDOMIZED COMPARISON OF NEOCART® TO MICROFRACTURE FOR THE REPAIR OF ARTICULAR CARTILAGE INJURIES IN THE KNEE

Indication studied:	Repair of Articular Cartilage Injuries of the Knee
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Sponsor:	Histogenics Corporation 830 Winter Street 3rd Floor Waltham, MA 02451 Tel: 781-547-7900

This study was performed in accordance with Good Clinical Practices (GCP), and according to International Conference on Harmonisation (ICH) Guidelines.

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4. ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation or Specialist Term	Explanation
ABI	Ankle-Brachial Index
ACI	Autologous Chondrocyte Implantation
ACL	Anterior Cruciate Ligament
AD	Adaptive Design
AE	Adverse Event
ALT	Alanine Aminotransferase
ASCVD	Atherosclerotic Cardiovascular Disease
AT	As Treated patient population
ATF	Audit Trail File
BMI	Body Mass Index
CBC	Complete Blood Count
CCI	Characterized Chondrocyte Implantation
CCS	Composite Clinical Success
CFR	Code of Federal Regulations
Chondroplasty	Reparative or plastic surgery of cartilage
cm	Centimeters
CPM	Continuous Passive Motion Therapy
CRF	Case Report Form
CT	Computerized Tomography
CT3	A bioadhesive glue that secures the NeoCart Implant in place and is comprised of polyethylene glycol (PEG) and methylated collagen to for a glue that is stronger than fibrin glue
CV	Cardiovascular
DSMB	Data Safety Monitoring Board: An independent data-monitoring committee that may be established by the Sponsor to assess, at intervals, the progress of a clinical trial, the safety data, and the critical efficacy endpoints, and to recommend to the Sponsor whether to continue, modify, or stop a trial (ICH E6).
Debridement	Excision of devitalized tissue and foreign matter from a wound
DVT	Deep Vein Thrombosis
ECG	Electrocardiogram
eCRF	Electronic Case Report Form
EQ-5D-5L	European Quality of Life, 5-Dimensional, 5 Level
FDA	United States Food and Drug Administration
HIV	Human Immunodeficiency Virus
HRQoL	Health Related Quality of Life
ICRS	International Cartilage Repair Society
IFU	Instructions for Use
IKDC	International Knee Documentation Committee
INR	International Normalized Ratio
IRB	Institutional Review Board
iSAP	Interim Statistical Analysis Plan
ITT	Intent-to-Treat
IUD	Intrauterine Device
IV	Intravenous

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Abbreviation or Specialist Term	Explanation
Kellgren-Lawrence	The Kellgren-Lawrence grading system is a radiological classification of knee osteoarthritis. It is based on x-rays and consists of normal, Grade I, Grade II, Grade III and Grade IV
kg	Kilogram
KOOS	Knee Injury Osteoarthritis Outcomes Score
LOCF	Last Observation Carried Forward
Loose bodies	Free-floating cartilage, bone or other material in the joint space
m	Meters
MACI	Matrix-induced autologous chondrocyte implantation
mg	Milligram
MI	Multiple Imputation
MITT	Modified Intention To Treat
ml	Milliliter
mm	Millimeter
MMRM	Mixed Model Repeated Measures
MRI	Magnetic Resonance Imaging
Non-drug therapies/treatments	Treatments, procedures, and/or products that are used or performed by the patient to address knee pain and that incurs a cost to the patient. Examples include: physical therapy, acupuncture, massage therapy, use of knee brace, etc.
OR	Outcomes Research
PCL	Posterior Cruciate Ligament
PE	Pulmonary Embolism
PEG	Polyethylene Glycol
PP	Per Protocol patient population
PT	Prothrombin Time
PTT	Partial Thromboplastin Time
PVNS	Pigmented Villonodular Synovitis
RNA	Ribonucleic Acid
ROC	Receiver Operating Curve characteristics analysis
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SF-36	Short Form 36 is a generic (non-disease specific) survey instrument containing 36 questions that is used to measure disease impact on patients' health-related quality of life
SF-12	Short Form 12 is a generic (non-disease-specific) survey instrument containing 12 questions that is used to measure disease impact on patients' health-related quality of life
s-GAG	Glycosaminoglycan
SOC	System Organ Class
SSR	Sample Size Re-estimation
T2	Time for 63% of transverse relaxation read on MRI
TED	Thromboembolic Disease
TEAE, TESA	Treatment Emergent Adverse (Serious Adverse) Event
TKR	Total Knee Replacement
UAE	Unexpected Adverse Event
VAS	Visual Analog Scale
WPAI	Work Productivity Activity Impairment

5. ETHICS

5.1. Independent Ethics Committee or Institutional Review Board

The study protocol and amendments ([Section 16.1.1](#)) and informed consent form (ICF) ([Section 16.1.3.2](#)) were reviewed and approved by the Institutional Review Board (IRB)/Independent Ethics Committee (IEC) at each participating site. A list of all IRBs/IECs used in this study is provided in [Section 16.1.3.1](#).

5.2. Ethical Conduct of the Study

The study was conducted in accordance with Good Clinical Practice (GCP) as required by the International Conference on Harmonization (ICH) guidelines and the FDA. Compliance with these requirements also constitutes conformity with the US 21 Code of Federal Regulations (CFR) Part 50 (Protection of Human Subjects), US 21 Part 56 (Institutional Review Boards), ICH Harmonised Tripartite Guideline, Guideline for GCP E6(R1), the Nuremburg Code, and the ethical principles of the Declaration of Helsinki.

5.3. Subject Information and Consent

Copies of the IRB-approved Informed Consent Form (ICF) were received and reviewed by Histogenics Corporation (Sponsor) prior to the enrollment of patients at an investigational site. The ICF document approved by the site's IRB was signed by the patient and the Investigator prior to enrollment. The Investigator provided a copy of the signed ICF to the patient, and the original was retained by the Investigator. A sample ICF is provided in [Section 16.1.3.2](#).

6. INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was a multi-national, multi-center study conducted in 30 centers in the United States (US) and in 3 centers in Canada. Fourteen (14) additional centers in the US and one in Canada were initiated but did not enroll any patients. Of these 5 consented one or more patients that screen failed (never randomized). The lists of all Investigators and other key study personnel, with their affiliations, a description of their role in the study, and their qualifications, are provided in [Section 16.1.4.1](#), with copies of their curricula vitae where relevant ([Section 16.1.4.2](#)). Investigators were trained at site initiation visits. Signed approval of the report from the Principal Investigator and the Sponsor's responsible medical officer is provided in [Section 16.1.5](#). The study administrative structure is described in [Table 1](#).

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Table 1: Study Administrative Structure

Role/Responsibility	Company / Institution
Study Management, Monitoring, Clinical Supply Management, and Report Author	Histogenics Corporation 830 Winter Street 3 rd Floor Waltham, MA 02451 USA
Data Management	Health Policy Associates <u>690 Canton Street #302</u> <u>Westwood, MA 02090</u>
Biostatistics	1. Greg Maislan PhD Biomedical Statistical Consulting 1357 Garden Road Wynnewood, PA 19096 USA 2. Gene Poggio PhD Biostatistical Consulting, Inc. 24 New England Executive Park, Suite 130 Burlington, MA 01803
MRI Core Lab	Image IQ 26801 Miles Road #103 Cleveland, OH 44128 USA
Central MRI Reading ^a	ImageIQ Core Lab 26801 Miles Road #103 Cleveland, OH 44128 USA

^a Includes both standard MRI analysis and T2 mapping.

7. INTRODUCTION

Cartilage is the specialized fibrous connective tissue of primarily mesenchymal origin that forms the skeleton of the human body during development that is replaced with bone. Mature cartilage forming cells (chondrocytes) secrete the components of the extracellular matrix (hyaluronan, chondroitin sulfates and keratan sulfate) and tropocollagen, which polymerizes extracellularly into fine collagen fibers.

Cartilage persists in the growing fetus, throughout development and into adulthood in further differentiated forms. The three most common are articular hyaline, elastic and fibrocartilage. Articular cartilage is hyaline cartilage that covers the surface of bones at the joint space acting as a shock absorber and enabling the bones to glide over each other with minimal friction. Articular cartilage is covered by the synovial membrane which secretes the non-Newtonian synovial fluid that keeps the joint lubricated. Damage to the articular cartilage may be due to acute injury or repetitive trauma, and patients often experience pain and loss of function for several years due to the lack of efficacious medical treatments.

The existence of a cartilage defect in the knee joint is an important clinical finding since full thickness defects do not heal spontaneously once skeletal maturity is reached (Chalmers, 2013) and can result in debilitating symptoms such as pain, swelling, catching and grinding with knee movement (Bhosale and Richardson, 2008). Although chondrocytes produce matrix components throughout life, their production is unable to keep pace with the repair requirements after acute damage to hyaline cartilage. If cartilage is injured after a period of active growth, the defects are usually filled by connective tissue or fibrous cartilage. The extracellular matrix of these repair tissues is only poorly integrated with the matrix of the damaged cartilage. In addition, these types of repair tissue have less resiliency than mature articular cartilage. Left untreated, such lesions may progress to degenerative osteoarthritis (Buckwalter and Mankin, 1998), which is the third leading cause of work disability in adult populations in the United States (Theis et al., 2017). By extension, disability and the often-associated opioid addiction crisis is an enormous contributor to the annual pharmacoeconomic burden and medical costs to society. The ultimate goal of treatment of cartilage injuries in the knee is for clinicians to be able to restore joint function to pre-injury levels in a timely, predictable and durable fashion.

There is a significant unmet medical need for knee cartilage defects resulting in symptoms of pain and reduced function. Although treatment options for cartilage defects in the knee have evolved over the past 25 years, there remains a need for predictable and durable treatments. With current surgical strategies, many patients remain symptomatic, or symptoms return within one to two years of the intervention. In the United States, based on reimbursement codes demonstrating incidence, it is believed that approximately 1.2 million patients each year have arthroscopies where focal cartilage defects are identified. Only approximately 500,000 patients ultimately receive treatment with the large majority being debridement and microfracture (Mfx). Despite the limitations of microfracture, there are still approximately 150,000 microfracture procedures done each year.

Due to the variable outcomes of current treatment options including the prolonged rehabilitation phase, many patients elect to tolerate knee pain and loss of function with as many as 60% of patients avoiding or delaying treatment. These cartilage injuries, along with other knee injuries, are believed to be one of the primary predictors of later stage osteoarthritis. These injuries can have a significant economic impact for both the patient and the health care system with time lost from work, progression of disease and potential eventual need for joint replacement. Better treatment options are particularly important given the younger demographics of the affected individuals in their prime earning, sporting, or childraising years.

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Clinical trials in the cartilage treatment space have been challenging to conduct due to patient perceptions regarding microfracture, difficult to recruit due to the specific inclusion / exclusion criteria for isolated defects, and until recent FDA guidance, lacked clarity on the appropriate endpoints for evaluation in clinical studies. Due to the paucity of adequate and well-controlled clinical studies in the United States, it is also difficult to make meaningful comparisons across treatment modalities.

The current treatment options can be broadly classified into three categories and are described in Table 2. Palliation (chondroplasty and debridement), repair (microfracture) or restorative (mosaicplasty, osteochondral auto or allograft, autologous chondrocyte implantation). While all have demonstrated benefit, especially in specific patient populations, there are many limitations and clinicians and patients have sought a better solution across the range of lesion sizes and body types.

Table 2 Current Treatment Options

Technique	Description	Patient population	Discussion
Debridement	Minimally invasive trimming and cleaning of the cartilage defect	N/A, does not treat the defect	Often first-line temporary pain relief during diagnostic arthroscopy
Microfracture	Minimally invasive drilling multiple holes in the cartilage defect to promote bleeding and a clot that then heals; single procedure	Optimal for lesions less than 2cm ² Commonly used up to 4 cm ²	Results are highly variable. Extensive rehabilitation time required for optimal results (Steadman protocol). Worse outcomes with larger lesion size, older patients, previous surgeries and higher BMI (Mithoefer, 2009). Replacement with fibrous cartilage not articular cartilage.
Mosaicplasty and Osteochondral autograft transplantation (OAT)	Damaged cartilage is removed and replaced with healthy cartilage taken from a non--weight bearing area of the joint; Single procedure	Option for single plug vs multiple. Mature hyaline cartilage at procedure, Recommended for small to med lesions (1.5-4cm ²)	Limited by amount of donor tissue, may need open surgery, minimal long-term data, donor site morbidity
Osteochondral Allograft Resurfacing	Similar to OATS with the exception that the tissue source is cadaveric donor bone and cartilage; Single procedure	Recommended for large lesions (4-10cm ²)	Limited availability, rare failure of the graft tissue to integrate with host cartilage
Autologous chondrocyte implantation (ACI) and Matrix-Applied Characterized Autologous Cultured Chondrocytes (MACI)	Harvest and laboratory culture of cartilage producing cells from the knee and implanting them into the defect; two surgical procedures	Originally approved for salvage procedure, data largely in larger lesions (average lesion ~5 cm ²)	First generation procedure of periosteal flap creation challenging, second generation improved. Long rehabilitation time, graft hypertrophy, delamination. Lack of predictable hyaline cartilage growth

7.1. Repairative Treatments

While chondroplasty and debridement can alleviate pain for some period of time by trimming and smoothing damaged cartilage in the knee, they are not curative procedures. The area of damaged cartilage is unable to regenerate and create a smooth hyaline covered joint. This limitation is what led

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attempts to either allow the body to self repair the defect or provide the necessary cells to replace the missing cartilage and integrate into the patient's native knee cartilage. These methods have been met with varying degrees of success. While all of the treatment options are perceived as less than ideal, industry and the FDA sought to determine the most appropriate control arm for clinical development. Debridement does not treat the underlying defect, and auto/allograft tissue results are variable and not consistently available. Microfracture had become the standard of care starting approximately 20 years ago, particularly for smaller lesions, and protocols were developed to provide for a fair basis of comparison to this repair procedure.

7.2. Microfracture

In 1997, Steadman, et al., introduced the concept of "microfracture" as a reparative approach to chondral defects [8]. As originally performed, the focal defect is debrided until the subchondral bone plate is visualized. Using an arthroscopic awl, a series of holes 3-4 mm apart are created (with 3 to 4 holes/cm²) such that bone marrow is released to create a fibrin clot filled with endogenous reparative-stimulating agents (e.g., mesenchymal stem cells, growth factors, and cell-signaling proteins). Mesenchymal stem cells within the clot begin to proliferate, and several weeks after the procedure, cell differentiation creates cells that histologically appear similar to cartilage progenitor cells [9]. As time elapses (in terms of months and years), fibrocartilage – not hyaline cartilage – comprises most of the former defect.

During the initial post-procedure timeframe, the clot may become dislodged leading to early failure. To prevent this, patients must be non-weightbearing for a minimum of 6 weeks and then slowly progress to increased activity over the next 6 to 8 months with high impact activities and sports initiated after the 1-year mark. This Steadman defined rehabilitation protocol is deemed to be critically important - this slow progression and the strict adherence to a rehabilitation protocol is essential for success of the procedure and yet in reality can be difficult for patients to maintain given the normal demands of work and life. Such rehabilitation protocol was utilized in the NeoCart Phase III trial to ensure best possible outcomes with microfracture.

Unfortunately, even when proper rehabilitation is followed the clot may not heal or it may heal with soft fibrocartilage that lacks the resilience of articular cartilage. This accounts for the high failure rate often seen with microfracture starting 2 to 4 years after treatment and recent literature suggests that the failure rate at 5 years may be as high as 60%, with the median time of failure being at 4 years ([Solheim, 2018](#)).

Over the last 20 years, clinicians now recognize, however, that the quality of clinical outcomes after microfracture is dependent upon several variables. Proper patient selection, surgical technique, and regimented physical therapy/rehabilitation all affect clinical outcome and patient satisfaction; each of these factors are addressed below.

7.2.1. Patient selection and success factors

Mithoefer, et al., [10] conducted a meta-analysis to systematically examine the clinical efficacy of microfracture by identifying studies that featured short- and long-term functional outcome scores, macroscopic and microscopic cartilage repair quality, and results from postoperative MRI assessments. The researchers identified 28 studies describing 3122 patients. By integrating the results of these studies, the authors identified a series of factors that affect the clinical outcome after microfracture. Their table [10] is reproduced below ([Table 3](#)).

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Table 3: Factors Affecting the Clinical Outcome After Microfracture

Factors	Better results with
Age	< 40 years old
Duration of Symptoms	< 12 months
Lesion Size	<4cm ²
Body Mass Index	<30 kg/m ²
Preoperative activity level	Tegner score > 4
Previous surgery	Primary microfracture
Repair cartilage volume	Good defect fill (.60%)

As seen above, assiduous patient selection is key to achieving positive outcomes after microfracture surgery. The researchers found, similar to the findings of Kreuz, et al. [11], that patients over 40 years of age tended to have worse outcomes than their younger counterparts. Lesions smaller than 4 cm² were associated with better outcomes; the study by Kreuz, et al. [11] indicated that defects >2 cm² tended to have worse outcomes, thus tightening their criteria for optimal patients. In this regard, a meta-analysis of Goyal, et al. [12] agrees with both groups of previously cited researchers: in their study, good clinical outcomes were associated with small lesions in younger patients with low post-operative demands.

In a 2016 review paper, Sommerfeldt, et al. [13], describe characteristics of a patient most likely to benefit from the microfracture procedure, “it [microfracture] is more likely to be successful when performed as a primary procedure for nonobese patients under the age of thirty to forty years with small (<2 to 4-cm²) femoral condylar defects that have been symptomatic for a short time (less than twelve to twenty-four months).”

7.2.2. Microfracture Surgical Technique

The underlying principle of microfracture is to access the pluripotent mesenchymal stem cells present in bone marrow – found just under subchondral bone – in order to stimulate the repair of cartilage defects. The goal, then, is to optimize the availability of these reparative cells within the defect. The techniques developed to achieve this goal are immensely varied: hole spacing (density), hole depth, method of hole creation (awls vs. drills), whether or not to create holes (and risk structural damage to subchondral bone). Chen, et al. [14], examined the creation of holes via compaction (i.e., micro-“fracture” via awls) vs. water-cooled drilling and the role of hole depth. Four micro-holes were created in the trochlear groove of New Zealand white rabbits: 1 hole ~2 mm in depth was created using an awl, 2 holes created by a drill to a depth of ~2 mm and the remaining hole drilled to a depth of ~6 mm; these series of 4 holes were replicated bilaterally in a total of 2 rabbits. The rabbits were sacrificed at 24 hours and the subchondral surfaces were examined histologically. The authors found that the compaction of bone using an awl created channels with restricted blood egress due residual bony remnants. The drilled channels were free of bony remnants and provided patent holes for blood egress. The authors had expected to find that drilling would create thermal necrosis in nearby osteocytes, but this did not occur. Surprisingly, they found disruption of nearby osteophytes during compaction, which they attributed to high shear forces created during the microfracture process. They concluded that the creation of microchannels via drilling was preferred over the use of awls in order to create patent holes that would encourage the flow of pluripotent cells.

While microfracture technique likely influences functional outcomes, a study by Kroell, et al. [15] demonstrates the difficulty in achieving uniformity of technique. In their research, the authors instruct four surgeons to create microfracture holes of a prescribed depth and density (i.e., spacing) in six pre-

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prepared chondral defects in fresh cadaveric knees. They observed, “all surgeons misjudged depth and inter-hole distance tending to make microfracture holes too deep (depth error 1.1 mm + 1.9) and too close together (inter-hole distance error: -0.8 mm + 0.4).” The authors concluded, “even experienced knee arthroscopy surgeons demonstrate inconsistency in surgical technique when performing microfracture.”

7.3. Restoration (Autologous Chondrocyte Implantation)

7.3.1. Carticel (Genzyme Corporation, now Vericel, Inc.)

Articular cartilage cannot repair itself and microfracture, while widespread in adoption, has significant limitations. In an effort to provide a true cartilage repair option, Genzyme Corporation developed the technique of harvesting a biopsy of a patient’s own healthy cartilage and amplifying the cells through ex-vivo culture before placing the cells back into the patient and securing them in the defect with a patient-harvested periosteal patch to promote integration. Carticel was originally available in 1995 as an unregulated therapy as the cells came from the patient’s own body. The FDA was in the process of implementing a new policy for biologics that was released in February 1997 and convened an advisory panel. The panel voted 11-0 with one abstention that Carticel provided a clinical benefit to patient with damage to the articular cartilage of the knee. Carticel received accelerated approval from the FDA in August of 1997 after the FDA instituted specific cell therapy guidelines. Under accelerated approval, the FDA required Genzyme to conduct confirmatory post-marketing studies which it completed in 2007, after which it approved new labeling that incorporated the safety and efficacy data from the STAR trial.

There have now been several published reports of long term results following ACI. The surgical procedure was described by Petersen et al. **YEAR?** showing functional improvement at 2 years and further improvement at long term follow-up between 5 and 11 years following implantation for 44 patients with osteochondritis dissecans or with defects on the femoral condyle alone or with anterior cruciate ligament (ACL) reconstruction. The modified Cincinnati Knee Score was significantly higher both at 2 years and at long term follow-up when compared to the baseline score. There was an increase in the Knee Score from 2 years and long term follow-up, but the significance of this increase was not discussed. Of note, there were 10 treatment failures (16%) of the 61 patients enrolled. Earlier data from what is believed to be part of the same group of patients was presented by Petersen et al. **YEAR?** These earlier data showed that the modified Cincinnati Knee Scores were statistically improved at baseline compared with the post-operative scores. These results are consistent with the later findings.¹⁹

Henderson et al. **YEAR?** described two series of consecutive patients treated with ACI who were followed for 12 and 24 months, respectively. There was a reduction in the Objective International Knee Documentation Committee (IKDC) scores in both groups at 3 months that essentially returned to baseline by 12 months. There was no statistical difference between the 12 and 24 month scores. The Subjective Knee Score was significantly higher at 12 months than at baseline, and this trend continued at 24 months, but the increase was not significant. Similarly, the ICRS Functional Knee Status was significantly higher at 12 months than at baseline, and even a bit higher at 24 months, but again, the 24 month Functional Knee Status was not significantly different than the 12 month scores.²¹. The Subjective Knee Score was significantly higher at 12 months than at baseline, and this trend continued at 24 months, but the increase was not significant. Similarly, the ICRS Functional Knee Status was significantly higher at 12 months than at baseline, and even a bit higher at 24 months, but again, the 24 month Functional Knee Status was not significantly different than the 12 month scores.

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Knutsen et al. **YEAR?** described a randomized trial of 80 patients that compared ACI to microfracture. Both groups had significantly higher Lysholm scores and reduced pain at one and two year follow-up when compared to the baseline measurements, but there was no statistical difference between the groups. The improvement in the Short Form-36 Health Survey (SF-36) score in the first two years was significantly higher for the microfracture group. All other results from the two methods yielded similar clinical results.

In a randomized trial comparing ACI to mosaicplasty, Bentley, et al. **YEAR?** found that the Cincinnati Rating Scores were better for ACI than mosaicplasty, but not statistically significant. The study included an arthroscopic assessment at one year where ICRS repair scores were assigned. Only 60 of the 100 patients (60%) returned for follow-up. There were 82% of the patients in the ACI group that had a repair score of good or excellent. This is significantly higher than the 34% of the patients in the mosaicplasty group, with at least a good repair score.²⁵ The study included an arthroscopic assessment at one year where ICRS repair scores were assigned. Only 60 of the 100 patients (60%) returned for follow-up.

In their 2017 meta-analysis, Schuette, et al. [21], guide their readers through a brief history of the evolution of ACI, a restorative approach to chondral defects. They write:

“ACI is a 2-step procedure in which chondrocytes are first arthroscopically harvested and cultured *in vitro*. The second procedure involves injecting the chondral defect with the cultured cells and then covering them with a periosteal patch (first-generation ACI) or collagen membrane (second-generation ACI). Issues negatively affecting clinical outcomes, such as periosteal patch hypertrophy associated with the first-generation ACI and extensive suturing and cell leakage associated with second generation ACI, have led to the development of third generation ACI, otherwise known as matrix-assisted autologous chondrocyte transplantation (MACT). In MACT, cultured chondrocytes are seeded into a matrix scaffold and then fixed to the chondral defect with fibrin glue. MACT was first introduced into clinical practice in Europe in 1998. Third-generation ACI was just recently approved by the Food and Drug Administration in the United States [MACI by Vericel, Inc., in December of 2016], with several additional clinical trials currently underway.”

With both Carticel and MACI, the body of clinical evidence for autologous chondrocyte implantation was collected in larger defects of approximately 5 cm², with concomitant procedures and previous microfracture failures, whereas current FDA guidance has required far stricter inclusion/exclusion criteria. Upon approval of MACI by FDA, Vericel, Inc., chose to conclude the commercialization of Carticel in Summer of 2017, and the product is no longer sold to clinicians. The underlying patient biopsies originally used for Carticel were transitioned to MACI upon commercialization in early 2017.

7.3.2. MACI (Vericel, Inc.)

In the MACI technique, in the initial arthroscopic procedure, chondrocytes are harvested from a non-weight-bearing articular surface and cultured *ex vivo* in a monolayer. These cultured chondrocytes are then seeded on one side of a membrane comprised of Type I and Type III porcine collagen. During the second procedure, performed arthroscopically or via a mini-arthrotomy, the chondrocyte-seeded side of the membrane is placed against the subchondral bone surface and affixed using a fibrin glue [22].

As referenced earlier, MACI received FDA approval for US commercialization in December of 2016. Key clinical evidence for MACI came from the SUMMIT [Demonstrate the Superiority of MACI Implant to Microfracture Treatment] multi-center study, whose results were published by Saris, et al.

Commented [AG2]: Cite 2011 Harris study (Osteoarthritis and Cartilage 19 (2011) 779e791)

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[23]. The investigators randomized patients with ≥ 1 focal chondral defect (Outerbridge grade III or IV and defect size ≥ 3 cm²) on either the femoral condyles or trochlea into either treatment with MACI or microfracture. Only patients with a Knee Injury and Osteoarthritis Outcome Score (KOOS) pain value < 55 were enrolled. The co-primary efficacy endpoint was the change in KOOS pain and function subscores from baseline to 2 years. Secondary and tertiary endpoints included: histological evaluation and MRI assessments of structural repair tissue, treatment failure, the remaining 3 KOOS subscales, and safety.

A total of 144 patients were enrolled, 72 in each treatment group; 137 patients (95%) completed the two-year follow-up (70 in the MACI group and 67 in the microfracture group). The mean defect size was equivalent between the two groups (4.9 cm² for MACI and 4.7 cm² for microfracture). The mean patient ages were 34.8 (MACI) and 32.9 (microfracture). At the two-year follow-up, the change over baseline in the KOOS pain and function subscores for MACI was significantly higher than the microfracture group (KOOS pain: MACI, 37.0 to 82.5 vs. microfracture, 35.5 to 70.9; KOOS function: MACI, 14.9 to 60.9 vs. microfracture, 12.6 to 48.7; $p = 0.001$). Significant improvements in the other KOOS subscales were seen in the MACI patients vs. the microfracture patients. Though repair tissue quality was “good” via histology/MRI assessments, no difference was seen between the two treatments. In this study, a “responder” was defined as someone that realized at least a 10-point improvement in both the KOOS pain and function subscales. At 2 years, 87.5% of patients in the MACI group were responders, while 68.1% of the microfracture patients were responders. Treatment-emergent adverse events (TEAEs) related to the study treatment were similar between the two groups (MACI: 34.7% and microfracture: 38.9%). The most common TEAEs related to the treatment were treatment failure, arthralgia, and joint swelling.

7.3.3. NeoCart (Histogenics, Inc.)

Like MACI, the NeoCart product begins by harvesting healthy chondrocytes from a non-weight-bearing portion of the knee. Unlike MACI, however, these chondrocytes are cultured on a three-dimensional “honeycomb-like” bovine-derived Type I collagen scaffold within a low oxygen-tension custom bioreactor designed to mimic the intra-articular environment [22]. The goal is to create a neocartilaginous implant that possesses increased glycosaminoglycan (GAG) and type-II collagen production. Thus, instead of implanting cells with the potential to differentiate and form hyaline cartilage, NeoCart “is a matured tissue-like implant containing hyaline GAGs designed to provide a more rapid and potentially sustained therapeutic effect for articular cartilage repair” [24]. The NeoCart implant is secured to the subchondral bone using a proprietary resorbable tissue adhesive (CT3, Histogenics, Waltham, MA).

7.3.3.1. Phase 1 Study:

The Phase I multi-center study was conducted in 2005 to assess the safety of NeoCart. It was an open label, single arm study of 8 patients with a single focal cartilage defect of the knee who were followed for five (5) years post implantation. The average age was 38 years, 63% were male. Baseline mean IKDC was 57 ± 25 and 75% had undergone previous knee surgery. There were only two related safety events with dislodgement of the NeoCart implant at 1 month and failure of the NeoCart implant to remain in place during the operative procedure. While the focus of the study was safety, efficacy data was also gathered.

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Throughout the 5-years post-implant, study subjects demonstrated improved ROM, function and pain scores. Average ROM was 128 ± 10 at baseline and increased at 6-months post-procedure (131 ± 8) and continued to improve throughout the study time course (5 years 137 ± 5). IKDC function scores also increased throughout the study, peaking at 36-months post-procedure at 78 ± 21 . This increase was greater than 20 points over the average baseline value (57 ± 25). IKDC decreased slightly at 4 and 5 years but remained improved over baseline values. One subject, 03-004, presented improved IKDC results only at 2 and 3 years, and returned to levels below baseline at 4 and 5 years. Average baseline VAS scores were 33 and decreased (improved) throughout the study, with an average decrease of 20 points from year 1 through year 5. With respect to MRI, 88% of subjects demonstrated good fill at the defect site and evidence of T2 stratification at 1 year.

7.3.3.2. Phase 2 Study

The Phase II NeoCart study was a prospective multi-center study conducted between 2006 and 2008 that compared the performance and safety profile of NeoCart versus microfracture in a controlled, randomized (2:1), multi-center clinical study. The objective of the study was to further evaluate the safety of NeoCart to fill articular cartilage defects in the knee, gather additional efficacy data, identify clinically meaningful endpoints, identify appropriate patient populations, and to obtain additional data to be used in the design of future clinical studies. A total of thirty-five patients were enrolled at 7 sites in the United States. Five patients were excluded during arthroscopy, leaving 30 patients, 21 in the NeoCart cohort and 9 in the microfracture cohort. The rehabilitation protocol, originally developed for microfracture recovery, was identical for both arms. Performance measurements included the following self-reported patient surveys: The International Knee Documentation Committee (IKDC) form (subjective); Knee Injury and Osteoarthritis Outcome Score (KOOS) pain, symptoms, activities of daily living (ADL), quality of life (QOL), and sports scales; Short Form-36 (SF-36) questionnaire; and a visual analog scale (VAS) pain scale. Clinical assessment and questionnaires were conducted at 3, 6, 12, and 24 months post-procedure. In addition, the researchers conducted a “responder analysis” by defining a “responder” as a patient that demonstrated a change in their KOOS pain score of ≥ 12 points and a change in the IKDC function score of ≥ 20 points.

Of the 30 treated patients, the patients were primarily male (90.5% in the NeoCart group and 66.7% in the microfracture group), white race (81.0% in the NeoCart group and 88.9% in the microfracture group) and had a mean age at baseline of 41.39 years in the NeoCart group and 38.84 in the microfracture group. The mean BMI was statistically significantly higher in the NeoCart group with a BMI of 28.88 kg/m² compared to 25.37 kg/m² in the microfracture group ($p=0.0096$; see

The location of injury for the majority of patients was the medial condyle (71.4% of patients in the NeoCart group and 66.7% in the microfracture group). The mean age of injury in the treated knee was similar in both groups (2.80 years in the NeoCart group and 2.22 years in the microfracture group). Mean lesion sizes prior to treatment and with treatment were also clinically similar between the treatment groups. A greater percentage of patients in the NeoCart group (71.4%) had a prior surgical history to the study knee compared to 44.4% of patients in the microfracture group. All other demographics and baseline characteristics were generally clinically balanced between the treatment groups

The safety analysis for the phase II study was favorable with no concerning issues. No clinically meaningful differences were noted in AE rates between the NeoCart and microfracture treatment groups. The most common TEAE assessed as related to NeoCart was Arthralgia; this was also the most common

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procedure-related TEAE in the microfracture group. There were no SAEs assessed as related to NeoCart. One death occurred in the microfracture group completely unrelated to the treatment and occurring at the very end of the 5-year study period.

There were no statistically significant differences in the mean change from baseline in range of motion between the NeoCart and microfracture groups at any time point. Within treatment groups, there was a statistically significant increase in the mean range of motion change from baseline in the NeoCart group at 12 months in both the passive (4.2; $p=0.0301$) and active (5.4; $p=0.0105$) parameters. Similarly, at the 5-year visit, the NeoCart group had statistically significant increases in the range of motion change from baseline in both the passive (12.1 degrees; $p=0.0003$) and active (11.9 degrees; $p=0.0010$) parameters. In the microfracture group, there were no statistically significant changes from baseline in the IKDC mean range of motion at 12 months and 5 years.

For IKDC subjective score changes there were statistically significant differences in the mean changes from baseline for subjective scores between the NeoCart and microfracture groups at the 1 year (14.90, $p=0.0090$) and 2 year (13.15, $p=0.0428$) time points; At 12 months, the NeoCart group had a mean IKDC subjective score increase from baseline of 31.25 ($p<0.0001$) which increased to 34.10 ($p<0.0001$) by the 5 year time point. The microfracture group also had improvements in the mean IKDC subjective scores. At 12 months, the microfracture group had a mean IKDC subjective score increase from baseline of 16.35 ($p=0.0002$), which increased to 37.07 ($p=0.0149$) by the 5 year time point. The KOOS instrument assessed the following 5 subscales: symptoms/stiffness, pain, daily living function, sports and recreational activities function, and quality of life. In general, there were statistically significant improvements from baseline seen in each of the subscales at the majority of time points in the NeoCart group. There were a few statistically significant increases from baseline at the protocol required time points noted in the KOOS subscale scores in the microfracture group. Overall, the increases from baseline in the KOOS scores were greater in the NeoCart group, with some statistically significant differences in mean change from baseline between the groups. The pain subscale indicates highly statistically significant improvements from baseline at each time point in the NeoCart group, as well as statistically significant differences in mean change from baseline between groups at 6 months, 12 months, 3 years, and 4 years.

A dual threshold responder analysis was an exploratory efficacy endpoint with the definition of a responder being a KOOS pain score improvement of 12 points and IKDC improvement of 20 points. At 12 months 66.7% of the NeoCart patients were responders and only 22.2% of the microfracture patients were ($P=0.0457$).

The NeoCart group had statistically significant improvements from baseline on the SF-36 in the mean physical component summary scores at all protocol-specified time points. The microfracture group had statistically significant increases from baseline on the SF-36 in the mean physical component summary scores at years 1 to 5. In the mental component summary of the SF-36, the only statistically significant increase from baseline was in the microfracture group at the 3-month time point. No differences between groups in the SF-36 mean physical and mental summary scores reached statistical significance.

Statistically significant mean improvements from baseline in the VAS pain assessment were seen in each of the three categories (lowest, average, and highest pain) at each time point in the NeoCart group. No statistically significant mean increases from baseline were seen in the VAS pain assessment in any of the categories (lowest, average, and highest pain) at any of the protocol-specified time points in the microfracture group. There were also several statistically significant differences noted between the

groups in the VAS pain assessment categories (lowest, average, and highest pain), with greater decreases in pain seen in the NeoCart group

7.4. The role of Physical Therapy/Rehabilitation

All patients undergoing treatment for cartilage damage of the knee irrespective of treatment modality should undergo a monitored rehabilitation program to ensure optimum results. In the article, “Functional Outcomes after Surgical Management of Articular Cartilage Lesions in the Knee: A Systematic Literature Review to Guide Postoperative Rehabilitation,” Schmitt, et al. [16] conclude from their research, “Significant quadriceps femoris strength deficits, gait deviations, and functional deficits persist for 5 to 7 years following ACI and microfracture surgical procedures. Future research regarding rehabilitation interventions to help mitigate these deficits is warranted.” As detailed in articles by Wilk, et al. [17] and Mithoefer, et al. [18], optimal post-operative rehabilitation of the knee after articular cartilage repair should be individualized to the patient: tailored to the patient’s age and BMI, his/her defect size, depth, and location, the repair technique used, duration of symptoms prior to intervention, and evolving quality of the cartilage [18]. As both sets of researchers instruct, the phases of post-operative physical therapy should mirror the biologic phase of wound repair: during Phase 1 (proliferative phase) of cartilage repair, the initial healing benefits from a reduction in swelling and a gradual restoration of passive range of motion and weightbearing [17]. During Phase 2 (transition phase), a patient proceeds from gradual to full weightbearing with a focus on joint function restoration. During the final phase (remodeling/maturation), the goal is to have the patient return to full activity. Recognize the inherent variability associated with post-operative rehabilitation: failure to individualize treatment to the patient’s condition, and failure to match the strength and motion restoration to the evolving healing of the cartilage defect, can all lead to considerable variability in patient outcomes.

Ebert, et al., [19] demonstrated that differences in rehabilitation regimens could alter the outcomes after chondral injury restoration using matrix-induced autologous chondrocyte implantation (MACI, discussed below). An accelerated weightbearing regimen (full weightbearing at 8 weeks post-procedure) was compared to a traditional regimen (full weightbearing at 11 weeks post-procedure) by measuring pain and various functional metrics at 3, 6, 12, and 24 months post-surgery. Patients in the accelerated weightbearing regimen reported significantly lower pain and demonstrated improved 6-minute walk distances compared to the those in the traditional rehabilitation regimen. While it is imperative to highlight that these patients received chondral restoration rather than microfracture – and recognizing that the needs of graft repair healing may be different – research by Ebert, et al., is meant to emphasize that differences in rehabilitation can affect surgical outcomes in treating chondral defects.

That said, research by Marder et al. [20], suggested that the impact of rehabilitation on functional outcomes can be secondary to other, more significant factors. In their study, patients with focal full-thickness chondral defects less than 2 cm² treated with microfracture entered one of two rehabilitation groups post-surgery: in one group, patients were treated with non-weight bearing and continuous passive motion (CPM) for six weeks, while in the second group, patients were allowed weight bearing as tolerated and did not use CPM. Outcomes were assessed using the Lysholm knee rating scale augmented by the Tegner method of activity evaluation. No significant difference was detected between the two groups when comparing the pre-operative and post-operative Lysholm scores. It is possible that the healing of small focal defects after microfracture is insensitive to differences in post-surgical rehabilitation.

8. STUDY OBJECTIVES

The objectives of this study are 1) to evaluate the safety and efficacy of, NeoCart, in the repair of cartilage defects of the femoral condyles and/or trochlea and 2) to compare NeoCart safety and efficacy to microfracture treatment.

9. INVESTIGATIONAL PLAN

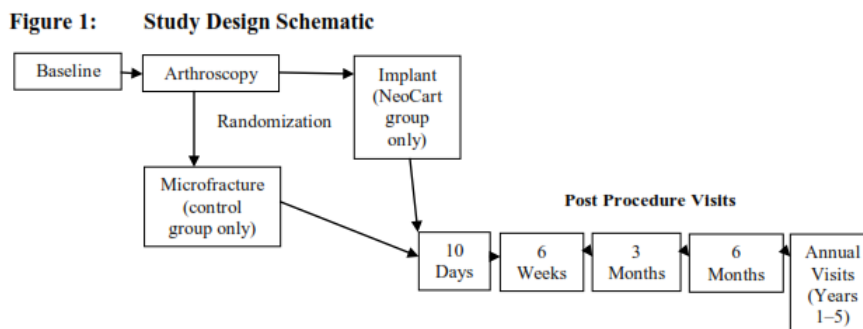
9.1. Overall Study Design and Plan: Description

This was a prospective, controlled, randomized, multi-center clinical study. The initial planned sample size of 245 was randomized 2:1 between NeoCart and microfracture at up to 40 investigational sites. Randomization was done at arthroscopy, at which time final patient eligibility was determined. Neither surgeon nor patient was blinded to treatment. The International Knee Documentation Committee (IKDC) Objective knee examination and the magnetic resonance imaging (MRI) reads were blinded.

The primary analysis of success is a responder analysis at 12 months in which the proportion of responders in the treatment arm is compared to the proportion of responders in the control arm. In this study, a responder is defined as a patient with an increase of at least 12 points on the KOOS pain assessment and at least 20 points on the IKDC Subjective assessment at the 12-month visit. The data was tested for superiority.

The study design is presented in [Figure 1](#).

Figure 1: Study Design Schematic



Commented [LK3]: Need to change annual visits to 1-3 years.

Abbreviations: D=Day; EOS=end of study; FU=follow-up

9.2. Discussion of the Study Design, Including the Choice of Control Groups

The NeoCart Phase III study was designed to meet the study objectives above, and was developed over a decade ago, before FDA guidance was published for products being investigated to treat cartilage defects of the knee. Histogenics engaged in multiple discussions with the FDA from 2007 through early 2009, in order to inform the Phase III study design. The agreed-upon Phase III clinical study protocol

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was submitted to the IND under a Special Protocol Assessment request and was approved by FDA on March 10, 2009. The trial was required to be a prospective, multi-center randomized trial of superiority comparing the success of NeoCart to that of microfracture. The primary endpoint was agreed upon to be one year with continuation through years 2 and 3. Although the trial was designed prior to the guidance document it complied with the requirement to demonstrate superiority and to have two different validated scores for pain and function endpoints to assess for meaningful improvement. The agreed upon endpoints were KOOS pain and IKDC subjective function scores which are very well accepted validated scoring systems for both pain and function of knee pathology. All KOOS and IKDC subscores were prospectively obtained at baseline and at all post-procedure visits

The guidance document does not require that these two different tests be combined into a dual threshold primary responder analysis which was done in the NeoCart trial. This added an additional hurdle as a patient was required to reach the success levels for both measures to be considered a positive responder.

To meet the objectives the primary endpoint for the Phase III study was defined as a test of superiority (one-sided Pearson's chi-square test, $\alpha=0.025$) vs microfracture, in terms of a composite clinical success (CCS) used in a dual threshold responder analysis for pain and function at 12-months post index surgery. A subject was considered a responder if they achieved both of the following outcomes:

- An improvement of at least 12 points in outcome compared to baseline on the KOOS pain assessment test, and
- An improvement of at least 20 points in outcome compared to baseline on the IKDC subjective test.

These improvement thresholds were set based upon the available data at the time and above what are currently considered minimum clinically important differences (MCID) for improvements on pain and function as established in the literature. The MCID had not been well established at the time of the writing of the protocol and the thresholds were determined with considerable input from the performance in the Phase II study where microfracture performed poorly.

The current standards for clinically meaningful IKDC scores start at a 6-point improvement as early as 6 months and increase to 15 or 16 points at later timeframes. For KOOS pain, the meaningful clinically important difference is seen in the 8 to 10-point range. Both thresholds in the NeoCart trial (KOOS of 12 points and IKDC of 20 points) were higher than what was and is now seen as a minimum clinically meaningful or important difference, or MCID ([Collins, 2011](#)).

Microfracture was used as the concurrent control for the NeoCart Phase III study as it is the most common surgical treatment modality, and current standard of care both at the time of the study design and persistent today despite the limitations of the procedure as noted for both efficacy and durability. Microfracture continues to be used as both active and as an historical control in other ongoing studies for treatment of knee cartilage defects based upon FDA guidance. Despite extensive use of microfracture commercially over the last two decades in broad patient populations, a treatment effect in the population evaluated in this Phase III study is not well established due to a paucity of adequate and well-controlled studies that are designed in accordance to the now available FDA guidance. The sponsor established inclusion and exclusion criteria and a study population based on FDA and clinical expert feedback to provide the microfracture treatment the best possible chance to succeed and attempt to institute clinical equipoise for the study, as there are well-known limitations to this treatment. Key inclusion and exclusion criteria were lesions less than 6cm², BMI less than 35, no concomitant knee surgical procedures, and no prior microfracture treatments allowed.

Commented [AG4]: We should build this out stepwise: Trials at that point had to be superiority, had to be compared against MFX and had to have pain & function endpoints.

We did that, and describe the prospective collection of KOOS and IKDC as starting point, per the now available guidance.

THEN describe how we developed the primary endpoint of the responder analysis.

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The FDA published the guidance “Preparation of IDEs and INDs for Products Intended to Repair or Replace Knee Cartilage” (“FDA Guidance”) for products indicated for treatment of knee cartilage in December 2011. Although the NeoCart Phase III study was initiated prior to the FDA guidance, the study design is consistent with the FDA guidance, including the requirement for improvement in both pain and function and a test of superiority versus an active concurrent control (i.e. microfracture). It includes an extension phase of an additional 3-years (assessment at 2-year and 3-year timepoints post-surgery) to evaluate durability. The FDA Guidance does not mandate a dual-threshold responder analysis.

9.3. Selection of Study Population

Patients were considered for enrollment into the study if they met all of the inclusion criteria and none of the exclusion criteria prior to randomization.

9.3.1. Inclusion Criteria

9.3.1.1. General Inclusion Criteria

Patients meeting all of the following criteria may have been included into study. The patient must have been:

- Able and willing to give informed consent.
- 18 years to 59 years of age. (increased from original age limit of 55 Rev 8 November 10, 2015).
- Symptomatic with knee pain indicative of an articular cartilage injury of the distal femur. Baseline KOOS pain score must be ≤ 80 and baseline IKDC Subjective score must be ≤ 70 .
- Medically able to undergo arthroscopy, arthrotomy for NeoCart implantation, and serial MRIs.
- Females must be post-menopausal or practicing an acceptable form of birth control.
- The following prior procedures are permitted after ≥ 3 months have passed since the procedure if, at the discretion of the investigator, pain is determined to be specifically referable to the cartilage defect and not residual from the prior injury or surgical procedure:
 - Partial meniscectomy ($<50\%$ radial width, preserving fully intact peripheral rim) on the study knee
 - Meniscal repair on the study knee
 - Ligament reconstruction on either knee

9.3.1.2. Arthroscopic Inclusion Criteria

Patients meeting all of the following criteria may have been included into study:

- At least one ICRS grade 3 or 4A lesion located on the femur (lateral/medial condyles and/or trochlea that would be treatable by microfracture. ICRS grade 4A lesions must have no substantive bone loss. Trochlear lesions must not be associated with clinically significant patellar malalignment.

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- One or two non-contiguous and non-opposing lesions may be treated as long as each lesion meets the arthroscopic inclusion and exclusion criteria and the total surface area of both treated lesions is ≤ 6 cm² (size of NeoCart). With respect to the individual lesion size, each treated lesion must be at least 1 cm in one linear dimension. The range of lesion sizes treated shall be between 0.5 cm²- 6 cm².
- Permitted concomitant procedures include:
 - Medial or lateral partial meniscectomy of less than 50% radial width, preserving a fully intact peripheral rim.
 - Removal of loose bodies or impending loose bodies.
 - Debridement/chondroplasty of non-study lesions.
 - Synovectomy.
 - Judicious excision of tissue from the study lesion(s) to the extent necessary to facilitate measurement and grading of the lesion(s) in order to complete the final qualification step prior to randomization.

9.3.2. Exclusion Criteria

9.3.2.1. Pre-Procedure Exclusion Criteria

Any patient who met any of the following criteria was not included into this study.

- Any previous surgical treatment of the study defect (such as abrasion arthroplasty, abrasion chondroplasty, multiple drilling or microfracture). Prior debridement or shaving of the articular cartilage/chondroplasty of the study lesion was permitted but must have been performed >3 months prior to the study arthroscopy.
- The following prior procedures within 3 months prior to study arthroscopy, in the judgment of the investigator, if pain was not specifically referable to the cartilage defect:
 - Partial meniscectomy or meniscal repair on the study knee
 - Ligament reconstruction on either knee
- Surgery on the contralateral knee within 3 months prior to study arthroscopy.
- Body Mass Index (BMI) > 35 kg/m².
- Radiographic evidence of osteoarthritis, Kellgren-Lawrence Grade 3 or greater.
- Malalignment greater than 3 degrees outside the mechanical axis from the contralateral knee, or malalignment requiring surgical correction.
- Symptomatic pathology of the contralateral knee.
- Any form of inflammatory arthritis.
- Ankylosing spondylitis.
- Synovoma, hemangioma, pigmented villonodular synovitis (PVNS), or other neoplasms in the knee.
- Any condition requiring chemotherapy.
- Positive pregnancy test or intention to become pregnant during the year following treatment.

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- Known history of allergy to bovine products or gentamycin.
- Any autoimmune disease.
- Human immunodeficiency virus (HIV), chronic hepatitis B or C viral infection.
- Insulin-dependent diabetes.
- Clinically significant or symptomatic vascular or neurologic disorder of the lower extremities.
- Current drug or alcohol abuse.
- Evidence suggesting inability to comply with the Protocol.
- Smoking habit of more than 1 pack of cigarettes per week and/or were frequent users (>1/week) of chewing tobacco. The number of smokers were limited to 10% of the enrolled patients.
- Any concomitant painful or disabling disease of the hips or lower limbs that would have interfered with the evaluation of the afflicted knee.
- Treatment with specific osteoarthritis drugs, such as chondroitin sulfate, diacerein, n-glucosamine, piacledine, and capsaicin, within the last 2 weeks.
- Corticosteroid therapy by systemic or intra-articular route within the last month of baseline or intramuscular or oral corticosteroids within the last 2 weeks.
- Intra-articular injections of hyaluronic acid into the afflicted knee within the last 6 months.
- Status as a participant in an investigational research trial within the last 3 months.

9.3.2.2. Arthroscopic Exclusion Criteria

Any patient who met any of the following criteria was not included into this study:

- Subchondral bone loss.
- Untreated ACL and/or posterior cruciate ligament (PCL) deficiency or ligamentous instability in the index knee.
- Study lesion requiring a concomitant procedure intended as a therapeutic intervention to the study lesion, e.g., debridement/chondroplasty, or that may have altered the mechanics of the joint, e.g., osteotomy. Permitted concomitant procedures are detailed in the inclusion criteria above.
- Meniscus whose rim was less than 50% normal radial width or whose peripheral rim was not fully intact.
- ICRS Grade 3 or 4 kissing lesions.
- ICRS Grade 3B or greater patellar or tibial lesions associated with pain specifically and reproducibly referable to the untreated lesions(s).

9.3.3. Removal of Subjects from Therapy or Assessment

Patients were informed that they could choose to withdraw from the study at any time.

9.4. Treatments

A description of the test treatments, microfracture and NeoCart, may be found in Section 9.5.4 Section 9.5.5, respectively.

9.4.1. Method of Assigning Subjects to Treatment Groups

Patients were assigned to treatment based upon a 2:1 randomization scheme. Randomization took place at the time of arthroscopy. Individual blinded patient randomization envelopes were created based upon this ~~schedule design~~ and distributed to the investigational sites. For each patient, site personnel would open the envelope during the arthroscopic procedure to determine the treatment assignment. Envelopes were to be opened sequentially within a strata and were created by Biostatistical Consulting Inc.

The Randomization List used to create envelopes was created using two strata (Stratum 1 (BMI < 28kg/m²) and Stratum 2 (BMI ≥ 28kg/m²). Stratum one was placed in blue envelopes and stratum two in black envelopes. Randomization was created by site and stratum using block sizes of six. The randomization scheme and codes are provided in Section 16.1.7.

9.4.2. Selection of Doses in the Study

Not applicable.

9.4.3. Drug Administration

Not applicable.

9.4.4. Blinding

Randomization was done at arthroscopy, at which time final patient eligibility was determined. During the arthroscopy procedure, the envelope was opened after final eligibility was confirmed by the investigator. Following the opening of the randomization envelope the surgeon, the study staff and the patient were not blinded to treatment. The IKDC Objective knee examination and the MRI reads were blinded.

9.4.5. Prior and Concomitant Therapy and Procedures

No prior surgical treatment of the study lesion was permitted including abrasion arthroplasty, abrasion chondroplasty, multiple drilling or microfracture. Further details are provided in the arthroscopic exclusion criteria section 9.3.2.1.

Prior debridement or shaving of the articular cartilage/chondroplasty of the study lesion is permitted but must have been performed >3 months prior to the study arthroscopy. Investigators were permitted to perform acceptable minor concomitant procedures during the arthroscopy. Details provided in the arthroscopic inclusion criteria section 9.3.1.2 .

Patients were not included into the study if the resulting meniscus rim was less than 50% of normal radial width, or there were unresolved complications during the procedure. No patients included in this study had ACL and/or PCL reconstruction concomitant with their biopsy or microfracture.

Debridement/chondroplasty of the study lesion was not permitted unless the patient was randomized to microfracture.

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9.4.6. Treatment Compliance

Both the NeoCart and microfracture procedures were performed by the same study investigators. Rehabilitation status was assessed at each visit. Compliance with participation in study visits was very high. For the 6 month visit 98.7 % of NeoCart patients and 96.0% of MF patients had complete data for each timepoint. For the 1-year ~~end-point~~ visit 158/159 (99.4%) of NeoCart patients and 72/73 (98.6%) of the MF patients were evaluable for the Primary Endpoint. Per protocol was slightly lower for each group due to deviations of the follow up visit being slightly out of the window, 88.3% for NeoCart at 1 year and 82.9% for MF. The out of window deviations had no impact on the primary and secondary endpoints.

9.5. Study Procedures

9.5.1. Screening and Baseline Procedures

The following procedures were performed after informed consent was obtained and within 3 months of the arthroscopy, unless specifically noted:

- a. Collect the patient's medical, surgical history and demographic information.
- b. Collect baseline pain medications taken in the one week prior to baseline.
- c. Collect baseline concomitant medications taken in the one week prior to baseline.
- d. Record previous treatments/procedures performed to the target knee in the 1 year prior to surgery.
- e. Perform a knee examination per the IKDC Objective Knee Evaluation Instructions (Protocol Attachment C).
- f. Obtain blood samples* for the following tests as per site standard of care:
 - Hematology: complete blood count (CBC) with differential; prothrombin time (PT), partial thromboplastin time (PTT), international normalized ratio (INR).
 - Metabolic Panel: (inclusive of) electrolytes, renal and liver function tests.
 - Viral Labs:
 - HIV screening test, hepatitis B (HBsAg, HBsAb and HBcAb tests) and hepatitis C (Hepatitis C antibody, and in the high-risk patients defined below, hepatitis C viral ribonucleic acid [RNA]) analyses.
 - Hepatitis C viral RNA test was required only in patients at high risk of recent infection due to:
 - ~~No b~~lood transfusions or organ implants allowed in the past three months.
 - ~~No c~~evidence of liver injury (abnormal alanine aminotransferase [ALT] levels).
 - ~~No report of a~~lcohol abuse and/or drug use and high-risk sexual activity in the past 3 months (Note: Patients who abuse alcohol/drugs fail entrance criteria).

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*Note: Blood tests must were required to be obtained within 30 days of the arthroscopy procedure except for the viral tests, which must were required to be obtained and reported within 7 days prior to the procedure.

- g. Perform a pregnancy test if the patient was female and capable of bearing children.
- h. Obtain a urine sample for urinalysis.
- i. Instruct the patient how to complete the KOOS, IKDC Subjective, VAS pain, Marx Activity Rating Scale, SF-12 Health Survey, European Quality of Life – 5 dimensional –5 level (EQ-5D-5L) Health Questionnaire, and Work Productivity and Activity Impairment (WPAI) Questionnaire and collect the completed forms. The pain medication forms must be completed on the same day the initial patient surveys were completed.
- j. Obtain radiographs within 3 months of the study arthroscopy in the following views:
 - Both legs, standing, full extension from femoral head to tibial malleolus, anteroposterior.
 - 45° flexion, weightbearing, posteroanterior.
 - Merchant view - 30° flexion.
- k. Obtain a study MRI within 3 months of study arthroscopy.

A patient was provisionally enrolled and scheduled for arthroscopic examination if he or she met all baseline inclusion criteria and did not meet any exclusion criteria. After provisional enrollment, the patient completed the initial patient diary employment question prior to the arthroscopy procedure.

9.5.2. Arthroscopy Procedure

The arthroscopy procedures were scheduled in order to allow ample time for lab processing, MRI image submission and core lab baseline letter return prior to the procedure. All required baseline results were available before the time of arthroscopy. Data was recorded during the procedure to document that the patient met the selection criteria. Data collection included assessments of all cartilage defects and status of all structures in the knee compartments.

9.5.3. Biopsy Procedure

The steps for the Biopsy Procedure were performed in accordance with the Instructions for Use (IFU), after patient randomization to the NeoCart cohort.

9.5.4. Microfracture Procedure

The following steps, described in more detail by Steadman, *et al.* [27], were performed after patient randomization to the microfracture cohort:

- a. Debride the lesion of all remaining unstable cartilage. All loose or marginally attached cartilage from the surrounding rim is to be debrided to form a stable, perpendicular edge of healthy, viable cartilage.
- b. A curette is used to carefully remove the calcified layer from the base of the lesion. The calcified cartilage comprises a layer between the deep zone of cartilage and the subchondral bone and this layer can increase in thickness with age. The calcified layer can be difficult to differentiate visually but can be distinguished more easily from the hard, underlying

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subchondral bone plate by tactile feedback. The use of a curette provides better manual feedback than an arthroscopic shaver and reduces the risk of excessive debridement.

- c. Make multiple holes in the exposed subchondral bone plate using an arthroscopic awl. A Linvatec Corp. awl is recommended. The holes should be made perpendicular to the surface.
- d. Microfracture holes should initially be made around the periphery immediately adjacent to the healthy stable cartilage, and then progressing toward the center of the lesion. The holes should be made 3-4 mm apart and should be to a depth of 2 to 4 mm. A minimum spacing should be maintained so that one hole does not break into another and damage the subchondral plate between holes. There should be visible marrow fat droplets and blood released from the microfracture holes.
- e. Evacuate all fluid and close the access sites without placement of intra-articular drains and dress the incision following standard hospital procedures.
- f. Post-surgical recovery care was conducted in accordance with standard hospital procedures and the patient is to be discharged at the discretion of the investigator.

9.5.5. NeoCart Implant Procedure

The NeoCart derived from the biopsy sample was intended to be implanted into the same patient. Therefore, utmost care was taken to match the identity of the patient to the patient indicated on the NeoCart label. An arthrotomy procedure was required for NeoCart implantation into the defect bed. The following steps were followed during the implant procedure:

1. The patient was prepared for arthrotomy using conventional preparation. Standard hospital procedures were followed to gain access to the defect.
2. Any fragments and debris from the defect were removed and the defect bed was smoothed. The zone of calcified cartilage was carefully scraped away taking care not to penetrate the subchondral bone. A tourniquet was not used during the procedure so that inadvertent bleeding from the subchondral bone could be ascertained and stopped prior to the implantation of the NeoCart.
3. The NeoCart was removed from the shipping container. The size and shape of the defect to be treated were determined and the NeoCart was trimmed to provide an implant piece that fit the defect dimensions. The trimmed NeoCart was then positioned into the defect to confirm the NeoCart fit.
4. A CT3 resorbable tissue adhesive kit was then opened and prepared. The CT3 was used within 20 minutes, or the material was discarded and new CT3 mixed.
5. A small amount of CT3 was evenly spread into the defect bed, and then the trimmed NeoCart implant was placed.
6. After NeoCart placement, moderate pressure was briefly applied to the implant to allow the CT3 to set. The NeoCart implant was then affixed to the defect. (The NeoCart implant was not to be repositioned after initial placement.)
7. A thin layer of CT3 was spread over the NeoCart and also extended the CT3 layer onto the native articular cartilage surface. The CT3 was to extend at least 2 mm past the NeoCart around the periphery of the defect. The surface of the CT3 was also to be as smooth as possible, without being proud in relation to the surrounding articular cartilage.
8. The Curing Component buffer was applied to the CT3 and the CT3 was allowed time to set. The setting time was accelerated by the use of the buffer. Any excess CT3 was trimmed to create a smooth surface.
9. Following the surface application of CT3, the leg was carefully extended to avoid excessive shear stresses on the NeoCart. No range of motion evaluations were performed, and additional flexing of the knee was to be avoided.

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10. The arthrotomy was closed and the incision dressed following standard hospital procedures. A brace was applied to immobilize the knee and locked at an angle appropriate to minimize joint movement at the Investigator's discretion, depending on the implant location. The patient was then fitted with anti-embolism stockings to minimize the risk of embolism.

11. Post-surgical recovery care was conducted in accordance with standard hospital procedures and the patient was discharged at the discretion of the Investigator

Rehabilitation

Rehabilitation was guided by the investigator in general phases described below and advanced according to each patient's specific issues (healing and function). The primary driver of the very prescriptive rehabilitation protocol was the microfracture procedure and the seminal work by Dr. Steadman regarding how to optimize success of microfracture. This rehabilitation plan was individualized, phased and included a detailed description of activities and limitations. It that incorporated specific strengthening and conditioning goals to be attained prior to advancing to the next phase of rehabilitation and to facilitate successful recovery from surgery, keeping in mind the patient's goals and previous activity level. Exercises were specified, and patients trained to complete the exercises, but the specific exercises and training on how to perform the exercises was done at the discretion of the investigator. Patients were contacted periodically for reinforcement and support in the performance of the exercises.

Use of Knee Brace:

Patients with condylar lesions in either study arm were not required to wear a knee brace but may have been prescribed a brace at the investigator's discretion. Patients with trochlear lesions should have been placed in a locking brace immediately and for at least 6 weeks post-operatively, with the unlocking schedule determined by the investigator. If the investigator prescribed a knee brace for patients, this was documented in the patient's file.

Prophylaxis for Thromboembolic Disease (TED):

For any patient placed into a knee immobilizer or partially- or fully-locked knee brace following any study procedure, two methods of thromboembolic disease (TED) prophylaxis, aspirin and a thigh-high or knee-high compression stocking for the operative extremity, was ordered for the duration of immobilization/bracing. The dosage of aspirin and the period of daily use of compression stockings was determined by the investigator.

Continuous passive motion therapy (CPM):

Continuous passive motion therapy (CPM) began on the day of the procedure for all patients not placed into a knee immobilizer or hinged knee brace with restricted motion. For patients placed into a knee brace or immobilizer, and at the discretion of the investigator, CPM began as early as the day of the procedure but, in any case, no later than the first post-operative visit. CPM continued for a minimum of 6 weeks and a maximum of 8 weeks. All patients were recommended to perform CPM for 6-10 hours per day, but investigators could alter the daily amount based on each patient's needs.

Weightbearing Status:

NeoCart patients with small condylar lesions that were less than 2cm² with a circumferentially stable margin of intact cartilage were limited to toe touch weightbearing until the first post-operative visit (10 days) and increased weightbearing as tolerated after that visit.

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NeoCart patients with condylar lesions that were greater than or equal to 2 cm² or with lesions that do not have a circumferentially stable margin of intact cartilage were limited to toe-touch weightbearing for 4-6 weeks after implant and then increased weightbearing as tolerated.

Microfracture patients with condylar lesions were limited to toe-touch weightbearing for a minimum of 6-8 weeks.

Both NeoCart and microfracture patients with trochlear lesions were allowed weightbearing as tolerated with a brace. If a patient had lesions treated on both the trochlea and condyle, the early weightbearing status defaulted to the most conservative, i.e., toe touch weightbearing for 6 weeks, while range of motion rehabilitation mirrored that typically prescribed for trochlear lesions.

Rehabilitation Stages/Phases:

Phase 1 – Protection, Edema Management, Range of Motion (RoM), Muscle Activation (initiated after the first post-operative visit through 6-8 weeks post-operatively)

Rehabilitation in this phase focused on gentle restoration of knee motion without the use of resistance (e.g., stationary bike with no load). For patients who begin weightbearing as tolerated in this phase (NeoCart patients with small condylar lesions and all patients with trochlear lesions), the investigator considered judicious and limited advancement of rehabilitation to assist with restoration of normal gait, avoiding strenuous motion and strengthening exercises aimed at preparation for functional activities.

Phase 2 - RoM, Gait training, Knee Control (initiated 6-8 weeks post-operatively dependent upon progress at 6 week follow-up visit and advancement over 6-8 weeks)

Rehabilitation in this phase focused on more advanced range of motion exercises and closed chain exercises to assist with resumption of normal gait and to prepare for functional activities (e.g., stationary biking with progressive resistance, water program, resistance tubing exercises).

Phase 3 - Strength and Balance (initiated 12-16 weeks post-operatively dependent upon progress at 3 month follow-up visit and continued with advancement to at least 6 months post-operatively)

Rehabilitation in this phase focused on improving strength, endurance and proprioception (e.g., free weights, machine weights).

Phase 4 - Sport Specific Training and Participation (initiated no earlier than 6 months post-operatively and dependent upon progress at 6 month follow-up visit)

Rehabilitation in this phase focused on more advanced strengthening, endurance and proprioceptive exercises and on return to patient-specific sporting activities (e.g., pivoting, cutting, jumping).

9.5.6. Study assessment and schedule

Patients were scheduled to return for follow-up visits at 10 days, 6 weeks, 3 months, 6 months, 12 months and annually thereafter until 3 years post-treatment (i.e., NeoCart patients' 10-day visit occurred 10 days post-NeoCart implantation, not 10 days post-arthroscopy). Patients began completion of a weekly diary measuring weightbearing status, compliance with the rehabilitation protocol, and health economic outcomes beginning 7 days post-procedure (microfracture or NeoCart implant), and continued completion of the diary weekly for 24 weeks. The procedures that were followed at each visit are summarized in [Table 4](#).

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Table 4: Schedule of Assessments

Study Procedures	Baseline	Arthroscopy ^a	Implant	10 days (±3 days)	6 weeks (±2 weeks)	3 months (±2 weeks)	6 months (±4 weeks)	1 year (±2 months)	Annual Years 2, 3 (±2 months)
Informed consent	X								
Medical history and knee examination	X								
X-ray	X								
Height and weight ^b	X					X	X	X	X
Pregnancy test (if applicable)	X								
Viral testing (HIV, Hepatitis B and C)	X								
Lab tests (blood chemistry, hematology, urinalysis and urine chemistry)	X				X				
Study MRI	X ^c								
Study MRI with T2 image acquisition							X	X	X
ICRS Cartilage Injury Evaluation ^d	X								
IKDC Objective and Subjective Knee Examination	X				X	X	X	X	X
KOOS knee survey	X					X	X	X	X
VAS pain assessment	X					X	X	X	X
Marx Activity Rating Scale ^e	X					X	X	X	X
SF-12 ^{**}	X					X	X	X	X
EQ-5D-5L [*]	X					X	X	X	X
WPAI ^f	X					X	X	X	X
Microfracture – control group only		X							
Cartilage harvest – NeoCart group only		X							
NeoCart implant – NeoCart group only			X						
Postoperative assessment				X					
Diary completion	X ^e	X-----X							
Medication use and non-surgical procedures	X			X	X	X	X	X	X
Adverse Event Documentation		X	X	X	X	X	X	X	X

^a Occurred within 3 months of the Baseline visit.

^b Weight and height were recorded at 1, 2, and 3 years. Weight was also recorded at Months 3 and 6.

^c Study MRI obtained within 3 months of arthroscopy.

^d Only page 2 of the ICRS Cartilage Injury Evaluation Package was completed.

^e Following provisional enrollment, the patient completed the patient diary employment question prior to the arthroscopy procedure. After the procedure (microfracture or NeoCart implant), the patient completed the diary approximately every 7 days for 24 weeks. Diary completion was performed either via a secure web portal or paper, depending on patient preference.

^f Not part of baseline or follow up evaluations prior to protocol version 7

^{**} SF 36 was used prior to protocol version 7

9.5.6.1. 10-day Follow-up Visit (±4 Days)

- a. Assess/establish patient rehabilitation.

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- b. Record any medication changes and all pain medications taken during week prior to visit.
- c. Record non-drug therapies/treatments.
- d. Record adverse events (AEs) (as applicable).

Note: Pain medication usage in this study may vary from no use of pain medications to multiple medications. Patients using multiple pain medications should be asked to document their pain medication use during the week prior to each visit. In order to optimize compliance, the method (pocket calendar, log, smart phone, etc.) used should be determined by the patient. The patient consent form should discuss this topic.

9.5.6.2. 6-week Follow-up Visit (± 2 Weeks)

- a. IKDC Objective knee examination.
- b. Obtain blood sample for hematology and comprehensive metabolic panel (refer to baseline visit for tests).
- c. Obtain a urine sample for urinalysis.
- d. Document rehabilitation progress.
- e. Record any medication changes and all pain medications taken during week prior to visit.
- f. Record non-drug therapies/treatments.
- g. Record AEs (as applicable).

9.5.6.3. 3-Month Follow-up Visit (± 2 Weeks)

- a. IKDC Objective knee examination.
- b. KOOS, IKDC Subjective, VAS pain, Marx Activity Rating Scale, SF-12 Health Survey, EQ-5D-5L Health Questionnaire, and WPAI Questionnaire.
- c. Document rehabilitation progress.
- d. Record any medication changes and all pain medications taken during week prior to visit.
- e. Obtain and record patient's weight.
- f. Record non-drug therapies/treatments.
- g. Record AEs.

9.5.6.4. 6-Month Follow-up Visit (± 4 Weeks)

- a. IKDC Objective knee examination.
- b. KOOS, IKDC Subjective, VAS pain, Marx Activity Rating Scale, SF-12 Health Survey, EQ-5D-5L Health Questionnaire, and WPAI Questionnaire.
- c. Obtain a study MRI (standard MRI and time for 63% of transverse relaxation read [T2] image acquisition to be performed at sites where T2 software is available).
- d. Document rehabilitation progress.
- e. Record any medication changes and all pain medications taken during week prior to visit.
- f. Obtain and record patient's weight.
- g. Record non-drug therapies/treatments.
- h. Record AEs.

9.5.6.5. Annual Visits (12, 24 and 36 months) (± 2 Months)

- a. IKDC Objective knee examination.
- b. KOOS, IKDC Subjective, VAS pain, Marx Activity Rating Scale, SF-12 Health Survey, EQ-5D-5L Health Questionnaire, and WPAI Questionnaire.

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- c. Obtain a study MRI (standard MRI and T2 image acquisition to be performed at sites where T2 software is available).
- d. Document rehabilitation progress.
- e. Record any medication changes and all pain medications taken during week prior to visit.
- f. Obtain and record patient's height and weight.
- g. Record non-drug therapies/treatments.
- h. Record AEs.

Randomized patients who withdrew from the study protocol for any reason (with the exception of withdrawal of consent) were encouraged to return for annual follow-up visits for a safety evaluation. Such data was included in the safety analysis.

9.5.7. Efficacy Parameters

The primary efficacy study endpoint was defined as a test of superiority (one-sided [Pearson's Chi-square](#) test, $\alpha=0.025$) vs microfracture, in terms of a composite clinical success (CCS) used in a dual-threshold responder analysis based on improvement in knee-specific pain and function at 1 year using the IKDC Subjective knee exam and the Pain domain score from the KOOS Knee Injury and Osteoarthritis Outcome Score.

Patients were required to demonstrate both an improvement in KOOS pain of ≥ 12 points and IKDC subjective score of ≥ 20 points to be a positive responder.

The IKDC instrument is a validated region-specific patient-oriented outcome measure of symptoms, function, and sports activity. A score of 100 represents no limitation with activities of daily living or sports activities and absence of symptoms. The normative score for this age group is about 90.

The KOOS has five separate scores: Pain, Symptoms, Activities of Daily Living (ADL), Sport/Recreation, and Quality of Life (QoL). Similar to IKDC, a KOOS Pain score of 100 represents no pain. The KOOS Pain assessment is preferred over a VAS Pain scale (to be evaluated in supporting analyses) because it evaluates pain levels using a pain frequency scale (never, rarely, sometimes, often, always) in relation to specific knee activities of daily living (e.g. walking, climbing, sitting, pivoting, etc.).

The dual threshold is a composite score inclusive of [the IKDC subjective](#) and [KOOS pain](#) values. ~~By The two sub-scores were used as the primary endpoint while~~[definition this means that](#) all individual IKDC and KOOS endpoints were prospectively collected both as exploratory analyses and for purposes of conducting the dual-threshold responder analysis.

The secondary efficacy endpoints included time to full weight-bearing, 'Treatment failure' (defined as >8 -point deterioration in KOOS Pain score at 1 year compared to baseline) and the presence of mature collagen layering as assessed by MRI T2 mapping at 1 year.

9.5.8. Pharmacokinetics Parameters

Not applicable.

9.5.9. Pharmacodynamics Parameters

Not applicable.

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9.5.10. Safety Parameters

All AE summaries were restricted to Treatment Emergent Adverse Events (TEAEs), which were defined as those AEs that occurred after the start of the arthroscopic procedure or worsened during the study. Verbatim terms were mapped to preferred terms and System Organ Classes using the Medical Dictionary for Regulatory Activities (MedDRA) dictionary (version 17)

9.5.10.1. Summaries of the following types are presented:

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- An overview of TEAEs: Patient incidence of any TEAEs, any serious TEAEs, any NeoCart related TEAEs, any procedure related TEAEs, any severe TEAEs, any TEAEs resulting in death, and any TEAEs involving the target knee. NeoCart and procedure related TEAEs are TEAEs with a possible or probable relationship to NeoCart or the procedure, respectively. TEAEs with missing severity were considered severe and TEAEs with missing relationship were considered related for this summary. Fisher's Exact Test was used to test for a difference in proportions between treatments.
- Patient incidence of all TEAEs by MedDRA System Organ Class and preferred term. Fisher's Exact Test was used to test for a difference in proportions between treatments for each System Organ Class.
- Patient incidence of all serious TEAEs by MedDRA System Organ Class and preferred term.
- Patient incidence of all TEAEs by MedDRA System Organ Class, preferred term, and highest severity: At each level of patient summarization, a patient was classified according to the highest severity if the patient reported more than one TEAE. TEAEs with missing severity were considered severe for this summary.

Patient incidence of all TEAEs by MedDRA System Organ Class, preferred term, and relationship to the NeoCart treatment and by MedDRA System Organ Class, preferred term, and relationship to the procedure: At each level of patient summarization, a patient was classified according to the closest relationship if the patient reported one or more events. TEAEs with a missing relationship were considered related for this summary Adverse Events

Any untoward medical occurrence in a clinical investigation subject administered a study treatment regardless of whether a causal relationship exists between the medical occurrence and the study treatment. An AE can therefore be any unfavorable and unintended sign, abnormal assessment (e.g. laboratory finding, vital sign, electrocardiogram (ECG), or disease temporally associated with the use of a study treatment, whether or not related to the study treatment.

9.5.10.1.1. Adverse Event Severity

The investigator assessed all AEs for severity (mild, moderate or severe: defined below), product-relatedness, procedure relatedness, whether it meets criteria for classification as an unexpected adverse event (UAE) (defined below). The investigator immediately notified the Sponsor when an AE is classified as an SAE or UAE.

The severity of an AE was categorized as mild, moderate or severe. Mild was defined as an awareness of sign/symptom that is easily tolerated. Moderate is defined as discomfort that interferes with usual activity. Severe is defined as an event that is incapacitating with inability to work or perform routine activities.

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A UAE is any AE the specificity or severity of which is not consistent with the current Investigator Brochure or with the risk information described in the current version of the Protocol for this study.

9.5.10.1.2. Study Treatment Causality

<<Define from protocol.>>

Commented [LK9]: Drug trial element, does not apply

9.5.10.2. Serious Adverse Event

An SAE is any adverse experience that results in any of the following outcomes: death, a life-threatening adverse experience, inpatient hospitalization or prolongation of existing hospitalization, a persistent or significant disability/incapacity, or a congenital anomaly/birth defect. Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered an SAE when, based upon appropriate medical judgement, they may jeopardize the patient and may require medical or surgical intervention to prevent one of the outcomes listed in the above definition.

9.5.10.3. Treatment Errors

There were no treatment errors

9.5.10.4. Adverse Events of Special Interest

<<If applicable?>> There were

Commented [LK10]: Add DVT data

9.5.10.5. Other Safety Related Information

<<If applicable?>>

9.5.10.6. Clinical Laboratory Assessments

Clinical laboratory tests were performed at the time points baseline and 6 weeks. For hematology, chemistry and urinalyses there were a few single isolated abnormal laboratory values but no clinically significant abnormal values of note in either the NeoCart or the microfracture treatment arms. Laboratory testing included the following: Electrolytes, renal function, liver function, hematology, coagulation, HIV, Hepatitis B and C and urinalysis. Additionally, a pregnancy test was performed on all females of childbearing potential. Laboratory values were only recorded in the eCRF if they were abnormal.

Commented [LK11]: Need to add data from reports, we only captured abnormalities in labs,

9.5.11. Appropriateness of Measurements

The assessments used in this study were routine and typical of the care for patients with articular cartilage injuries of the knee. All surveys used to assess efficacy were validated assessments and are typically used in this patient population. The KOOS pain and IKDC function measurements were noted as examples of clinically meaningful primary endpoints that may be used to assess efficacy in the 2011 FDA guidance. Furthermore, two separate well recognized validated instruments were used for measurement of joint pain and function.

Commented [AG12]: I would link this to FDA guidance – all are validated scales measuring pain/function and are noted as appropriate in the guidance

Commented [LK13R12]: Sumitra: Need more detail in this section on why measures were relevant, validated etc.

Commented [AG14R12]: Added in verbiage from FDA guidance

9.5.12. Study Stopping Criteria

Safety findings were periodically reviewed by the safety monitor as well as an independent DSMB (Data Safety Monitoring Board) including any protocol amendments, all adverse events and study progress. Study Stopping Rules were defined in the Protocol and listed in the table (Table 5) below. The DSMB met at regular scheduled intervals and reviewed all safety events. Had the DSMB identified that any of the stopping criteria were met, the Sponsor was to initiate a comprehensive safety review and promptly notify the FDA, principal investigators and all reviewing IRB. There were no safety events during the study that warranted stopping.

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Table 5: Study Stopping Rules

Potential Major Safety Event	Study Stopping Criteria: rate and # of events in NeoCart arm (n=163 patients) that would have triggered automatic study hold	References
1. Death	If 1 occurs (probably to definitely related to the implant). Rate: 0.6% If 2 occurs (possibly related to the implant). Rate: 1.2%	Wood (2006) – 3 deaths/294 ACI patients (2 motor vehicle, 1 intracranial hemorrhage), insufficient data to assess potential association to ACI ⁴⁵
2. Vascular Injury to the lower extremity.	Definitely related to the NeoCart biopsy or implantation procedure: defined as loss of arterial supply to limb, pulselessness, ankle-brachial index (ABI) of <0.9. Rate> 1% (2 events).	Wood (2006) 0/294 ACI patients ⁴⁵
3. Neurological injury to the lower extremity.	Definitely related to the NeoCart biopsy or implantation procedure: defined as a persistent sensory or (> 6 mos.) motor deficit at or distal to the surgical site that is worse than baseline status. Rate of >2% (4 events).	Bernard (1994) refers to rates of nerve damage between .05% - 0.6% and cited work by NC Small (1986) 30 (1%) cases of irreversible injury to saphenous or peroneal nerve in more than 3000 refixations of the meniscus ⁴⁷ Wood (2006) reports
4. Deep vein thrombosis with pulmonary embolism (PE).	Definitely related to the NeoCart biopsy or implantation procedure, occurring within 1 month of treatment in which PE has been confirmed by spiral computerized tomography (CT) scan and patient treated with anticoagulation therapy. Rate of >2% (4 events).	Wood (2006) – 2/294 ACI patients with Pulmonary Embolism. (0.7%) 5/294 ACI patients with deep vein thrombosis (DVT) (1.7%). ⁴⁵
5. Septic Arthritis/significant infection.	Definitely related to NeoCart biopsy or implantation procedure and/or related to the NeoCart implant confirmed by positive intraarticular cultures requiring surgery/drainage and/or intravenous (IV) antibiotics. Rate of >5% (8 events).	Wood (2006) – 11/294 ACI patients with joint infections (3.7%) ⁴⁵ Fong (2004) – 7/472 post operative deep articular infections detected after arthroscopic-assisted ACL reconstruction (1%) ⁴⁸
6. High rate of secondary surgical interventions.	Definitely related to the NeoCart implant. Rate of >25% (25 events) within 1 year of treatment.	Knutsen (2004) 4/40 microfracture patients (10%). ⁴⁰ Wood (2006) – 273/294 ACI patients had surgical revisions subsequent to index treatment involving 389 total procedures. Of these, 187 (48%) were cartilage repair procedures directly related to the graft.

9.6. Data Quality Assurance

Data required according to the protocol were recorded in the web-based ~~eCRFs~~ EDC (electronic data capture system) on iMedNet platform created and managed by Health Policy Associates. On 1/8/16, the data was migrated to this system as the study EDC was originally hosted by ProSoft. All data were imported into iMedNet and SDV status and signatures were applied in accordance with how the records were entered into Prosoft at the time of transfer.

Auto queries were generated in the EDC system for data that did not meet specifications or when required fields were skipped. Inconsistencies in the data were queried to the investigators via the electronic data capture system; answers to queries or changes to the data were made by an authorized member of the investigator's staff. Changes are documented via the Audit Trail File (ATF).

Manual queries (to be answered by site staff) were raised during source data verification, medical or safety review and data management review. Additionally, to monitor and data manager manual queries, HPA programed edit checks to look for various inconsistencies based on review of exports or team feedback. Plausibility checks were performed to ensure correctness and completeness of the data. Some examples include missing forms, missing fields, unexpected values, or inconsistency across forms. The result of running these checks is a list of known issues, which HPA reviewed and worked with clinical and data teams to resolve. In addition to monitor and data manager manual queries, HPA programed edit checks to look for various inconsistencies based on review of exports or team feedback. Some examples include missing forms, missing fields, unexpected values, or inconsistency across forms. The result of running these checks is a list of known issues, which HPA reviewed and worked with clinical and data teams to resolve. Once the sites respond to raised queries in the database, the queries were reviewed and closed as appropriate.

Laboratory and antibody data were received per the protocol schedule and abnormal values were entered electronically and merged to the eCRF data (but not uploaded into the EDC system).

The sponsor's data management function was responsible for data processing, in accordance with the sponsor's data management procedures.

After all data through the 1 year primary endpoint were entered and all queries were resolved, the database was closed on **August 23rd, 2018**. Unblinding took place on **August 28th, 2018**.

9.7. Statistical Methods Planned in the Protocol and Determination of Sample Size

The statistical analysis of the data obtained from this study was performed using SAS® version 9.4. All two-sided statistical tests was performed at the 0.05 significance level, and all one-sided tests was performed at the 0.025 significance level, unless otherwise indicated.

All relevant data collected in this study was documented using summary tables and/or subject data listings. Continuous variables was summarized using descriptive statistics, specifically the mean, median, standard deviation, minimum and maximum. Categorical variables was summarized by frequencies and percentages. All results are presented by treatment group.

All variable options collected on the eCRFs was included even if all treatment groups have counts of zero. Patients' missing data is counted in frequencies as 'Missing', or indicated as missing by the inclusion of n in all summary statistics. Sorting order of variable reporting is either alphabetical or specified by the sponsor.

Commented [AG15]: This heading seems off – this is a massive section of content and it is NOT just methods. Team, please get together and determine how this should be organized. This really seems like the overall statistical plan

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Data listings is sorted by treatment group and patient ID number. All date fields are presented in a format of ddMonyyyy (i.e., 01Jan2012) in the listings. All data appearing in summary tables along with any other relevant data are included in the data listings.

9.7.1. Statistical and Analytical Plans

The Statistical Analysis Plan (SAP) for the NeoCart Phase III clinical study, titled “Statistical Analysis Plan for a Randomized Comparison of NeoCart® to Microfracture for the Repair of Articular Cartilage Injuries in the Knee” (dated May 22, 2013), which was submitted to the IND (#A80) on May 29, 2013 as an attachment to Revision 6 of the Clinical Trial Protocol #8-01. Since approval of the Special Protocol Assessment in March 2009, each major revision of the protocol and related SAP has been reviewed by the agency and a response received that the SPA was maintained. A revision was submitted August 2018 prior to the database lock to clarify discussions that had been conducted with the FDA and to help provide guidance for the SAS programming of the data analyses and outputs of the tables, figures and listings (TFLs) for the clinical study report for the Biologics License Application (BLA), and to delete references to interim analyses that were not conducted.

The primary efficacy analysis was a test of superiority in terms of a composite clinical success (CCS) endpoint used in a responder analysis at 1 year post-treatment surgery. Clinical success required an improvement from baseline of at least 12 points in the KOOS pain score and an improvement from baseline of at least 20 points in the IKDC Subjective score. The response rate of the NeoCart arm was compared to the response rate of the microfracture arm.

The primary efficacy endpoint was the responder rate at 1 year. A patient was considered a responder if he/she achieved the following outcomes:

- A clinically significant improvement of at least 12 points at 1 year compared to baseline on the KOOS pain assessment test.
- A clinically significant improvement of at least 20 points at 1 year compared to baseline on the IKDC subjective test.

The secondary efficacy endpoints were:

- Time to full weight-bearing;
- ‘Treatment failure’, defined as >8 point deterioration in KOOS pain score at 1 year compared to baseline; and
- Presence of mature collagen layering as assessed by MRI T2 mapping at 1 year

A sequential gatekeeping testing strategy was used for the secondary efficacy endpoints. The first test was conducted using an $\alpha=0.05$ level, and only if the first test was significant, the second and third tests were each conducted at the 0.025-level (all levels were two-sided). This procedure was designed to control type 1 error to no more than $\alpha=0.05$ (2-sided) for the set of secondary efficacy endpoints.

A number of supporting efficacy endpoints were evaluated, including: responder status at each study visit, responder status for each component of the composite endpoint, and the following quantitative clinical scores including KOOS subscale scores, IKDC subjective scores, SF-12 scores, and VAS pain scores.

Commented [AG16]: Took this right from SAP

Commented [AG17]: This is the wrong content (below). This should be a summary of the initial SAP, interim and any changes. We are saying the same thing multiple times on primary efficacy. We MUST align on this section

Commented [LK18]: Heather and Jocelyn, Not sure what happened here. I know you were working on this. Hopefully you have the correct version copied.

Commented [AG19]:

Commented [AG20R19]:

Commented [AG21]: Per note, this belongs elsewhere. This whole section should be deleted

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Covariate analyses of the KOOS pain and IKDC subjective scores were conducted. The following covariates were used: age, gender, lesion size, BMI, lesion location, acute vs. chronic, lesion grade (3 vs. 4A), and smoking status.

The primary endpoint for the study was evaluated at 12-months. Patients were followed for up to 36-months for extended longer-term evaluation. The following outcomes were assessed for longer term follow-up.

- Responder rates at 24 and 36 months.
- Changes from baseline to 3, 6, 12, 24 and 36 months in KOOS sub scores
- Change from baseline to 3, 6, 12, 24 and 36 months in IKDC subjective knee exam score
- Change from baseline to 3, 6, 12, 24 and 36 months in VAS pain score (highest)
- Changes from baseline to 3, 6, 12, 24 and 36 months in the SF-36 domain and summary scores

Safety variables included analysis of abnormal laboratory evaluations (hematology, blood chemistry, and urinalysis) and AEs.

9.7.2. Determination of Sample Size

The initial sample size was selected to be 245 subjects (approximately 163 NeoCart subjects and 82 Mfx subjects). With a sample size of 163 NeoCart subjects and 82 Mfx subjects, the power is 86%, assuming a one-sided test for superiority based on the chi-square test, a one-sided alpha level of 0.025, and true responder rates of 0.70 and 0.50 for the NeoCart and Mfx treatments, respectively.

Specifically, a two group χ^2 test with a 0.025 one-sided significance level had 86% power to detect the difference between an Mfx responder rate of $\pi_{Mfx} = 0.50$ and a NeoCart responder rate of $\pi_{NeoCart} = 0.70$ (odds ratio of 2.33) when the sample sizes are 163 and 82, respectively (a total sample size of 245) [Elashoff, 2007i]. With an assumed 7% lost to follow-up rate by 1 year, the expected number of subjects was 228 (152 NeoCart subjects and 76 Mfx subjects), and the power of the test for superiority was 83%.

9.7.3. Analysis Populations

The following populations were defined for data analyses and/or presentation. Table 6 highlights the patient numbers within each of these analysis sets.

9.7.3.1. Intent-to-Treat (ITT) Population

This population is defined within the SAP as all patients randomized to receive either NeoCart or Microfracture, whether or not they ultimately received their assigned treatment. The primary endpoint for efficacy is based on this study population. To summarize: the ITT Population includes all patients who were consented, enrolled in the trial and randomized at arthroscopy irrespective of treatment received, major protocol deviations or ability to gather all the data at the specified time points.

9.7.3.2. Modified Intent-to-Treat (mITT) Population

This population was not originally defined in the original SAP but discussed with the FDA prior to the submission of the revised SAP in July 2018. Whereas the ITT population includes all patients randomized to either test or control, regardless of whether they received treatment or not, the mITT

Commented [JB22]: Waiting on feedback from Histogenics on how to define lesion location/size/grade and acute vs chronic

Commented [AG23R22]: I thought this was handled this summer at time of other SAP

Commented [JB24]: This will be limited to shift tables as only abnormalities were entered into the eDC

Commented [JB25]: Changes pending, particularly on secondary gated analysis

Will need to discuss imputations on ITT.MITT/AT populations

Commented [LK26R25]: Sumitra: Need to ensure that this is aligned w/ SAP....reviewer will review together. analyses performed need to show traceability to the SAP.

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group excludes those patients that did not receive treatment due to circumstances outside the control of the study (study halt due to financial reasons and patient withdrawal after biopsy but before implantation of the NeoCart). The mITT Population includes all patients who were randomized to the NeoCart arm and received the implant, but with the 7 patients who either withdrew or were not implanted removed from the analysis. For the ITT analysis these 7 patients were considered as non-responders. Thus, the mITT is equivalent to the ITT set, but excludes the patients who never received treatment. There are no changes in the control population between the ITT and mITT analyses as they received the microfracture treatment at the same time as randomization.

9.7.3.3. As Treated (AT) Population

This population set is defined within the SAP as those patients that received treatment; each patient in the AT population is sorted into either the test or control group depending on the actual treatment he/she received. The safety analyses are conducted on the AT population set. Per-Protocol (PP) Population

The PP population, as defined in the SAP, is the subset of patients within the AT population whom have either completed their 12 month follow-up or have withdrawn prior to 12 months due to treatment failure. Patients within the PP population are devoid of major protocol deviations

Table 6: Numbers of Patients Within Each Analysis Set

	NeoCart	Microfracture	Notes
ITT	170	79	Patients signing consent, meeting the inclusion/exclusion criteria, and randomized to a treatment arm
mITT	163	79	Removed 7 patients who were never treated with NeoCart
AT	162	80	Patient 10-006 randomized to NeoCart switched from NeoCart arm to Microfracture arm
PP	155	65	Subset of AT population that met the 1-year follow-up time period and had no major protocol deviations

9.7.4. Study Population Analyses

9.7.4.1. Disposition of Patients

The number and percent of randomized patients completing the study through year 3 are summarized, both overall and by treatment group. The number of patients signing the informed consent and the number of screen failures (consented but did not randomized patients) are presented. The number and percent of randomized patients who did not complete the study are presented, along with the primary reason for early termination, both overall and by treatment group.

9.7.4.2. Protocol Deviations

Deviations will be presented as major and minor. The details of all major deviations will be provided, and the categories and numbers of minor deviations will also be provided.

Commented [LK27]: Should this be 1 year primary endpoint at this stage since we only have complete data for 1 year?

Commented [AG28R27]: We should agree this is for 1 year only for efficacy and through 12/31 for safety,

Commented [LK29R27]: Aaron Although the primary EP is at Year 1, the disposition should not be censored at year 1.

Commented [LK30R27]: Sumitra: Agree – the agency is pretty interested in topic of disposition so for these data would include all up to cut-off date. Also, if there are patients who have 2 year data before 12/31, we need to include that in whatever efficacy analyses we have for 2 yrs. You can't pick and choose which items you choose to analyze....the FDA will expect that you've analyzed whatever your SAP directed for any data received up to data cut-off for the BLA

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9.7.4.3. Demographic and Other Baseline Characteristics

Descriptive statistics are presented for the continuous variables age, height, weight, and BMI; frequencies and percentages are calculated for the categorical variables gender, race and whether any pain medications were taken in the week prior to baseline. Any patient who selected multiple races or other are categorized as other in analyses.

Smoking status (cigarette smoking/tobacco use status and other nicotine use), is summarized using frequencies and percentages and defined as:

- Any patient with current smoking or tobacco use is classified as a Current Smoker/Tobacco User
- Any patient with previous smoking or tobacco use is classified as a Past Smoker/Tobacco User
- Any patient with no current or former history of smoking or tobacco use is classified as Never Smoked/Used Tobacco

Medical history, baseline concomitant medications, surgical history for the treatment knee, and surgical history for the contralateral knee is summarized using frequencies and percentages. Concomitant medications are categorized by medication class per instructions of the sponsor's medical reviewer.

The baseline ICRS cartilage evaluation and baseline X-ray results are summarized using descriptive statistics for continuous variables and counts and percentages for categorical variables. Patients missing categorical responses for variables are counted as Missing.

- Time since Injury is reported in years to two decimal places; for partial dates, calculations to the nearest month is reported to one decimal place and for years only to no decimal places.

Baseline x-ray results are summarized including mechanical axis alignment and leg angle in the treatment and contralateral knees, presence or absence of osteophytes, sclerosis, subchondral cysts, and well as results from the Kellgren-Lawrence radiographic grading scale.

Baseline MRI results for potential study lesions and non-study lesions are summarized by treatment group at the lesion level using counts and percentages (see Section 8.4). Laboratory tests for positive, negative, or missing infectious disease results at baseline are summarized by treatment group using counts and percentages.

An adjusted chi-square test was conducted as a supportive analysis. The adjusted chi-square test was obtained from a logistic regression model with composite clinical success as the outcome variable. Model covariates included treatment group and a set of *a priori* defined covariates. These included the same covariates that were previously specified for analyses of KOOS and IKDC mean changes over time including age, gender, total lesion size at arthroscopy, total lesion size at treatment, BMI, number of study lesions, injury type, most severe lesion ICRS grade, and smoking status.

9.7.5. Efficacy Analyses

9.7.5.1. Primary Efficacy Endpoint

The primary efficacy endpoint was the responder rate at 12 months. A patient was considered a responder if he/she achieved the following outcomes:

Commented [LK31]: placeholder Measurements of Treatment Compliance
Analysis of Efficacy
Statistical/Analytical Issues
Adjustments for Covariates

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- A clinically significant improvement of at least 12 points in Month 12 outcome compared to baseline on the KOOS pain assessment test, and
- A clinically significant improvement of at least 20 points in Month 12 outcome compared to baseline on the IKDC subjective test.

The primary analysis was conducted on the ITT Population. Data from the mITT population is also presented to evaluate the true effect of the procedure on patients.

Patients with no observations past baseline were considered failures for this endpoint.

9.7.5.2. Secondary Efficacy Endpoints

The secondary efficacy endpoints are:

- Time to full weight-bearing
- Treatment failure' defined as >8 point deterioration in KOOS Pain score at 1 year compared to baseline
- Presence of mature collagen layering as assessed by MRI T2 mapping at 1 year.

9.7.5.3. Covariate Analysis

Covariate analyses of the KOOS pain and IKDC Subjective scores were conducted. The following covariates were used: age, gender, lesion size, BMI, lesion location, acute vs chronic, lesion grade (3 vs 4A), and smoking status. This same list of *a priori* specified covariates was used in several modeling approaches as detailed in the SAP. An adjusted chi-square test was conducted as a supportive analysis. The adjusted chi-square test was obtained from a logistic regression model with composite clinical success as the outcome variable. -

9.7.5.4. Test for site heterogeneity in treatment effect

For the ITT analysis, a standard Breslow-Day test of homogeneity of the odds ratios across sites was performed. Sites with less than five patients were combined into a single pooled site. If the Breslow-Day p-value was less than 0.05 and if there were a number of sites enrolling a relatively small number of patients, site were assessed as a random effect. Specifically, SAS Proc GLMMIX was to be used to construct a test for superiority accounting for random site effects reflecting site-to-site variation in treatment group response rate differences.

9.7.5.5. Additional Planned Efficacy Analyses

9.7.5.5.1. Responder Analyses

Responder rates based on each of the two CCS component scores (KOOS pain and IKDC subjective score) were summarized by treatment group using frequencies and percentages. Exact (Clopper-Pearson) 95% confidence intervals were presented by treatment group for the true population proportions. For each endpoint, a two-sided 95% confidence interval based on the normal approximation was presented for the difference in proportions between treatments.

9.7.5.5.2. Exploratory Endpoint Analysis of Interest

<< A-In addition to the primary and secondary endpoints, a number of supporting efficacy endpoints were pre-specified in the final statistical analysis plan and evaluated. These include responder status for each component of the composite endpoint and the following quantitative clinical scores including KOOS sub scores and IKDC scores, VAS pain scores, IKDC Subjective scores, Marx Activity Rating Scale and SF-12 Health Survey scores. Descriptive analyses of mean values over time and MMRM analyses for mean changes from baseline were used to assess quantitative supporting efficacy endpoints. Descriptive statistical analyses, including counts and percentages and, where appropriate, confidence intervals for group differences in percentages is used for categorical supporting efficacy endpoints. A secondary goal is to evaluate the comparative effects of the two treatments on healthcare resource utilization, Health-Related Quality of Life (HRQoL) and work productivity. In support of planned pharmacoeconomic and HRQoL endpoints, health care resource utilization (e.g., unscheduled/non-protocol-required office visits, additional hospital days, and pain medications) and influence of the treatment on activities of daily living, assessment were performed using validated questionnaires, whenever possible. For example, for HRQoL impact on productivity, the Marx Activity Rating Scale, SF-12 Health Survey, EQ-5D-5L Health Questionnaire, and WPAI Questionnaire were self-administered by patients with a reasonable frequency throughout the study.

Lastly, an additional efficacy endpoint was summarized using summary statistics and difference in least squares mean values, with a p-value for difference in composite (KOOS Pain and Function) endpoints assessed jointly at 1 year based on multivariate analysis of variance.

Tests of means and change from baseline at 1 year and 2-years for the KOOS Pain and Function (Sports/Recreation Activities [SRA]) domains, including the least squares means, difference in Pain and Function scores between treatment groups, and the p-value for the difference in co-endpoints based on multivariate analysis of variance (MANCOVA).

An analysis to statistically compare the proportion of subjects with at least a 10 point improvement in both in KOOS pain and KOOS function (SRA) at all follow-up time points was also be conducted.

Commented [AG32]: This is from 7.5.2 of SAP of this summer, these are the MACI and other mean change endpoints, which are critical. This whole section needs to be cleaned up.

9.7.5.5.3. Analyses of Quantitative Endpoints

Descriptive statistics for mean values over time by group including n, mean, SD, median, minimum, and maximum values were summarized. Unadjusted group differences in means were described using estimated mean differences, 95% confidence intervals, and standardized mean differences.

Descriptive statistics over time for mean changes from baseline were similarly described. Adjusted group differences in mean changes from baseline with p- values and 95% confidence intervals were determined using the mixed model repeated measures (MMRM) [28] approach with parameters estimated by maximum likelihood (see section 6.5.2).

In addition to the above analyses, the following specific analyses were performed:

- A repeated measures model was analyzed for the KOOS pain scores through time 3-, 6-, and 12-months using MMRM.
- Individual tests of mean changes from baseline for the 3-, 6-, and 12-month KOOS pain scores were conducted based on contrasts from MMRM.

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- An analysis of the integrated (time-weighted) change from baseline KOOS pain scores over the 3-, 6-, and 12-month horizon were conducted. The MMRM model applied a weighting of 1:1:2, respectively for the time points (i.e., weighted by the size of the time interval).
- A repeated measures model was analyzed for the IKDC subjective scores through time 3-, 6-, and 12-months using MMRM.
- Individual tests of means for the IKDC subjective score at 3-, 6-, and 12-months were conducted based on contrasts from MMRM.
- An analysis of the integrated (time-weighted) change from baseline IKDC subjective scores over the 3-, 6-, and 12-month horizon was conducted. The MMRM model applied a weighting of 1:1:2, respectively for the time points (i.e., weighted by the size of the time interval).
- Covariate analyses of the KOOS pain and IKDC subjective scores were conducted. The following covariates were used: age, gender, lesion size, BMI, lesion location, acute vs. chronic, lesion grade (3 vs. 4A), and smoking status. This same list of *a priori* specified covariates was used in several modeling approaches as detailed below.
- Similar outcome measures for pain and symptoms were analyzed for the consistency across measures. For pain: KOOS pain, VAS pain, and IKDC subjective pain sub scores were compared. KOOS symptoms and IKDC subjective symptoms sub scores were compared. Graphical comparisons as well as simple linear regressions and correlation analyses were utilized.

Descriptive statistics were generated for the data sourced from the IKDC objective form including counts and percentages for categorical variables and mean, standard deviation, median, minimum, and maximum for quantitative variables.

9.7.5.5.4. Analysis of Responder Status

For a patient to be a responder the patient must have not undergone a surgery on the index knee for a failed NeoCart or microfracture index treatment; and among patients with no failed treatment, a change from baseline in KOOS pain of at least 12 points and a change from baseline in IKDC subjective scores of at least 20 points.

The responder rates defined this way were summarized by treatment group using frequencies and percentages. Exact (Clopper-Pearson) 95% confidence intervals are presented by treatment group for the true responder rates.

The CCS endpoints were analyzed using two-sided chi-square tests with 0.05 significance levels to test for a difference in responder rates between treatments. Two-sided 95% confidence intervals based on the normal approximation are presented for the difference in responder rates between treatments.

Commented [AB33]: This section needs to be reconciled with Section 9.7.5.5.1.

Clinical Scores

The statistical methods for clinical scores that were described for follow-up up to Month 12 were used to evaluate clinical data out to Month 36 including univariate comparisons of mean values and MMRM analyses of changes from baseline.

9.7.5.6. Analysis of MRI Data

All MRI images collected under this protocol were submitted to the MRI Core Lab. The MRI Core Lab was blinded to the randomization arm. Analysis included standard MRI review to assess the quantity of the defect repair during the course of the trial. These MRI findings are divided into anatomic assessments that were made using generally accepted, well-established MRI sequences, and more advanced sequences that allow for the direct, qualitative assessment of the area of defect repair. Standard MRI sequences provide high contrast between fat suppressed bone and cartilage; however, they are not fluid sensitive and are degraded in the postoperative setting due to susceptibility artifact. Fast spin echo sequences with high in-plane resolution provide good cartilage to fluid contrast, making them more sensitive to partial thickness lesions.

In addition to standard MRI review, an advanced T2 MRI image acquisition sequence was used at qualified MRI centers. T2 mapping permits direct, qualitative and quantitative assessment of the collagen component of the extracellular matrix of the NeoCart implant and adjacent articular cartilage. T2 relaxation time is a reproducible MRI parameter that reflects the internuclear reaction, or dephasing, that occurs as a result of transverse relaxation of the excited hydrogen dipoles. The stratification of the T2 signal is indicative of a normal arrangement of the type II collagen fibers within the cartilage repair.

9.7.5.7. Supporting Efficacy Endpoints

A number of supporting efficacy endpoints were evaluated. These include responder status for each component of the composite endpoint and the following quantitative clinical scores including KOOS and IKDC domain scores, VAS pain score, IKDC Subjective scores, SF-12 scores, Marx Activity Rating scores, EQ-5D-FL scores, and WPAI scores. As the Marx, EQ-5D-5L, and WPAI questionnaires were introduced in revision 7 of the Clinical Study Protocol the number of patients with baseline and follow-up data is reduced; analysis from these questionnaires is exploratory due to this decreased study population. Revision 7 of the Clinical Study Protocol was submitted to the IND on December 12, 2014 (#A089). Covariate analyses of the KOOS Pain and IKDC Subjective scores were conducted. The following covariates were used: age, gender, total lesion size at arthroscopy, total lesion size at treatment, BMI, number of study lesions, injury type, most severe lesion ICRS grade, and smoking status.

9.7.5.8. Extended Follow-up

The primary endpoint for the study is evaluated at 1-year. Subjects were followed for up to 3 years for extended longer-term evaluation. Identical to the primary endpoint, for a patient to be a responder for CCS after 1 year (i.e. at 2 years and 3 years) the patient must not have undergone a secondary re-operation for a failed index treatment, and among patients with no failed treatment, an improvement from baseline in KOOS pain score of at least 12 points and an improvement from baseline in IKDC subjective score of at least 20 points.

9.7.6. Handling of Dropouts or Missing Data

Multiple imputation was specified to be used in sensitivity analyses to determine Month 12 responder status when Month 12 CCS was missing. In this approach, “Prediction is based on the patient level of success and failure—not on the two components of success separately.”

However, statistical efficiencies can be obtained by using linear regression based multiple imputation [29-32] on the underlying continuous variables of change in KOOS and change IKDC subjective scores separately, rather than on an endpoint defined by collapsing multiple underlying continuous variables. Therefore, missing Month 12 KOOS and IKDC were individually imputed as continuous variables in parallel imputations and then combined to form the CCS. This approach takes advantage of potentially stronger associations among continuous variables. Also, multiple imputation of the individual scores is closer to the LOCF approach specified for the primary analysis. In the primary analysis, LOCF was applied to KOOS pain and IKDC subjective score individually, and then each score is to be dichotomized in order to form the composite clinical endpoint.

The original plan included varying certain model parameters to reflect extremes concerning the impact of missing values on group differences. This type of analysis was retained but was accomplished using conventional tipping point analyses.

9.7.6.1. Responder analysis

When computing responder status in primary analyses, e.g., 1-year composite clinical success in the intent-to-treat (ITT) (as well as in the As-Treated (AT) analyses), the last-observation carried-forward (LOCF) method were used to impute the components of missing KOOS and IKDC. That is, subjects missing 1-year values had their 1 year values replaced by their Month 6 values, if Month 6 was available. If Month 6 is not available, then the Month 3 values were used. If there are no values past baseline, the subject was considered a Month 12 nonresponder in primary efficacy analyses in the ITT analysis set.

Sensitivity analyses based on multiple imputation and on ‘tipping point’ analyses will also be performed.

9.7.6.2. Analysis of mean changes

Supporting analyses involving group comparisons in mean changes will involve the use of mixed model repeated measures (MMRM) which allows inclusion of all available data including intermediate values of KOOS Pain and IKDC Subjective score changes as well as relevant baseline covariates to inform the contrast of interest.

Covariates will include age, gender, total lesion size at arthroscopy, total lesion size at treatment, BMI, number of study lesions, injury type, most severe lesion ICRS grade, and smoking status. Last-observation-carried-forward imputation will also be conducted for the specified secondary efficacy endpoints. If a patient has no results post-baseline for T2 mapping, they were excluded from analysis. For all other endpoints and outcomes, there was no imputation of missing data with the exception of analysis done on the questionnaires added in version 7 of the protocol and the new exploratory analyses for KOOS Pain and Function at 1 and 2 years.

No data imputation is necessary for time to full weight-bearing because this endpoint was analyzed using survival analysis methods and subjects who terminate the study early were censored as of the time of their last observation. As time to weight-bearing is collected via an unmonitored patient diary,

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additional censoring of patients will occur if the date of full weight-bearing occurs on or prior to the date of study treatment (microfracture procedure or NeoCart implantation). These patients were censored at the treatment date.

9.7.6.3. Multiple imputation

Multiple imputation was used as an alternative to LOCF for imputing missing 1-year responder status in sensitivity analyses for the primary efficacy endpoint. [Section 8](#) provides details concerning the methods to be used. Three questionnaires were added to the study with the approval of version 7 of the protocol (December 12, 2014). Due to the timing of the addition of these questionnaires, patients enrolled under earlier versions of the protocol are missing these questionnaires at baseline and any other visits attended prior to version 7. For this questionnaire data, missing values were imputed up to 1-year using regression based on available timepoints through PROC MI. Multiply imputed inferences from the unadjusted mean differences and the MMRM mean differences were pooled together using PROC MIANALYZE in order to create combined results for each. As more patients were enrolled under versions 7 and 8 of the protocol and fewer patients enrolled under those versions have reached 2-3-year visit windows at the time of submission, extended follow-up visit responses will not be analyzed for Marx activity rating, EQ-5D-5L, or WPAI.

9.7.7. Interim Analyses

The original study design included a formal interim analysis to be conducted based on 80-90% of the originally planned sample for the purpose of sample size re-estimation. There was no provision for early stopping for superiority. The interim analysis for sample size re-estimation was not performed as the rate of lost to follow-up (the only nuisance parameter) was lower than anticipated (2.4% by Jan 2017 rather than 7%), and the sponsor decided not to conduct the interim analysis for sample size re-estimation.

Additionally, the original study design included a pre-planned administrative analysis to be conducted “for the sponsor’s financial planning purposes with results potentially provided to a small number of individuals responsible for company financing”. This was specified to occur when “~75% of patients have been enrolled”. No specific statistical analysis was specified for this ‘administrative analysis’.

The rigor of this planned “administrative analysis” had been enhanced by pre-planning of a futility analysis based on the determination of conditional power. To this end, a futility analysis was planned take place after 50% of the initially planned sample size (i.e., 123 patients) was due for their month 12 follow-up visit.

However, the iSAP stipulated that the futility analysis “may not be conducted if enrollment is completed or nearly completed by the time that 50% of patients are evaluable for the Month 12 study success criterion.” At the time that approximately 50% of the patients were evaluable for the success criterion, the study enrollment was almost complete, and the sponsor decided not to conduct the administrative futility analysis.

9.7.8. Data Monitoring Committee

A Data Safety Monitoring Committee (DSMB) was convened as the primary data and safety advisory group for the Phase III clinical study. The members of the committee consist of three voting members appointed by the Sponsor that are experts in the fields of Orthopedic Surgery and/or Musculoskeletal Radiology and a non-voting Statistician. The primary responsibility of the DSMB is to protect the

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safety of the study participants. This included evaluation of the progress of the trial, review of the data and all safety events, making recommendations regarding implementation of the stopping rules if any of the criteria are met and possible communication with the FDA regarding any safety concerns. A formal charter of the DSMB was created and a majority of 2 voting members was required for any proposal, motion or recommendation to be made. Meetings occurred two times per year with review of all adverse events and documentation of decisions. No emergency meetings were required for any reason. Meeting will continue throughout the duration of the extended 3 year post procedure follow up for the study.

9.7.9. Pharmacokinetic and Pharmacodynamic Analyses

Not applicable.

9.7.10. Safety Analysis

9.7.10.1. Extent of Exposure

The Safety Population includes all patients who were consented, enrolled and randomized into the clinical study. Patients who were randomized into the NeoCart arm but not implanted with the NeoCart construct were followed for safety events through standard post-operative follow-up for arthroscopy. Safety events were classified by the AT population to ensure that the one patient who was randomized to the NeoCart arm but declined a repeat biopsy and underwent microfracture would have potential events captured under the correct treatment.

9.7.10.2. Adverse Events

For each safety analysis in which the outcome is an event (possibly composite), treatment groups were compared using counts and percentages.

Only treatment-emergent AEs were analyzed. AEs were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 17. The number and percentage of patients with at least one adverse event, at least one serious adverse event (SAE), and at least one procedure related adverse event were presented by treatment group. The number and percentage of patients with at least one NeoCart related adverse event were presented for the NeoCart treatment group. AEs that are definitely, probably, or possibly related to the procedure, or for which the relationship to the procedure is missing, were considered procedure related, and AEs that are definitely, probably, or possibly related to NeoCart, or for which the relationship to NeoCart is missing, were considered NeoCart related. AEs and SAEs are summarized at the patient level by MedDRA system organ class (SOC) and preferred term using frequencies and percentages. AEs will also be tabulated at the event level by severity and by relationship to the procedure for each treatment group, and by relationship to NeoCart for the NeoCart treatment group.

All Adverse Events were captured during the study independent of relatedness to the treatment arm or procedure. Events were assessed by the principal investigator for causality and relatedness to either the procedure or the implant or both. Summaries were restricted to Treatment Emergent Adverse Events (TEAEs), which were defined as those AEs that occurred after the start of the arthroscopic procedure or worsened during the study. Verbatim terms were mapped to preferred terms and System Organ Classes using the Medical Dictionary for Regulatory Activities (MedDRA) dictionary (version 17).

Summaries of the following types are presented:

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- An overview of TEAEs: Patient incidence of any TEAEs, any serious TEAEs, any NeoCart related TEAEs, any procedure related TEAEs, any severe TEAEs, any TEAEs resulting in death and any TEAEs involving the target knee. NeoCart and procedure related TEAEs are TEAEs with a possible or probable relationship to NeoCart or the procedure, respectively. TEAEs with missing severity were considered severe and TEAEs with missing relationship were considered related for this summary. The Fisher's exact test was used to test for a difference in proportions between treatments.
- Patient incidence of all TEAEs by version 17 MedDRA System Organ Class and preferred term: The Fisher's exact test was used to test for a difference in proportions between treatments for each System Organ Class.
- Patient incidence of all serious TEAEs by MedDRA System Organ Class and preferred term.
- Patient incidence of all TEAEs by MedDRA System Organ Class, preferred term, and highest severity: At each level of patient summarization, a patient was classified according to the highest severity if the patient reported more than one TEAE. TEAEs with missing severity were considered severe for this summary.
- Patient incidence of all TEAEs by MedDRA System Organ Class, preferred term, and relationship to the NeoCart treatment and by MedDRA System Organ Class, preferred term, and relationship to the procedure: At each level of patient summarization, a patient was classified according to the closest relationship if the patient reported one or more events. TEAEs with a missing relationship were considered related for this summary.

9.7.10.3. Clinical Laboratory Evaluations

For continuous laboratory parameters, the change from baseline (Day 0) to the 6-week visit are summarized by treatment group using descriptive statistics. For urinalysis parameters with a categorical classification (e.g., negative, trace, positive), the number and percentage of patients in each category are presented by treatment group. For each laboratory test, the numbers and percentages of patients with clinically significant laboratory results are presented by visit and treatment group.

9.8. Changes in the Conduct of the Study or Planned Analyses

9.8.1. Changes in the Conduct of the Study

The study was initiated under Protocol Revision 5 (02Mar2010). The following summarizes the significant changes made to subsequent versions of the protocol, as well as the rationales for the changes.

No subjects were enrolled in the Phase III clinical study using protocol versions 0 to 4. The first subject was treated under protocol version 5 (02Mar2010). Below is a summary of the protocol modifications as noted in each protocol version. All protocol revisions prior to version 5 were submitted to the FDA but were not executed by the clinical sites. The following summarizes the significant changes made to the following protocol versions, as well the rationales for the changes.

Protocol Version 5 (02Mar2010)

- Added to the study design section that the IKDC objective knee exam will be blinded. This was stated elsewhere in the original protocol but it was felt it should be also stated in the Study

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Design section. Reason for change is that it is stated elsewhere in the original protocol, but it was felt it should be also stated.

- Overview: Inclusion of trochlea lesions was removed as an inclusion and added as an exclusion to the study:
 - Arthroscopy Inclusion Criteria:
 - Trochlea lesions no longer allowed
- Changed text to the Arthroscopic exclusion criteria text on pages 7 and 13 from: “more than slight symptoms of anterior knee pain referable to the patellofemoral joint and who have ICRS Grade 3B, C, D, or Grade 4 patellar lesions.” to: “ICRS Grade 3B or greater patella, trochlea or tibial lesions, the size of which exceeds the area of the study lesion(s) and/or is associated with pain specifically and reproducibly referable to the untreated lesions(s).” Added trochlea reference since trochlea no longer a treated surface. Also the original criterion inadvertently omitted the tibia and reference to lesion size.
- Rehabilitation plan text was modified on the following pages 8, 18 and 19. Removed references specific to the trochlea rehabilitation, since trochlea lesions are not treated. Use of stationary bicycle to begin after the 10 day post-operative visit. Original protocol incorrectly stated that NeoCart subjects begin stationary cycling at 4 to 6 weeks. Deleted statement that “modification of existing physical therapy is not made until after the 6 month MRI”. The original language was incorrect. The physical therapy plan is modified on an ongoing basis depending on patient symptoms, range of motion, strength etc.
- Follow-up Evaluations text was modified on the following pages 19, 20 and 21. Changed text from “Record medications taken by patient” to “Record any medication changes and all pain medications taken during week prior to visit. The collection of pain medication use data one week prior to the visit was requested by CBER at the time of SPA approval. The company agreed to the change and modified the screening visit section but neglected to specify the requirement in the follow-up evaluations section.
- Follow-up Evaluations text modified on page 20. Added note related to pain medication recordkeeping. “Note: Pain medication usage in this study may vary from no use of pain medications to multiple medications. Patients using multiple pain medications should be asked to document their pain medication use during the week prior to each visit in order to accurately convey that information. In order to optimize compliance, the recording method (pocket calendar, log, smartphone etc) used should be determined by the patient. The patient consent form should discuss this topic. Reason for the modification” To ensure topic is discussed with patient.
- The following text was added to the inform consent. An excess number of cells are sometimes generated during NeoCart production. The informed consent includes a request that the patient provide his/her permission to allow Histogenics to perform cartilage related research with the excess cells. Patients who do not provide permission may participate in the study. A provision for the taking of pictures of the knee during implant or arthroscopy is included. Photographs are not a protocol requirement. This provision allows for the taking of a photograph in those rare instances in which a photographic image of the knee would be helpful in the documentation of an observation. Reason for text addition, refers to language that has been commonly added in the study consents. This language now added to informed consent template which is provided as Attachment B of the protocol.
- The Statistical plan was updated on pages 2 and 4 (Attachment D of the Protocol). The original statistical plan stated that “a randomized block design, stratified by age (< 42 yrs.), BMI (> 28

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kg/m²), and lesion location (condyle vs. trochlea) will be used.” We now propose that there be two stratification factors (investigative site and BMI). Reason for the change, because with the removal of the treatment of the trochlea, the lesion location stratification factor is no longer relevant. Investigative site was selected as the new stratification factor (to replace lesion location) to ensure that sites with lower enrollment (6/site) will have at least a few subjects in the control (microfracture) arm. Age was dropped as a stratification factor because three stratification factors were considered excessive given the potential for some sites enrolling as few as 6 subjects.

- Page 4 of Statistical plan was modified. The original statistical plan specified an integrated time weighted analysis for the KOOS scores but did not reference to the company’s intention to run a time weighted analysis for the IKDC subjective score as well. The revised plan now specifies the analysis for the IKDC as well “An analysis of the integrated (time-weighted) change from baseline IKDC subjective scores over the 3-, 6-, and 12-month horizon will be conducted. The initial model will be equally weighing of the observations and a weighing of 1:1:2, respectively for the time points (weighted by the size of the time interval).” Reason for the change was that in the original plan, this analysis was inadvertently omitted.
- IKDC objective instructions (Attachment E of the protocol) was modified to added the following text after “stocking present?” question, “if no, is checked, please complete, protocol deviation CRF”, thigh circumference measurement and a note that hop test is not performed at 12 weeks. Reason for this change was to track compliance, added field to assess muscle strength, removed hop test at 3 months as it is inconsistent with rehabilitation status.
- Clarifications when referring to site T2 MRIs –changed from “T2 mapping” to “T2 image acquisition.” Reason for this change is that the investigative site does not create the T2 map. It collects the T2 image data. The MRI core lab creates the T2 map.

Protocol Version 6 (29May2013)

(New text in **Bold**)

SUMMARY OF KEY CHANGES

The key revisions to Protocol # 8-01 and the SAP are summarized below. As well, the Informed Consent Form and the Summary of Procedure Requirements were revised for consistency with the Protocol. A detailed summary of changes by protocol section is attached.

Histogenics is proposing revisions to the protocol to improve the study based on feedback from clinical sites to optimize patient enrollment and clarify protocol interpretation and compliance.

- Increase in the number of clinical sites from 30 to 40 for added availability of patients and patient flow
- Clarifications to inclusion/exclusion criteria to help clinical sites interpret and comply with protocol requirements
- Assessment and documentation of patient rehabilitation
- Clarifications to study stopping criteria for consistency with literature sources (note: no changes to rates or number of events as approved in Version 5 of the protocol)
- Clarify and add to risk analysis an option for either second biopsy or conventional treatment in the event of a manufacturing process failure or defect site increasing in size between arthroscopy and implantation (informed consent template revised to reflect same)

In addition to the proposed revisions to the protocol, Histogenics is also submitting a detailed statistical analysis plan (SAP) for the NeoCart Phase III Protocol #8-01. The proposed SAP was developed in accordance with the document entitled "NeoCart Trial Design Statistical Considerations", authored by Berry Consultants, LLC, dated March 1, 2010 and submitted to FDA as part of the SPA protocol.

The comprehensive SAP included with this protocol revision provides details and clarification to the above referenced Statistical Considerations document. There has been no change in the primary Month 12 efficacy endpoint and no change in the essential aspects of the analysis plan to be used to test for superiority. The key proposed revisions include the following:

- In addition to the KOOS and IKDC subjective score improvement criteria, the clinical composite success (CCS) criteria for Month 24 and Month 36 will require that the patient has not undergone a secondary re-operation for a failed index treatment to insure KOOS and IKDC scores are not reflecting successful secondary treatment for a primary treatment failure
- Details of the specific type of multiple imputation methodology to be used in sensitivity analyses of the primary superiority (Month 12) comparison
- Addition of a formal futility analysis based on conditional power to the previously specified administrative analysis. This futility analysis will be performed when 50% of the initial sample size is theoretically due for Month 12 follow-up.

Protocol Version 7 (06Nov2014)(New text in **BOLD**)**SUMMARY OF KEY CHANGES**

Key changes to Protocol # 8-01 are summarized below and a detailed summary table of changes by protocol section is attached. To allow for administrative efficiency, the Informed Consent Form (ICF) and the Statistical Analysis Plan (SAP) have been removed as attachments to the protocol.

Histogenics is proposing revisions to the protocol to improve the study based on feedback from clinical sites and Histogenics' DSMB. As well, changes have been made to collect additional health outcomes data and the statistical plan has been revised to address patients for whom the NeoCart implant is not available.

- Clarifications to prior surgical treatment of the study defect that are allowed and defining wash-outs for other surgical procedures.
- Modifications to the postoperative therapy regimen to both standardize treatment in both arms of the study and to further reduce risk factors for DVT/PE and arthrofibrosis in patients immobilized post-operatively.
 - The use of a knee immobilizer is no longer a requirement for patients receiving NeoCart. Investigator discretion allowed for post-operative knee immobilization for both treatment groups.
 - For patients not placed in a knee immobilizer, CPM will begin on the day of the study procedure. For patients that are placed in a knee immobilizer, CPM may begin as early as the day of the study procedure but, in any case, no later than the first post-operative visit.
 - For any patient that is placed into a knee immobilizer following the study procedure, aspirin and TED hose will be ordered for, at a minimum, the duration of the immobilization period.
- Additional PRO health outcomes surveys (Marx Activity Rating Scale, SF-12, EQ-5D-5L, and WPAI) to potentially demonstrate improved Quality of Life (QOL) outcomes for patients, as well as NeoCart cost-effectiveness.
- Analysis population redefined to modified Intent-to-treat (mITT) patient population to include all patients randomized, except patients randomized to the NeoCart group for whom the NeoCart implant was not available. The primary efficacy analysis for the superiority claim will be based on the mITT population.

To date, four patients randomized to the NeoCart group did not receive the NeoCart due to various manufacturing-related issues (particulates, voluntary study suspension and failed lot release testing). Since the lack of NeoCart availability is not considered a negative reflection on the efficacy of actual treatment with NeoCart, it is thought that these patients should not be included in the efficacy analyses.

Protocol Version 8.0 (10Nov2015)(New text in **BOLD**)**SUMMARY OF KEY CHANGES**

Key changes to Protocol # 8-01 are summarized below and a detailed summary table of changes by protocol section is attached.

Histogenics is proposing revisions to the protocol to improve the study by enabling study investigators to identify a broader pool of patients that meet the eligibility criteria and to improve the rate of enrollment into the study. Histogenics has carefully and critically revisited the I/E criteria by engaging our investigators in the review of the various screen failure and disqualification modes as well as a review of the literature and the I/E criteria of other studies of cartilage repair products undergoing investigation in US trials.

1. Include lesions of the femoral trochlea and update the rehabilitation plan accordingly
2. Reduce exclusion limits for prior procedures (ligament reconstruction and meniscal repair) from within 6 months prior to study arthroscopy to within 3 months prior to study arthroscopy
3. Remove exclusion of asymptomatic lesions present in non-study locations that are larger than the study lesion.
4. Increase upper limit of age from 55 to 59

9.8.2. **Changes in the Planned Analyses**

The original statistical analysis plan (SAP) for the NeoCart Phase III clinical study, titled “Statistical Analysis Plan for a Randomized Comparison of NeoCart® to Microfracture for the Repair of Articular Cartilage Injuries in the Knee” (dated May 22, 2013), was submitted to the IND (#A80) on May 29, 2013 as an attachment to Revision 6 of the Clinical Trial Protocol #8-01. Since approval of the Special Protocol Assessment in March 2009, each major revision of the protocol and related SAP has been reviewed by the agency and a response received that the SPA was maintained.

The most recent revision was submitted in August 2018 prior to the initial review of the data. This revision was needed to add elements discussed in previous FDA correspondence, provide proper guidance for SAS programming of the Phase III clinical study data analyses and outputs of the tables, figures, and listings (TFLs) for the clinical study report for the Biologics License Application (BLA), and to delete references to interim analyses that were not conducted. No changes were made in the revision, or have been made in any SAP revision, to the primary or secondary endpoints since the approval of the Special Protocol Assessment (see Regulatory History Section 1.3 for more information). The primary population analysis set has also been maintained. Additionally, this revision does not impact the conduct of the Phase III clinical study protocol or how data are collected, and the clinical study protocol is not being amended. Thus, the Special Protocol Assessment is not expected to be impacted by this revision. The ICH E9 guideline, “Statistical Principles for Clinical Trials” has been followed throughout the development of the SAP to its finalization in this document. Regarding revision timing of statistical analysis plans and clinical study database lock, ICH E9 states “The [statistical] plan should be reviewed and possibly updated as a result of the blind review of the data...and should be finalized before breaking the blind.” In accordance with this guideline, the SAP is being finalized and submitted to the Food and Drug Administration (FDA) as an IND amendment prior to the database lock of the Phase III clinical study. The changes included in this revision fall into the following 3 categories and more detailed rationale for each change and corresponding section reference is provided in below.

Commented [AG34]: I think this needs to be clearer – a good part of the content is here but needs to flow – Sumitra should lead on this:

- Original SAP- March 2010
- Next SAP – May 2013
- iSAP updates – A93 & A94 submissions to FDA (June July 2015), that were responses to may 2013
- Then August 2018
-

Three sensitivity analyses are added based on a July 19, 2017 Type C meeting with FDA to ensure that these analyses are conducted for the BLA submission: The analyses are as follows: 1) addition of a modified intent-to-treat (ITT) analysis population; 2) addition of an analysis to statistically compare the difference in the least mean squares between NeoCart and Microfracture arms in changes to KOOS Pain and KOOS Function (SRA) from baseline values at the 1 year and 2 year timepoints; 3) addition of an analysis to statistically compare the proportion of subjects with at least a 10 point improvement from baseline in both in KOOS pain and KOOS function (SRA) at 1 year and 2 year timepoints.

- Text in the previous SAP version (May 22, 2013) describing interim analyses to be conducted, but which were not conducted is being deleted (e.g. interim analyses for re-estimation of sample size and futility). The previous version of the SAP also did not identify the analyses as being optional. Based on discussions with the Agency, an interim statistical plan (iSAP) was later developed and submitted to the IND (#A094,, June 29, 2015). The iSAP provided detailed information on the interim analyses and specified the parameters for triggering the interim analyses.
- Clarification of analyses to be performed for programming purposes. An SAP should provide guidance to the clinical SAS programming team for the analysis and the production of TFLs for the clinical study report. This revision provides more detailed explanation of the analyses stated in the previous SAP version and adds guidance for performing analyses that were not previously included (e.g. how to analyze several patient surveys added in a later protocol version, and analysis of the IKDC objective evaluation).

9.8.3. Changes Following Study Unblinding and Post-hoc Analyses

There were no changes to the protocol V8.0 either before the database lock or post unblinding of the study results.

There were a number of additional analyses that were conducted upon requests from the FDA these included the analysis of the dual threshold response and the individual KOOS pain and IKDC function scores based upon lesion size as determined at the time of treatment.

Commented [AB35]: Were any analyses conducted after unblinding?

Commented [LK36R35]: Yes the cuts by lesion size, very important section

Commented [AG37R35]: Yes, we need to add in "Per FDA request, we also conducted X analyses based on size."

Commented [LK38R35]: Aaron: Important: We also did subgroup analyses. (The SAP specifically stated "No subgroup analyses will be performed.")

Commented [AG39R35]: We can leave for now, subject to completion upon feedback from FDA

10. STUDY SUBJECTS

10.1. Disposition of Patients

Figure 2 details the disposition of patients from evaluation and entry into the trial, randomization and the time point at which any patients were lost to the study including the reason if known. There were no patient deaths or withdrawal from the study due to adverse events of any severity.

Figure 2: Disposition of Subjects

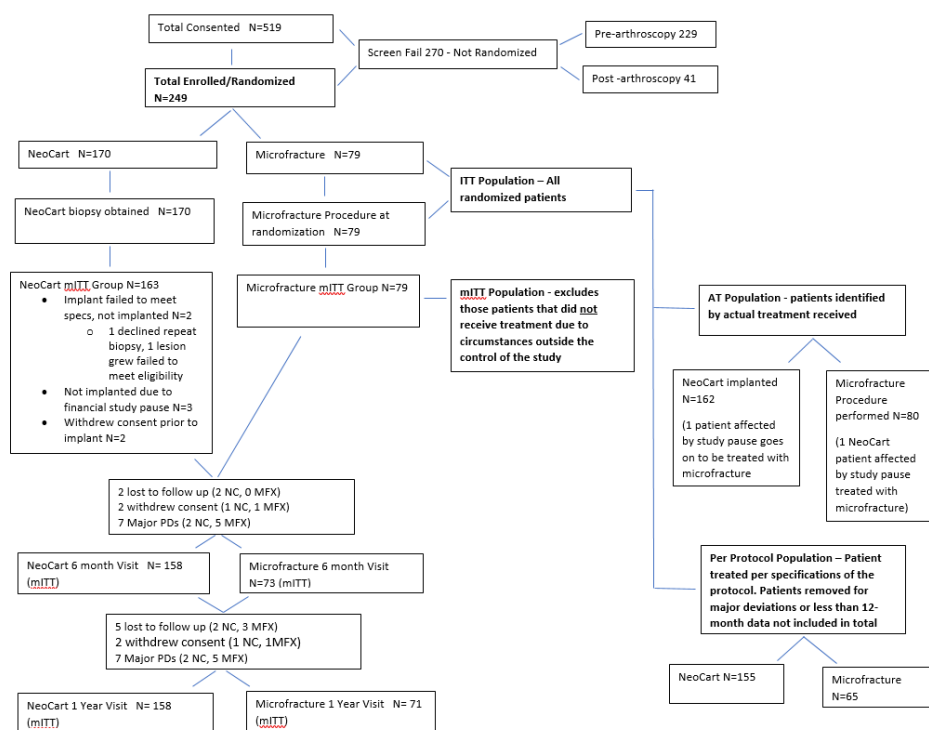


Table 7: Analysis Populations

Disposition	NeoCart n (%)	Microfracture n (%)	Total n (%)
Consented Patients			519
Screen Failures			270
Randomized/ITT Population	170 (68.3)	79 (31.7)	249 (100.0)
Modified ITT Population	163 (67.4)	79 (32.6)	242 (97.2)
As-Treated Population	162 (66.9)	80 (33.1)	242 (97.2)
Per Protocol Population	155 (70.5)	65 (29.5)	220 (88.4)

Source: Table 14.1.1.1

A total of 518 patients were consented; of these 269 were determined to be screen fails. A total of 148 patients were identified prior to the diagnostic procedure with 9.8% not meeting the inclusion criteria and 18.7% having one or more exclusion criteria. A total of 53 patients were excluded at the time of arthroscopy with 5.2% not meeting the inclusion criteria and another 5% having one or more exclusion criteria. There was a total of 33 sites that enrolled patients (30 in the US and 3f in Canada), 19 sites enrolled 5 or more patients in the study (see Table 10 for an accounting of the factors that contributed to screen failures). Thus, 249 patients entered the study and were randomized to either the treatment or control arm, with 170 Randomized NeoCart patients, and 79 Microfracture patients, which constitute the Intent to Treat (ITT) study population. During the final stages of enrollment, the sponsor enrolled four (4) additional patients than anticipated due to the moderate screen failure rate. Patients who had already consented once 245 patients were enrolled were still allowed to proceed to be enrolled, if eligible in the final week of enrollment which took place in June 2017.

Table 8 details the patient accountability for patients enrolled in the study. Of the 249 patients who were randomized in the study (the ITT population), 235 patients completed their 1 year follow-up ($235/249 = 94.4\%$ retention rate); if one considers those patients that received treatment initially (the mITT population), then the overall 1-year retention rate was 97.1% ($235/242$). Table 9 lists the patient enrollment by site, categorized by treatment arm and study population. The factors that led to premature discontinuation from the study are provided, as well

A listing of the factors that contributed to the overall screen failures ($n = 269$) is provided in Table 10. The two criteria that contributed to the most screen failures were the pre-procedure inclusion criterion, "Symptomatic with knee pain indicative of an articular cartilage injury of the distal femur. Baseline KOOS knee pain score must be ≤ 80 . Baseline IKDC Subjective score must be ≤ 70 ," and the pre-procedure exclusion criterion, "MRI prior to treatment indicates subject does not have suitable potential lesions or is otherwise contraindicated for treatment"; each accounted for 39 screen failures (14.5%).

Commented [AG40]: We agreed to only report 1 year data, its premature to report discontinued b/c its incomplete, I would modify Table 8

Table 8: Patient Accountability

Disposition	NeoCart n (%)	Microfracture n (%)	Total n (%)
Completed Follow-up through 1 Year	159 (93.5)	75 (94.9)	234 (94.0)

Commented [LK41]: Need to reconcile the one patient in NeoCart that is not above

Note: Percentages are calculated based on the number of randomized patients in each treatment group or total, as appropriate.
. 15 patients were discontinued prior to 1 year: 11 NeoCart - 3 discontinued during pause in study, 4 withdrew consent, 1 lesion worsened post-arthroscopy, 2 lost to follow-up, 1 other; 4 Microfracture - 3 lost to follow-up, 1 withdrew consent.
Source: [Table 14.1.1.2](#)

Table 9: Patient Enrollment by Site

Site	NeoCart						Microfracture			
	Consented	Screen Failure	ITT	MITT	AT	PP	ITT	MITT	AT	PP
02	17	4	9	9	9	9	4	4	4	3
03	39	10	21	19	19	18	8	8	8	6
05	8	5	1	1	1	1	2	2	2	2
06	1	1	0	0	0	0	0	0	0	0
07	17	17	0	0	0	0	0	0	0	0
08	26	14	8	8	8	7	4	4	4	3
09	19	5	8	8	8	8	6	6	6	6
10	9	1	5	5	4	4	3	3	4	2
11	2	1	1	1	1	1	0	0	0	0
13	11	4	5	5	5	5	2	2	2	1
14	4	3	1	0	0	0	0	0	0	0
15	4	2	1	1	1	1	1	1	1	1
17	9	5	4	4	4	4	0	0	0	0
19	7	2	4	4	4	2	1	1	1	1
21	7	2	4	4	4	4	1	1	1	1
23	7	1	5	4	4	4	1	1	1	1
25	5	2	1	1	1	1	2	2	2	1
26	13	8	3	3	3	3	2	2	2	2
27	1	1	0	0	0	0	0	0	0	0
28	21	4	12	12	12	11	5	5	5	2
29	5	1	3	3	3	3	1	1	1	1
30	17	13	2	1	1	1	2	2	2	1
31	2	1	1	1	1	1	0	0	0	0
32	1	1	0	0	0	0	0	0	0	0
33	7	1	4	4	4	4	2	2	2	1
34	126	97	19	18	18	18	10	10	10	10
35	28	21	5	4	4	4	2	2	2	1
36	10	0	8	8	8	8	2	2	2	2
38	26	12	9	9	9	9	5	5	5	5
39	18	8	6	6	6	5	4	4	4	4
40	15	9	4	4	4	3	2	2	2	2
41	8	2	5	5	5	5	1	1	1	1
42	3	1	2	2	2	2	0	0	0	0
43	2	2	0	0	0	0	0	0	0	0
44	1	1	0	0	0	0	0	0	0	0
45	9	5	4	4	4	4	0	0	0	0
47	6	1	2	2	2	2	3	3	3	2
48	2	0	1	1	1	1	1	1	1	1
50	6	2	2	2	2	2	2	2	2	2
Total	519	270	170	163	162	155	79	79	80	65

Source: [Table 14.1.1.3](#)

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Table 10: Inclusion/Exclusion Failures

Criteria	Total ³ (N = 270)	
	n	(%) ¹
Pre-Procedure Inclusion Criteria		
Able and willing to give informed consent	13	(4.8)
18 years to 55/59 years of age	0	(0.0)
Symptomatic with knee pain indicative of an articular cartilage injury of the distal femur.	39	(14.4)
Baseline KOOS knee pain score must be ≤ 80. Baseline IKDC Subjective score must be ≤ 70		
Medically able to undergo arthroscopy, arthrotomy for NeoCart implantation, and serial MRIs	1	(0.4)
Females must be post-menopausal or practicing an acceptable form of birth control	0	(0.0)
Pre-Procedure Exclusion Criteria		
Any previous surgical treatment of the study defect (abrasion arthroplasty/chondroplasty, multiple drilling or microfracture). Debridement of the study lesion <3 months prior to the study arthroscopy	3	(1.1)
Procedures within 3 months prior to study arthroscopy, in the judgment of the investigator, if pain is not specifically referable to the cartilage defect	1	(0.4)
Body Mass Index (BMI) > 35 kg/m ²	11	(4.1)
Radiographic evidence of osteoarthritis; Kellgren-Lawrence Grade 3 or greater	28	(10.4)
Malalignment greater than 3 degrees outside the mechanical axis from the contralateral knee, or malalignment requiring surgical correction	23	(8.5)
Symptomatic pathology of the contralateral knee	7	(2.6)
Surgery (with the exception of ligament reconstruction or meniscal repair addressed separately) on the contralateral knee within the 3 months prior to the scheduled arthroscopy	1	(0.4)
Any form of inflammatory arthritis	5	(1.9)
Ankylosing spondylitis	1	(0.4)
Synovoma, hemangioma, pigmented villonodular synovitis (PVNS), or other neoplasms in the knee	0	(0.0)
Any condition requiring chemotherapy	0	(0.0)
Positive pregnancy test or intention to become pregnant during the year following treatment ²	1	(0.4)
Known history of allergy to bovine products or gentamycin	0	(0.0)
Any autoimmune disease	5	(1.9)
Human immunodeficiency virus (HIV), chronic hepatitis B or C viral infection	3	(1.1)
Insulin-dependent diabetes	1	(0.4)
Clinically significant or symptomatic vascular or neurologic disorder of the lower extremities	2	(0.7)
Current drug or alcohol abuse	2	(0.7)
Evidence suggesting inability to comply with the Protocol	15	(5.6)
Ligament reconstruction on either knee within six months prior to study arthroscopy	1	(0.4)
Any concomitant painful or disabling disease of the hips or lower limbs that would interfere with the evaluation of the afflicted knee	7	(2.6)
Smoking habit of more than 1 pack of cigarettes per week and/or are frequent users (>1/week) of chewing tobacco. The number of smokers was limited to 10% of the enrolled subjects	1	(0.4)
Treatment with specific osteoarthritis drugs, such as chondroitin sulfate, diacerein, n-glucosamine, piascledine, and capsaicin, within the last 2 weeks	4	(1.5)
Corticosteroid therapy by systemic or intra-articular route within the last month of baseline or intramuscular or oral corticosteroids within the last 2 weeks	6	(2.2)
Intra-articular injections of hyaluronic acid into the afflicted knee within the last six months	7	(2.6)
Status as a participant in an investigational research trial within the last 3 months	1	(0.4)

Criteria	Total ³ (N = 270)	
	n	(%) ¹
MRI prior to treatment indicates subject does not have suitable potential lesions or is otherwise contraindicated for treatment	40	(14.8)
Unable to proceed due to study pause	14	(5.2)
Initial informed consent given but withdrawn	28	(10.4)
Subject lost to follow up	5	(1.9)
Arthroscopy Inclusion Criteria		
At least one ICRS grade 3 or 4A lesion located on the femur (lateral/medial condyles and/or trochlea) that would be treatable by microfracture, with no substantive bone loss or malignment.	12	(4.4)
1 or 2 non-contiguous/non-opposing lesions may be treated if each meets criteria and the total surface area is <= 6 cm2. Each must be >=1 cm in one dimension, range between 0.5 cm2 - 6 cm2	22	(8.1)
Arthroscopy Exclusion Criteria		
Subchondral bone loss	1	(0.4)
Untreated anterior cruciate ligament (ACL) and/or posterior cruciate ligament (PCL) deficiency or ligamentous instability in the index knee	0	(0.0)
Study lesion requiring a concomitant procedure intended as a therapeutic intervention, or that may alter the mechanics of the joint. Permitted procedures are detailed in the inclusion criteria.	7	(2.6)
Meniscus whose rim is less than 50% normal radial width or whose peripheral rim is not fully intact	6	(2.2)
ICRS Grade 3 or 4 kissing lesions	14	(5.2)
ICRS Grade 3B+ patella, trochlea or tibial lesions, size exceeding the area of the study lesion(s) and/or is associated with pain specifically/reproducibly referable to the untreated lesions(s)	10	(3.7)

(1) Percentages are calculated based on the number of patients who were consented but not randomized. A patient may have failed more than one criterion, so the percentages may sum to more than 100%.

(2) Patient 30-013 is not marked as a failure for positive pregnancy test but no pregnancy test was conducted and patient was of child-bearing potential. Patient has been marked as a major protocol deviation for inclusion/exclusion failure on this criteria.

(3) Only includes patients identified as screen failures who were not randomized. 03-031, 28-004, 28-006, and 28-013 failed criteria but were randomized - these patients are not in the PP population due to these major deviations. 28-008 and 41-005 had recent steroid use, the former has a minor deviation and the latter received a waiver.

Source: [Table 14.1.1.4](#)

10.2. Protocol deviations

There were a total of 9 major protocol deviations: 8 at Baseline and 1 at Arthroscopy (There was suspected contamination of the NeoCart graft the patient refused to undergo a second biopsy and converted to the microfracture treatment arm) and 381 minor deviations (see Table 11). The bulk of the deviations were missed or out of window assessments.

Table 11: Protocol deviations

	NeoCart (N = 170)		Microfracture (N = 79)		Total (N = 249)	
Protocol Deviation	n	(%)	n	(%)	n	(%)
Any Protocol Deviation	141	(82.9)	70	(88.6)	211	(84.7)
Major Deviations						
Randomization procedure	0	(0.0)	0	(0.0)	0	(0.0)
Inclusion/Exclusion criteria	2	(1.2)	6	(7.6)	8	(3.2)
Other	1	(0.6)	0	(0.0)	1	(0.4)
Minor Deviations						
Randomization procedure	4	(2.4)	2	(2.5)	6	(2.4)
Informed consent procedure	4	(2.4)	0	(0.0)	4	(1.6)
Inclusion/Exclusion criteria	5	(2.9)	1	(1.3)	6	(2.4)
Missed/Out-of-Window Follow-Up Visit	65	(38.2)	34	(43.0)	99	(39.8)
Missed assessment	120	(70.6)	55	(69.6)	175	(70.3)
Blinding/Unblinding	13	(7.6)	7	(8.9)	20	(8.0)
Other	59	(34.7)	18	(22.8)	77	(30.9)

Note: Counts are patient level; patients are only counted once for each deviation type. The percentages are based on the number of patients in each treatment group or overall, as appropriate. A patient can have more than one type of protocol deviation, so the percentages may sum to more than 100%.

Source: Study NeoCart 8-01, 14.1.2 Protocol deviations.sas, uses ADDV, Listing(s) 16.2.2

10.3. Demographics and Baseline Characteristics

10.3.1. Demographics

Summaries of subject demographic and baseline disease characteristics are provided in [Table 12](#) and [Table 13](#) for the ITT and mITT populations, respectively. The study inclusion and exclusion criteria were chosen in agreement with the FDA to select for a subject population that would minimize bias against microfracture, the control procedure, where limitations are known in clinical practice. There were no appreciable differences between the patients in each arm of the study. Of the 249 treated patients 66.05% in the NeoCart group and 60.0% in the microfracture arm were male. The mean age in the NeoCart group was 38.6 years and 38.9 in the microfracture group. The majority of patients categorized themselves as white (83.3% NeoCart/80.0% microfracture) with the next largest category [Black](#) or African American 8.6% NeoCart and 8.8% microfracture). Patients weighed between 105 and 286 pounds.

Baseline characteristics were generally balanced between the treatment groups. The mean age is 38.8 years and identical for the ITT population in both treatment arms. The mean lesion size was 2.4cm² in the NeoCart arm and 2.1cm² in the microfracture arm. The distribution between right and left knee for the index procedure was similar both between sides and treatment groups. There were slightly more Medial Femoral Condyle lesions in the NeoCart group (62.4%) than microfracture (48.1%). The mean age of injury NeoCart 1.94 years and microfracture 2.42 years

Activity level prior to injury for the entire study population was very high with 31.2% classifying as high competitive sportsman/woman and 47.1% well trained and frequently sporting in Neocart

arm and similar 30.4% and 44.3% in the microfracture group. With the vast majority in both arms decreasing activity to primarily non-sporting (56.5%/68.4% NeoCart/microfracture) or sporting sometimes 35.3%/27.8% respectively at the time of entry into the study. There was also a very large percentage of both treatment groups that were functionally restricted and unable to do many of the things they would like due to knee pain (NeoCart 77.6%, microfracture 73.4%) at the time of entry into the study (Table 14.1.6).

Demographics for the AT and PP populations are found in Section 14.1.4, Table 14.1.3.1.3 and Table 14.1.3.1.4, respectively.

Commented [AB42]: I cannot find this table in the TFLs provided by Jocelyn.

Table 12: Demographics (ITT Population)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)	Total (N = 249)
Age (years)¹	n	170	79	249
	Mean	38.8	38.8	38.8
	Median	39	38	39
	SD	8.5	9.7	8.9
	Min-Max	18 - 55	21 - 59	18 - 59
Sex				
Female	n (%)	59 (34.7)	31 (39.2)	90 (36.1)
Male	n (%)	111 (65.3)	48 (60.8)	159 (63.9)
Race²				
American Indian or Alaska Native	n (%)	0 (0.0)	1 (1.3)	1 (0.4)
Asian	n (%)	2 (1.2)	4 (5.1)	6 (2.4)
Black or African American	n (%)	14 (8.2)	7 (8.9)	21 (8.4)
Native Hawaiian or Other Pacific Islander	n (%)	1 (0.6)	0 (0.0)	1 (0.4)
White	n (%)	142 (83.5)	63 (79.7)	205 (82.3)
Other	n (%)	6 (3.5)	3 (3.8)	9 (3.6)
Unknown	n (%)	5 (2.9)	1 (1.3)	6 (2.4)
Ethnicity				
Hispanic or Latino	n (%)	16 (9.4)	5 (6.3)	21 (8.4)
Not Hispanic or Latino	n (%)	154 (90.6)	74 (93.7)	228 (91.6)
Height (inches)				
	n	168	79	247
	Mean	69.7	69.2	69.5
	Median	70	69	70
	SD	3.9	3.9	3.9
	Min-Max	60 - 85	59 - 79	59 - 85
Weight (lbs)				
	n	168	79	247
	Mean	189.5	182.5	187.3
	Median	190.5	182	190
	SD	33.5	35.5	34.3
	Min-Max	114.4 - 286	105 - 275	105 - 286
BMI (kg/m²)				
	n	170	79	249
	Mean	27.3	26.7	27.1
	Median	27.5	26.6	27.3
	SD	3.6	4.0	3.7
	Min-Max	20.3 - 34.9	18.7 - 36.4	18.7 - 36.4
Pain Medications Taken in Past Week				
Yes	n (%)	71 (41.8)	36 (45.6)	107 (43.0)
No	n (%)	99 (58.2)	43 (54.4)	142 (57.0)

(1) Age as collected at date of informed consent

(2) Patients who checked more than one race are categorized as 'Other'

NOTE: Two patients (03-017, 03-019) had source documents misplaced and therefore some baseline data is missing.

Source: [14.1.3.1.1](#)

Table 13: Demographics (mITT Population)

Parameter		Statistic	NeoCart (N = 163)	Microfracture (N = 79)	Total (N = 242)
Age (years) ¹		n	163	79	242
		Mean	38.7	38.8	38.7
		Median	39	38	38.5
		SD	8.5	9.7	8.9
		Min-Max	18 - 55	21 - 59	18 - 59
Sex	Female	n (%)	56 (34.4)	31 (39.2)	87 (36.0)
	Male	n (%)	107 (65.6)	48 (60.8)	155 (64.0)
Race ²	American Indian or Alaska Native	n (%)	0 (0.0)	1 (1.3)	1 (0.4)
	Asian	n (%)	2 (1.2)	4 (5.1)	6 (2.5)
	Black or African American	n (%)	14 (8.6)	7 (8.9)	21 (8.7)
	Native Hawaiian or Other Pacific Islander	n (%)	1 (0.6)	0 (0.0)	1 (0.4)
	White	n (%)	136 (83.4)	63 (79.7)	199 (82.2)
	Other	n (%)	6 (3.7)	3 (3.8)	9 (3.7)
	Unknown	n (%)	4 (2.5)	1 (1.3)	5 (2.1)
Ethnicity	Hispanic or Latino	n (%)	14 (8.6)	5 (6.3)	19 (7.9)
	Not Hispanic or Latino	n (%)	149 (91.4)	74 (93.7)	223 (92.1)
Height (inches)		n	163	79	242
		Mean	69.8	69.2	69.6
		Median	70	69	70
		SD	3.9	3.9	3.9
		Min-Max	60 - 85	59 - 79	59 - 85
Weight (lbs)		n	163	79	242
		Mean	189.6	182.5	187.2
		Median	190.7	182	190
		SD	33.5	35.5	34.3
		Min-Max	114.4 - 286	105 - 275	105 - 286
BMI (kg/m ²)		n	163	79	242
		Mean	27.3	26.7	27.1
		Median	27.5	26.6	27.3
		SD	3.6	4.0	3.7
		Min-Max	20.3 - 34.9	18.7 - 36.4	18.7 - 36.4
Pain Medications Taken in Past Week					
	Yes	n (%)	68 (41.7)	36 (45.6)	104 (43.0)
	No	n (%)	95 (58.3)	43 (54.4)	138 (57.0)

(1) Age as collected at date of informed consent

(2) Patients who checked more than one race are categorized as 'Other'

NOTE: Two patients (03-017, 03-019) had source documents misplaced and therefore some baseline data is missing.

Source: [Table 14.1.3.1.2](#)

10.3.2. Baseline Disease Characteristics

This section details the baseline condition(s) of the study patients as it relates to their "primary disease", namely, the knee condition that led to enrollment into the study. [Table 14](#) provides a listing of the previous knee surgeries (in the index and contralateral knee) undergone for patients in the treatment and control arms for the ITT population. The incidence of all previous knee surgeries is similar between the two arms in this population. An accounting of the previous knee surgeries for the mITT, AT, and PP populations are located in [Section 14.1.4](#), [Table 14.1.5.2](#), [Table 14.1.5.3](#), and [Table 14.1.5.4](#), respectively.

Table 14: Knee Surgical History (ITT Population)

Parameter	NeoCart (N = 170)		Microfracture (N = 79)	
	Treatment Knee n (%)	Contralateral Knee n (%)	Treatment Knee n (%)	Contralateral Knee n (%)
	170 (100.0) 0 (0.0)	170 (100.0) 0 (0.0)	79 (100.0) 0 (0.0)	79 (100.0) 0 (0.0)
Bone Realignment - Tibial Tubercle Transfer	2 (1.2)	1 (0.6)	0 (0.0)	0 (0.0)
Patellectomy	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Proximal Anterior	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Proximal Distal	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Proximal Lateral	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Proximal Medial	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Trochlear Plasty	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Cartilage Resurfacing and Reconstructive Surgery	38 (22.4)	14 (8.2)	15 (19.0)	3 (3.8)
Abrasion Arthroplasty	0 (0.0)	1 (0.6)	0 (0.0)	1 (1.3)
Autologous Chondrocyte Implantation with Membrane	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Autologous Chondrocyte Implantation with Periosteum	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Carbon Fibre Resurfacing	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Debridement	32 (18.8)	9 (5.3)	10 (12.7)	1 (1.3)
Microfracture	1 (0.6)	5 (2.9)	0 (0.0)	2 (2.5)
Multiple Osteochondral	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Autologous Grafts				
Osteochondral Allograft	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Perichondral Resurfacing	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Periosteal Resurfacing	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Subchondral Drillings	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Other Type of Treatment	6 (3.5)	4 (2.4)	4 (5.1)	2 (2.5)
Other, Non-Study Lesion	13 (7.6)	0 (0.0)	3 (3.8)	0 (0.0)
Treatment (Treatment Knee)				
Extensor Mechanism Surgery	1 (0.6)	3 (1.8)	0 (0.0)	2 (2.5)
Patella Tendon Repair	1 (0.6)	3 (1.8)	0 (0.0)	2 (2.5)
Quadriceps Tendon Repair	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)

Parameter	NeoCart (N = 170)		Microfracture (N = 79)	
	Treatment Knee n (%)	Contralateral Knee n (%)	Treatment Knee n (%)	Contralateral Knee n (%)
Ligament Surgery	26 (15.3)	15 (8.8)	13 (16.5)	4 (5.1)
ACL Repair	12 (7.1)	11 (6.5)	7 (8.9)	3 (3.8)
Extraarticular ACL Reconstruction	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Intraarticular ACL Reconstruction	13 (7.6)	5 (2.9)	7 (8.9)	1 (1.3)
Intraarticular PCL Reconstruction	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Lateral Collateral Ligament Repair/Reconstruction	0 (0.0)	0 (0.0)	1 (1.3)	0 (0.0)
Medial Collateral Ligament Repair/Reconstruction	1 (0.6)	2 (1.2)	0 (0.0)	0 (0.0)
PCL Repair	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Posterolateral Corner Reconstruction	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Meniscal Surgery	35 (20.6)	18 (10.6)	21 (26.6)	9 (11.4)
Lateral Meniscal Repair	0 (0.0)	3 (1.8)	1 (1.3)	2 (2.5)
Lateral Meniscal Transplant	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Lateral Meniscectomy	13 (7.6)	4 (2.4)	7 (8.9)	1 (1.3)
Medial Meniscal Repair	4 (2.4)	5 (2.9)	5 (6.3)	2 (2.5)
Medial Meniscal Transplant	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Medial Meniscectomy	23 (13.5)	8 (4.7)	9 (11.4)	4 (5.1)
Unknown	0 (0.0)	1 (0.6)	0 (0.0)	1 (1.3)
Osteotomy Surgery	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Femur	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Tibia	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Varus	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Valgus	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Other	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Patellofemoral Surgery - Soft Tissue Realignment	3 (1.8)	1 (0.6)	2 (2.5)	1 (1.3)
Lateral Release	2 (1.2)	1 (0.6)	2 (2.5)	1 (1.3)
Medial Imbrication	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Patellofemoral Plica	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)

Note: Patients may have more than one type of surgery so the percentages may sum to more than 100%.

Source: [Table 14.1.5.1](#)

The baseline ICRS cartilage evaluation for the ITT population is presented in [Table 15](#). In general, the patients enrolled (in both study arms) had healthy knees and very high activity levels prior to the onset of the injury that led them to study enrollment. A total of 89% of the patients in the treatment arm characterized their contralateral knee as "Normal" or "Nearly Normal"

compared to 96% of the patients in the control arm. In addition, prior to their injury, 78% of the patients in the treatment arm described their activity level as either "highly competitive sportsman/women" or "well-trained and frequently sporting"; 75% of the patients in the control arm described themselves in one of these two categories, as well. When asked if they faced any restrictions to knee function prior to their injury, only 1.2% of treatment patients mentioned any restrictions, while 2.5% of the control arm patients (in the ITT population) mention restrictions. Conversely, at baseline, 95% of patients in both study arms felt some form of functional restriction due to their knee; this speaks to the level of debility patients felt upon entering the study. The mean time between injury onset and study enrollment was 1.94 years (median = 0.55 years) for the treatment arm and 2.42 years (median = 0.69 years) for the control arm. Baseline ICRS cartilage evaluation results for the mITT, AT, and PP populations are located in [Section 14.1.4](#), Table 14.1.6.2, Table 14.1.6.3, and Table 14.1.6.4, respectively.

Table 15: Baseline ICRS Cartilage Evaluation (ITT Population)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Treatment Knee			
Left	n (%)	90 (52.9)	38 (48.1)
Right	n (%)	76 (44.7)	40 (50.6)
Condition of Contralateral Knee			
Normal	n (%)	127 (74.7)	57 (72.2)
Nearly Normal	n (%)	25 (14.7)	19 (24.1)
Abnormal	n (%)	13 (7.6)	2 (2.5)
Severely abnormal	n (%)	1 (0.6)	0 (0.0)
Time Since Injury (years)			
	n	165	78
	Mean	1.94	2.42
	Median	0.55	0.69
	SD	3.41	4.12
	Min - Max	0 - 21	0 - 22
Etiology/Cause of Injury or Symptoms			
Non-traumatic, gradual onset	n (%)	64 (37.6)	33 (41.8)
Non-traumatic, sudden onset	n (%)	43 (25.3)	22 (27.8)
Traumatic, non-contact onset	n (%)	32 (18.8)	11 (13.9)
Traumatic, contact onset	n (%)	27 (15.9)	12 (15.2)
Activity at Time of Onset of Symptoms			
Activity of daily living (ADL)	n (%)	72 (42.4)	24 (30.4)
Traffic accident	n (%)	3 (1.8)	0 (0.0)
Work injury	n (%)	4 (2.4)	5 (6.3)
Other	n (%)	87 (51.2)	49 (62.0)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Activity Level Prior to Injury			
High competitive sportsman/woman	n (%)	53 (31.2)	24 (30.4)
Well-trained and frequently sporting	n (%)	80 (47.1)	35 (44.3)
Sporting sometimes	n (%)	26 (15.3)	17 (21.5)
Non-sporting	n (%)	7 (4.1)	2 (2.5)
Activity Level Now			
High competitive sportsman/woman	n (%)	0 (0.0)	2 (2.5)
Well-trained and frequently sporting	n (%)	10 (5.9)	0 (0.0)
Sporting sometimes	n (%)	60 (35.3)	22 (27.8)
Non-sporting	n (%)	96 (56.5)	54 (68.4)
Functional Status Prior to Injury			
I can do everything that I want to do with my joint	n (%)	132 (77.6)	71 (89.9)
I can do nearly everything that I want to do with my joint	n (%)	32 (18.8)	5 (6.3)
I am restricted and a lot of things that I want to do with my joint are not possible	n (%)	2 (1.2)	2 (2.5)
I am very restricted and I can do almost nothing with my joint without severe pain and disability	n (%)	0 (0.0)	0 (0.0)
Functional Status Now			
I can do everything that I want to do with my joint	n (%)	0 (0.0)	1 (1.3)
I can do nearly everything that I want to do with my joint	n (%)	4 (2.4)	2 (2.5)
I am restricted and a lot of things that I want to do with my joint are not possible	n (%)	132 (77.6)	58 (73.4)
I am very restricted and I can do almost nothing with my joint without severe pain and disability	n (%)	30 (17.6)	17 (21.5)

(1) Time since injury = (Date of ICRS Cartilage Evaluation - Date of Onset of Injury)/365.25. For partially reported dates, difference in months (/12) and years or years are calculated.

Source: [Table 14.1.6.1](#)

Results from the Baseline IKDC objective exam for both study arms in the ITT population are presented in [Table 16](#).

Table 16: Baseline IKDC objective exam (ITT Population)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Treatment Knee			
Left	n (%)	91 (53.5)	39 (49.4)
Right	n (%)	76 (44.7)	40 (50.6)
Missing	n (%)	2 (1.2)	0 (0.0)
Generalized Laxity			
Tight	n (%)	2 (1.2)	2 (2.5)
Normal	n (%)	153 (90.0)	69 (87.3)

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Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Lax	n (%)	10 (5.9)	8 (10.1)
Missing	n (%)	4 (2.4)	0 (0.0)
Alignment			
Obvious varus	n (%)	15 (8.8)	5 (6.3)
Normal	n (%)	148 (87.1)	72 (91.1)
Obvious valgus	n (%)	4 (2.4)	2 (2.5)
Missing	n (%)	2 (1.2)	0 (0.0)
Patella Position			
Obvious baja	n (%)	0 (0.0)	1 (1.3)
Normal	n (%)	158 (92.9)	77 (97.5)
Obvious alta	n (%)	7 (4.1)	1 (1.3)
Missing	n (%)	4 (2.4)	0 (0.0)
Patella Subluxation/Dislocation			
Centered	n (%)	157 (92.4)	78 (98.7)
Subluxable	n (%)	8 (4.7)	0 (0.0)
Subluxed	n (%)	1 (0.6)	0 (0.0)
Dislocated	n (%)	0 (0.0)	0 (0.0)
Missing	n (%)	3 (1.8)	1 (1.3)
Passive Range of Motion, Treatment Knee¹			
	n	167	79
	Mean	136.01	138.46
	Median	137	140
	SD	10.99	11.58
	Min - Max	86 - 157	100 - 164
Active Range of Motion, Treatment Knee¹			
	n	166	79
	Mean	133.22	135.00
	Median	135	135
	SD	11.08	11.25
	Min - Max	76 - 154	95 - 160
Passive Range of Motion, Contralateral Knee¹			
	n	167	79
	Mean	140.68	142.97
	Median	142	143
	SD	9.99	9.82
	Min - Max	100 - 165	110 - 166
Active Range of Motion, Contralateral Knee¹			
	n	165	79
	Mean	138.52	140.03
	Median	140	140
	SD	9.03	9.46
	Min - Max	100 - 156	107 - 160
Effusion			
None	n (%)	93 (54.7)	41 (51.9)
Mild	n (%)	64 (37.6)	35 (44.3)
Moderate	n (%)	10 (5.9)	3 (3.8)
Severe	n (%)	0 (0.0)	0 (0.0)
Missing	n (%)	2 (1.2)	0 (0.0)

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Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Treatment knee ligamentously stable			
Yes	n (%)	160 (94.1)	79 (100.0)
No	n (%)	3 (1.8)	0 (0.0)
Missing	n (%)	7 (4.1)	0 (0.0)
Crepitus Anterior Compartment			
None	n (%)	112 (65.9)	49 (62.0)
Moderate	n (%)	31 (18.2)	16 (20.3)
Mild pain	n (%)	18 (10.6)	12 (15.2)
Crepitation with > mild pain	n (%)	6 (3.5)	2 (2.5)
Missing	n (%)	2 (1.2)	0 (0.0)
Crepitus Lateral Compartment			
None	n (%)	147 (86.5)	65 (82.3)
Moderate	n (%)	11 (6.5)	5 (6.3)
Mild pain	n (%)	6 (3.5)	6 (7.6)
Crepitation with > mild pain	n (%)	3 (1.8)	2 (2.5)
Missing	n (%)	2 (1.2)	1 (1.3)
Crepitus Medial Compartment			
None	n (%)	136 (80.0)	66 (83.5)
Moderate	n (%)	15 (8.8)	4 (5.1)
Mild pain	n (%)	11 (6.5)	7 (8.9)
Crepitation with > mild pain	n (%)	5 (2.9)	2 (2.5)
Missing	n (%)	2 (1.2)	0 (0.0)
Functional Test, Treatment Knee²			
	n	127	66
	Mean	38.45	34.59
	Median	37.7	34.2
	SD	16.64	16.54
	Min - Max	1.8 - 100	1.7 - 84
Functional Test, Contralateral Knee²			
	n	139	69
	Mean	47.04	44.63
	Median	48	45.7
	SD	16.08	16.47
	Min - Max	8.5 - 100	3.7 - 86.3

(1) Range of Motion derived as Hyperextension + Flexion - Zero

(2) Functional Test Score derived using the average per knee of up to 3 trials.

Source: [Table 14.1.9.1](#)

Results from the arthroscopic evaluation for patients in both study arms within the ITT population are presented in Table 17.

The majority of patients had a single study lesion in the index knee (NeoCart 84.1%, microfracture 91.1%). The most common additional procedure performed was partial meniscectomy (24.7% NeoCart and 26.6% microfracture). The median % of meniscus remaining in both arms was 90%. The arthroscopic evaluation results for the mITT, AT, and PP populations are located in Appendix.

Prior surgical history of the index knee with either meniscal or ligament repair had occurred in 30% of the NeoCart patients and 33% of the microfracture patients.

There was no evidence of osteoarthritis in the overwhelming majority of the patients in either group and no impairment of the joint space or sclerosis. Both treatment groups demonstrated a normal varus axis alignment within 2-3 degrees of the mechanical axis of the treatment knee (NeoCart 52.9%, microfracture 54.4%) with approximately a third in each arm having minor valgus alignment.

Table 17: Arthroscopic Non-Study Lesions (ITT Population)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
Total Number of Non-Study Lesions			
0	n (%)	52 (30.6)	30 (38.0)
1	n (%)	53 (31.2)	18 (22.8)
2	n (%)	37 (21.8)	19 (24.1)
3	n (%)	18 (10.6)	5 (6.3)
4	n (%)	9 (5.3)	7 (8.9)
5	n (%)	1 (0.6)	0 (0.0)
Lesion Surface Location¹			
Medial Femoral Condyle	n (%)	20 (9.0)	8 (8.1)
Lateral Femoral Condyle	n (%)	14 (6.3)	6 (6.1)
Patella	n (%)	80 (36.0)	34 (34.3)
Tibia	n (%)	67 (30.2)	30 (30.3)
Trochlea	n (%)	41 (18.5)	21 (21.2)
Lesion Location¹			
Sagittal			
Anterior	n (%)	16 (7.2)	3 (3.0)
Distal	n (%)	7 (3.2)	6 (6.1)
Middle	n (%)	147 (66.2)	69 (69.7)
Posterior	n (%)	30 (13.5)	14 (14.1)
Proximal	n (%)	10 (4.5)	7 (7.1)
Missing	n (%)	12 (5.4)	0 (0.0)
Coronal/Axial			
Lateral	n (%)	39 (17.6)	17 (17.2)
Central	n (%)	113 (50.9)	58 (58.6)
Medial	n (%)	58 (26.1)	24 (24.2)
Missing	n (%)	12 (5.4)	0 (0.0)
ICRS Cartilage Lesion Grade¹			
1	n (%)	54 (24.3)	23 (23.2)
2	n (%)	129 (58.1)	63 (63.6)
3A	n (%)	21 (9.5)	9 (9.1)
3B	n (%)	7 (3.2)	2 (2.0)
3C	n (%)	3 (1.4)	1 (1.0)
3D	n (%)	1 (0.5)	1 (1.0)
4A	n (%)	6 (2.7)	0 (0.0)

Parameter	Statistic	NeoCart (N = 170)	Microfracture (N = 79)
4B	n (%)	0 (0.0)	0 (0.0)
Missing	n (%)	1 (0.5)	0 (0.0)
Treatment Done¹			
Debridement	n (%)	54 (24.3)	27 (27.3)
Other	n (%)	0 (0.0)	4 (4.0)
None	n (%)	168 (75.7)	68 (68.7)
Pre-Treatment Lesion Shape¹			
Oval	n (%)	101 (45.5)	39 (39.4)
Rectangular	n (%)	25 (11.3)	12 (12.1)
Round	n (%)	68 (30.6)	38 (38.4)
Square	n (%)	2 (0.9)	2 (2.0)
Stellate	n (%)	12 (5.4)	6 (6.1)
Trapezoid	n (%)	0 (0.0)	1 (1.0)
Triangular	n (%)	2 (0.9)	1 (1.0)
Missing	n (%)	12 (5.4)	0 (0.0)
Pre-Treatment Lesion Area Estimate (cm2)¹			
	n	221	97
	Mean	1.04	1.14
	Median	0.8	1
	SD	0.91	1.17
	Min - Max	0.01 - 4.8	0 - 6
Total Pre-Treatment Lesion Area Estimate (cm2)²			
	n	118	48
	Mean	1.95	2.30
	Median	1.23	1.5
	SD	1.84	2.82
	Min - Max	0.04 - 8.5	0.002 - 16.35
Kissing Lesion^{1,3}			
Yes	n (%)	18 (8.1)	13 (13.1)
No	n (%)	202 (91.0)	86 (86.9)
Missing	n (%)	2 (0.9)	0 (0.0)

(1) Lesion level, so patients may appear more than once.

(2) Patient level, patients only appear once. For patients with more than one non-study lesion, the total lesion area is the sum of the areas of all lesions.

(3) Kissing lesions permitted if under grade 3B to 4.

Source: [Table 14.1.11.4.1](#)

While Table 17 focused on the study lesion as characterized during the arthroscopic evaluation in the ITT population, [Table 18](#) presents the lesion characteristics in both arms of the As-Treated population. The median total post-treatment estimated lesion size in the NeoCart arm in the AT population (2.27 cm²) is approximately 25% larger than the median total post-treatment estimated lesion size (1.8 cm²) in the Microfracture arm in the AT population. Given that the median post-treatment lesion size in the Microfracture arm was 180 mm², and the median number of microfracture holes was 12 (per lesion), then each microfracture hole encompassed 15 mm² of

area, on average. The lesion characteristics at treatment for the ITT, mITT, and PP populations are located in [Section 14.1.4](#).

Table 18: Lesion Parameters at Treatment (AT Population)

Parameter	Statistic	NeoCart (N = 162)	Microfracture (N = 80)
Number of Treated Study Lesions			
1	n (%)	139 (85.8)	73 (91.3)
2	n (%)	23 (14.2)	7 (8.8)
Post-Treatment Lesion Shape¹			
Oval	n (%)	85 (45.9)	51 (58.6)
Rectangular	n (%)	66 (35.7)	17 (19.5)
Round	n (%)	14 (7.6)	16 (18.4)
Square	n (%)	5 (2.7)	0 (0.0)
Stellate	n (%)	0 (0.0)	1 (1.1)
Trapezoid	n (%)	14 (7.6)	1 (1.1)
Triangular	n (%)	1 (0.5)	1 (1.1)
Post-Treatment Lesion Area Estimate (cm2)¹			
	n	185	87
	Mean	2.17	2.04
	Median	1.98	1.76
	SD	1.26	1.15
	Min - Max	0.24 - 6	0.5 - 6
Total Post-Treatment Lesion Area Estimate (cm2)²			
	n	162	80
	Mean	2.48	2.22
	Median	2.27	1.8
	SD	1.33	1.29
	Min - Max	0.5 - 6	0.5 - 6.5
Post-Treatment Lesion Depth – minimum (mm)¹			
	n	185	86
	Mean	2.29	2.33
	Median	2	2
	SD	0.70	0.68
	Min - Max	0.5 - 5	0.5 - 4
Post-Treatment Lesion Depth – maximum (mm)¹			
	n	185	86
	Mean	2.93	3.12
	Median	3	3
	SD	0.73	0.81
	Min - Max	1 - 5	1 - 5

Parameter	Statistic	NeoCart (N = 162)	Microfracture (N = 80)
Was Calcified Layer Removed?¹			
Yes	n (%)	179 (96.8)	
No	n (%)	6 (3.2)	
If Yes, Calcified Layer Remaining¹			
	n	177	
	Mean	1.02	
	Median	0	
	SD	7.77	
	Min - Max	0 - 75	
Any Bleeding from the Lesion Bed?¹			
Yes	n (%)	94 (50.8)	
No	n (%)	91 (49.2)	
Method Used to Stop Bleeding, if Required^{1,3}			
Dry gauze with pressure	n (%)	55 (29.7)	
Epinephrine	n (%)	50 (27.0)	
Gel foam	n (%)	5 (2.7)	
Other	n (%)	15 (8.1)	
Number of Microfracture Holes¹			
	n		87
	Mean		16.68
	Median		12
	SD		15.46
	Min - Max		2 - 99

(1) Lesion level, so patients may appear more than once.

(2) Patient level, patients only appear once. For patients with more than one non-study lesion, the total lesion area is the sum of the areas of all lesions.

(3) Multiple methods could be used to stop bleeding

Source: [Table 14.1.12.3](#)

The next three tables focus on the NeoCart implant procedure, specifically, in the AT population. [Table 19](#) describes attributes of the NeoCart procedure. The median number of 57 days elapsed between the arthroscopic evaluation and the index procedure. The mean procedure length was 74 minutes, which included the time to ensure intra-operative inclusion and exclusion criteria were measured and documented. In 93% of the cases, only one adhesive kit was needed to secure the implant. The attributes of the NeoCart implant procedure for the ITT, mITT, and PP population are located in [Section 14.1.4](#), [Table 14.1.x](#), [Table 14.1.x](#), and [Table 14.1.x](#), respectively.

Table 19: NeoCart Implant Procedure (AT Population)

Parameter	Statistic	NeoCart (N = 162)
Time Since Arthroscopy (days)	n	162
	Mean	57.90
	Median	57
	SD	8.88
	Min - Max	43 - 99
Incision Location		
Lateral Parapatellar	n (%)	48 (29.6)
Medial Parapatellar	n (%)	104 (64.2)
Other	n (%)	10 (6.2)
Incision Length (mm)	n	162
	Mean	5.89
	Median	6
	SD	1.96
	Min - Max	0.8 - 13
Duration of Procedure (minutes)	n	161
	Mean	73.79
	Median	71
	SD	25.36
	Min - Max	30 - 150
Duration of Tourniquet Use (Down-Up) (minutes)	n	44
	Mean	37.95
	Median	35
	SD	23.58
	Min - Max	2 - 133
Duration of CT3 Application (Last Applied-Start Mix) (minutes)	n	146
	Mean	7.80
	Median	6
	SD	4.47
	Min - Max	3 - 28

Parameter	Statistic	NeoCart (N = 162)
Second CT3 Kit Needed?		
Yes	n (%)	11 (6.8)
No	n (%)	151 (93.2)
Duration of Second CT3 Application, if required (Last Applied-Start Mix) (minutes)	n	11
	Mean	2.55
	Median	1
	SD	2.38
	Min - Max	1 - 8
Angle of Immobilization Brace (degrees)	n	161
	Mean	0.31
	Median	0
	SD	2.07
	Min - Max	0 - 20

Source: [Table 14.1.12.1](#)

The NeoCart implant characteristics are detailed in [Table 20](#) for the AT population. In 89% of the cases, the surgeons reported complete contact of the implant with the native cartilage; 11% had a gap at the border that did not exceed 1 mm. In 94% of the cases, healthy cartilage was said to have surrounded the implant. The implant was rarely reported to be higher than the native cartilage. In general, the implant was found to be either the same height as the native cartilage or lower than the native cartilage (instances where "all" the implant was lower than the native cartilage occurred in 19/162 = 12% of the cases). The NeoCart implant characteristics for the ITT, mITT, and PP populations are located in [Section 14.1.4](#), [Table 14.1.11.3.2](#), [Table 14.1.11.3.3](#), and [Table 14.1.11.3.4](#), respectively.

Table 20: NeoCart Implant Characteristics (AT Population)

Parameter	Statistic	NeoCart (N = 162)
NeoCart Fit Edges		
Complete contact with native cartilage	n (%)	164 (88.6)
Gap at border does not exceed 1mm	n (%)	20 (10.8)
3/4 of implant contacting native cartilage, 1/4 with a notable border > 1mm width	n (%)	1 (0.5)
1/2 of implant contacting native, 1/2 with a notable border > 1mm width	n (%)	0 (0.0)
From no contact to 1/4 of implant contacting native cartilage	n (%)	0 (0.0)
Healthy Cartilage Surrounding Implant?		
Yes	n (%)	174 (94.1)
No	n (%)	11 (5.9)
Average ICRS Grade of Surrounding Cartilage, if Not Healthy		
	n	11
	Mean	1.18
	Median	1
	SD	0.40
	Min - Max	1 - 2
Portion of Implant Same Height as Native Cartilage		
All	n (%)	124 (67.0)
3/4	n (%)	31 (16.8)
1/2	n (%)	8 (4.3)
1/4	n (%)	2 (1.1)
None	n (%)	20 (10.8)
Portion of Implant Higher than Native Cartilage		
All	n (%)	1 (0.5)
3/4	n (%)	0 (0.0)
1/2	n (%)	2 (1.1)
1/4	n (%)	10 (5.4)
None	n (%)	172 (93.0)
Maximum Height Above Native Cartilage (mm)		
	n	185
	Mean	0.05
	Median	0
	SD	0.39
	Min - Max	0 - 5
Portion of Implant Lower than Native Cartilage		
All	n (%)	19 (10.3)
3/4	n (%)	2 (1.1)
1/2	n (%)	6 (3.2)
1/4	n (%)	21 (11.4)
None	n (%)	137 (74.1)
Maximum Depth Below Native Cartilage (mm)		
	n	182
	Mean	0.18
	Median	0
	SD	0.63
	Min - Max	0 - 5

Source: [Table 14.1.12.4](#)

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The pain management details surrounding the NeoCart implant procedure for the AT population are seen in [Table 21](#). These results are quite comparable to those found during the arthroscopic evaluation in the NeoCart arm of the ITT population. The pain management details arising from the NeoCart implant procedure in the ITT, mITT, and PP populations are located in Appendix.

Table 21: NeoCart Implant Procedure Pain Management (AT Population)

Parameter	NeoCart (N = 162) n (%)
Intra-Operative Anesthesia	
General	142 (87.7)
Spinal/Epidural	14 (8.6)
Regional/MAC (monitored anesthesia care)	18 (11.1)
Post-Operative Pain Management	
Local anesthesia injection	68 (42.0)
Femoral/Regional Block	68 (42.0)
Pain Medication PO	156 (96.3)
Cold Therapy	102 (63.0)
Other	8 (4.9)
None	0 (0.0)

Source: [Table 14.1.12.2](#)

10.3.3. Previous and Concurrent Diseases

The baseline medical history of the ITT population is shown in [Table 22](#). Overall, approximately two-thirds of the patients reported a history of “Musculoskeletal” conditions while one-third reported “Immunological” conditions. Medical History for the mITT, AT and PP populations is located in Appendix.

Table 22: Medical History (ITT Population)

Body System	NeoCart (N = 170)	Microfracture (N = 79)	Total (N = 249)
	n (%)	n (%)	n (%)
Cancer/Neoplasm	7 (4.1)	2 (2.5)	9 (3.6)
Cardiovascular	29 (17.1)	13 (16.5)	42 (16.9)
Dermatological	15 (8.8)	10 (12.7)	25 (10.0)
Endocrine/Metabolic	12 (7.1)	9 (11.4)	21 (8.4)
Gastrointestinal	27 (15.9)	15 (19.0)	42 (16.9)
Genitourinary	17 (10.0)	6 (7.6)	23 (9.2)
Hematological	8 (4.7)	1 (1.3)	9 (3.6)
Immunological	59 (34.7)	23 (29.1)	82 (32.9)
Musculoskeletal	110 (64.7)	55 (69.6)	165 (66.3)
Neurological	29 (17.1)	14 (17.7)	43 (17.3)
Psychiatric/Psychological	26 (15.3)	20 (25.3)	46 (18.5)
Respiratory	27 (15.9)	6 (7.6)	33 (13.3)
Other	48 (28.2)	22 (27.8)	70 (28.1)

Source: [Table 14.1.4.1.1](#)

Smoking has been identified as a negative predictor for poor wound healing in general and specifically to be a negative influence on knee ligament and cartilage repair. (cite Kanneganti P et al, The effect of smoking on ligament and cartilage surgery in the knee: a systematic review, AJSM 2012, Dec: 40(12): 2872-2878.) The baseline smoking status of each arm within the ITT population is shown in [Table 23](#). In the NeoCart arm, 8.2% of patients reported as current smokers/tobacco users; the balance either never smoked (77.6%) or were past smokers (14.1%). In the Microfracture arm, 5.1% of patients self-reported as current smokers/tobacco users, with the balance having never smoked (81.0%) or as previous smokers (13.9%). The baseline smoking status for the mITT, AT, and PP populations are located in Appendix.

Table 23: Baseline Smoking Status (ITT Population)

Parameter	NeoCart (N = 170)	Microfracture (N = 79)	Total (N = 249)
	n (%)	n (%)	n (%)
Cigarette Smoking Status			
Current Smoker/Tobacco User	14 (8.2)	4 (5.1)	18 (7.2)
Never Smoked/Used Tobacco	132 (77.6)	64 (81.0)	196 (78.7)
Past Smoker/Tobacco User	24 (14.1)	11 (13.9)	35 (14.1)
Other Nicotine Use¹			
Chewing Tobacco	3 (1.8)	1 (1.3)	4 (1.6)
Cigar	4 (2.4)	2 (2.5)	6 (2.4)
Nicotine gum	0 (0.0)	0 (0.0)	0 (0.0)
Nicotine patch	0 (0.0)	0 (0.0)	0 (0.0)
Pipe	1 (0.6)	0 (0.0)	1 (0.4)
Vapor Cigarettes	1 (0.6)	0 (0.0)	1 (0.4)
Unknown	2 (1.2)	0 (0.0)	2 (0.8)

(1) Patients may check more than one Other Nicotine Use category, so the percentages may sum to more than 100%.
Source: [Table 14.1.3.2.1](#)

10.3.4. Previous and Concomitant Therapies

Patients were excluded from the study if they had any previous surgical treatment of the index lesion other than debridement. They were also excluded if they had untreated ACL and/or PCL deficiency or ligamentous instability of the index knee.

10.4. Measurements of Treatment Compliance

Need to check with Andrea on how to present rehab and survey compliance if possible.

Commented [LK43]: Aaron: Important: This text belongs in the disposition section. I would think in the context of this study, compliance would relate to compliance with rehabilitation requirements.

Commented [AG44R43]: Agree, we should decide what we are measuring – rehab, filling out

11. EFFICACY EVALUATION

11.1. Analyses of the primary efficacy variable

11.1.1. Primary Endpoint

The primary endpoint for the Phase III study was defined as a test of superiority (one-sided Pearson's chi-square test, $\alpha=0.025$) vs microfracture, in terms of a composite clinical success (CCS) used in a dual threshold responder analysis for pain and function at 1 year post index surgery. A subject was considered a responder if they achieved both of the following outcomes:

- An improvement of at least 12 points in outcome compared to baseline on the KOOS pain assessment test, and
- An improvement of at least 20 points in outcome compared to baseline on the IKDC subjective test.

The following tables (24, 25) detail the primary endpoint results for the different analysis groups with the difference in both percentage and statistical evaluations using the discussed analyses. The SAP specified the Pearson's Asymptotic P value (chi-square) where for the single sided superiority test statistical significance was almost reached in the ITT population ($P=0.0744$) and for those patients who actually received the NeoCart implant in the mITT population the results of improvement over microfracture was highly statistically significant ($P=0.0257$)

Table 24: Primary Efficacy Endpoint (ITT Population)

		NeoCart (N = 170)		Microfracture (N = 79)	
Primary Efficacy Endpoint		n / N (%)	95% CI ¹	n / N (%)	95% CI ¹
Clinically significant improvements of ≥12 points in KOOS pain score and ≥20 points in IKDC subjective score at 1 year		121 / 170 (71.2)	63.7 - 77.9	49 / 79 (62.0)	50.4 - 72.7
NeoCart – Microfracture					
Difference in %	95% CI ²	P-Value ³ Pearson's Exact P-Value 2-sided (1-sided)	Pearson's Asymptotic P- Value ³ 2-sided (1-sided)	Fisher's Exact P- Value 2-sided (1-sided)	
9.2	-3.5 - 21.8	0.1877 (0.0939)	0.1487 (0.0744)	0.1877 (0.0978)	

(1) Clopper-Pearson exact confidence interval for a binomial proportion.

NeoCart – Microfracture				
Difference in %	95% CI ²	P-Value ³ Pearson's Exact P-Value 2-sided (1-sided)	Pearson's Asymptotic P- Value ³ 2-sided (1-sided)	Fisher's Exact P- Value 2-sided (1-sided)

(2) Confidence interval calculated based on the normal approximation.

(3) One-sided normal approximation test of superiority of NeoCart over Microfracture

Source: [Table 14.2.1.1](#)

Table 25: Primary Efficacy Endpoint (mITT Population)

Primary Efficacy Endpoint	NeoCart (N = 170)	Microfracture (N = 79)		
	n / N (%)	95% CI ¹	n / N (%)	95% CI ¹
Clinically significant improvements of ≥12 points in KOOS pain score and ≥20 points in IKDC subjective score at 1 year	121 / 163 (74.2)	66.8–80.8	49 / 79 (62.0)	50.4 - 72.7

NeoCart - Microfracture				
Difference in %	95% CI ²	P-Value ³ Pearson's Exact P-Value 2-sided (1-sided)	Pearson's Asymptotic P- Value ³ 2-sided (1-sided)	Fisher's Exact P- Value 2-sided (1-sided)
12.2	-0.4 - 24.8	0.0714 (0.0357)	0.0514 (0.0257)	0.0714 (0.0371)

(1) Clopper-Pearson exact confidence interval for a binomial proportion.

(2) Confidence interval calculated based on the normal approximation.

(3) One-sided normal approximation test of superiority of NeoCart over Microfracture

Source: [Table 14.2.1.2](#)

With a pre-specified difference of approximately 15-20%, if two patients in the microfracture arm had not met the dual responder threshold statistical significance would have been achieved. Besides the Primary Efficacy Endpoint, for the remainder of the data tables, this report will provide the MITT results with the ITT, As Treated and Per Protocol tables provided in the Appendix.

11.1.2. Primary Endpoints by Subgroup Analysis

Post-hoc analyses were conducted to determine patient populations where certain meaningful improvements or statistically significant superiority results were identified, particularly where microfracture is known to have certain limitations or variability in success, such as in heavier patients, or those with larger lesions. These analyses were all conducted with prospectively collected data. The same data that was used to calculate the dual threshold responder rates.

11.1.2.1. Lesion size

With regard to lesion size, the NeoCart arm performed similarly across the entire study population (Table 26). However, there was a clear differentiation of microfracture in smaller lesion sizes only, with NeoCart showing statistically significant superiority in lesions greater than 2.2 cm² for the ITT population and greater than 1.6 cm² for the mITT population. The mean lesion size for the study was 2.2 cm², and for all lesions greater than that, NeoCart patients had a composite success rate of 71.7.% while the microfracture success rate was only 51.5% (p=0.0050) at 1 year for the ITT population. Evaluation of the earlier timepoint of 6 months demonstrated potential earlier pain relief and functional improvements, for lesions greater than 1.8 cm² in the NeoCart arm. , At 6 months, NeoCart patients had a composite success rate of 59.5% while the microfracture success rate was only 40.5% (p=0.0177) at 1 year for the ITT population.

Similar to the relative performance in lesion sizes between NeoCart and microfracture, literature suggests that patient who have increased BMI demonstrate worse results with microfracture. Tables X & Y illustrate the impact of weight (BMI) on the 1-year primary efficacy endpoint with NeoCart performing equally across the different weight categories with a highly statistically significant improvement when compared to microfracture for those patients with a BMI of 28 or more. As noted above, microfracture is known to perform better in lower weight and younger patients. This has relevance in the United States as a BMI of greater than 24.9 is considered obese. Clinicians must take into account the limitations of a treatment that is impacted by weight in such a negative fashion, and regularly exclude such patients from receiving microfracture in their clinical practice.

In an effort to better understand if there was a specific lesion size cut-off where NeoCart and MF performed equally well below the mean lesion size of 2.2cm² the data was analyzed in 0.2cm² increments and is presented in Table 26. For the ITT population statistical significance of superiority of the NeoCart arm is reached at the 2.2cm lesion size. In the mITT population which represents those patients who actually received the NeoCart treatment there is a high degree of statistical superiority across all lesion sizes.

Table 26: Dual Threshold Responder Analysis by Lesion Size

Lesion size cm ²	N ITT NeoCart	N ITT Mfx	Positive dual threshold responder Difference in %	P value Pearson's chi square 1 sided superiority	N MITT NeoCart	N ITT Mfx	Positive dual threshold responder Difference in %	P value Pearson's chi square 1 sided superiority
≥1.6	119	50	13.7	0.0429	113	50	17.5	0.0137
<1.6	51	29	2.1	0.4189	50	29	3.6	0.3619
≥1.8	111	42	15.5	0.0354	106	42	18.8	0.0133
<1.8	59	37	2.60	0.3909	57	37	5.2	0.2898
≥2.0	105	36	16.7	0.0343	100	36	20.2	0.0131
<2.0	65	43	4.1	0.3216	63	43	6.4	0.2307
≥2.2	92	33	20.2	0.0174	87	33	24.3	0.0050
<2.2	78	46	0.9	0.4556	76	46	2.8	0.3700
≥2.4	83	28	18.7	0.0337	78	28	23.4	0.0099
<2.4	87	51	3.4	0.3364	85	51	5.1	0.2653
≥2.6	73	22	24.0	0.0170	71	22	26.1	0.0100
<2.6	97	57	2.4	0.3785	92	57	6.2	0.2116

Source: Table 14.2.19.1.1, Table 14.2.19.1.2, Table 14.2.19.2.1, Table 14.2.19.2.2, Table 14.2.19.3.1, Table 14.2.19.3.2, Table 14.2.19.4.1, Table 14.2.19.4.2, Table 14.2.19.5.1, Table 14.2.19.5.2, Table 14.2.19.6.1, Table 14.2.19.6.2

11.1.2.2. Body mass index

. For the mITT population the NeoCart arm had similar rates of success of 77.8% above a BMI of 28 and 71.4% below. For microfracture performance at the lower weight range was good with a dual threshold positive response of 68.1%. However, this dropped down to only 53.1% when the BMI was greater than or equal to 28.

11.2. Analyses of the secondary efficacy variables

There were three secondary endpoints:

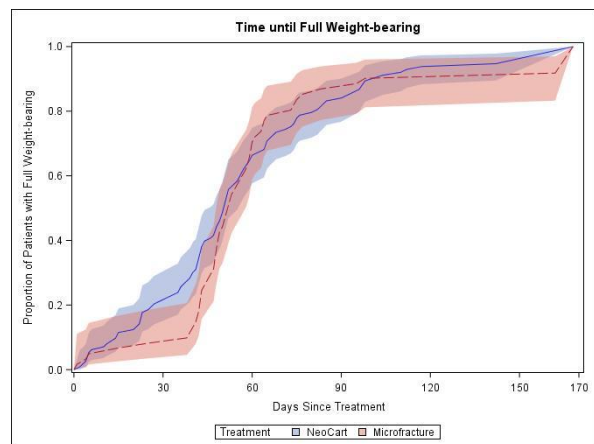
1. Time to full weight-bearing
2. Treatment failure defined as > 8-point deterioration in KOOS pain score at 1-year compared to baseline.
3. Mature collagen layering as assessed by MRI T2 mapping at 1 year.

11.2.1. Time to full weight-bearing

The time to full weight bearing (median days) at 1 year was 51 for NeoCart and 51 for Microfracture(p<0.6833). While the Kaplan-Meier curve demonstrated that more patients were able to full weight bear earlier with NeoCart there was no statistically significant difference between the groups. These values may not represent a true picture of time to weight bearing for several reasons, first the protocol required a very proscriptive rehabilitation plan to limit patients from being full weight bearing early as this variable has been shown to be associated with worse outcomes in the microfracture group. This is an absolute necessity for the microfracture arm success to prevent dislodgement of the mesenchymal stem cell rich clot. For the NeoCart arm the articular cartilage graft is secure in the defect with the adhesive preventing dislodgement,

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however the company retained the closest possible rehabilitation protocol between arms to reduce potential bias. In addition, this data was obtained from patient diaries. Patients entered data with varying degrees of completeness limiting the ability to draw conclusions from the data.



11.2.2. Treatment failure

For Treatment failures the percentage was very low and equal between the two treatment arms with 4 patients in the NeoCart arm and 2 patients in the microfracture arm resulting in identical rates of 2.5% and a p value of < 0.9709 . The Phase III trial also demonstrated several additional findings when reviewing those patients who demonstrated declines in pain and function scores following surgery. There were a few patients that did not respond favorably in both treatment arms. There was a very large difference between the treatment arms who reported worse outcomes after the procedure with 12.7% of microfracture compared to 2.5% in the NeoCart arm on the IKDC function scales. KOOS pain scales demonstrated similar failure rates of 2.5% in each arm. Getting patients back to good functional performance is key both from a quality of life point of view and an ability to perform daily activities including driving a car, climbing stairs and being able to work.

11.2.3. MRI Data

MRIs were obtained at the timepoints noted in Table 4 (Baseline, 6months, 1 year and then annually. The baseline MRI included the following views: Both legs, standing, full extension from femoral head to tibial malleolus, antero-posterior, 45° flexion, weight bearing, postero-anterior and merchant view-30° flexion. The post procedure exam was a standard MRI and time for 63% of transverse relaxation read (T2) image acquisition to be performed at sites where T2 software was available.

All MRI images were submitted to the MRI Core Laboratory. Analysis included standard MRI review to assess the quality of the NeoCart implant and microfracture during the course of the

trial. These MRI findings were divided into anatomic assessments that were made using generally accepted, well-established MRI sequences and more advanced sequences that allowed for the direct, qualitative assessment of both the area of NeoCart implant or microfracture.

In addition to standard MRI review, an advanced T2 mapping MRI sequence was obtained to evaluate for qualitative and quantitative assessment of the collagen component of the extracellular matrix of the NeoCart implant and adjacent articular cartilage. Standard MRI sequences provided high contrast between fat suppressed bone and cartilage; however, they are generally not fluid sensitive and are degraded in the postoperative setting due to susceptibility artifact. Fast spin echo sequences with high in-plane resolution provide good cartilage to fluid contrast, making them more sensitive to partial thickness lesions.

T2 relaxation time is a reproducible MRI parameter that reflects the internuclear reaction, or dephasing, that occurs as a result of transverse relaxation of the excited hydrogen dipoles. The stratification of T2 reflects the arrangement of the type II collagen fibers within the matrix. T2 mapping of articular cartilage at clinical field strengths reflects collagen fiber. Standardized regions of interest T2 mapping analysis was performed in the basilar and superficial halves of the area of NeoCart implant or microfracture, comparing this to T2 values in the adjacent cartilage that appears normal on standardized fast spin echo techniques. Particular attention was paid to the presence or absence of the zonal stratification of T2 of the area of the NeoCart implant or microfracture compared to the patient's native hyaline cartilage. This allowed the Core Lab to possibly determine whether the NeoCart implant formed repair cartilage with collagen orientation and T2 values similar to hyaline cartilage and the nature of the cartilage that formed as the result of the microfracture procedure.

At one year the MRI imaging results were similar between the two treatment arms with slightly more lesions being graded isointense in the NeoCart arm than the microfracture arm (37.5% vs 24.7%) and slightly more lesions being graded hypointense in the microfracture arm 27.2% vs 15.8%. Integration of the defect area was similar as well as the percentage with 67-100% fill (58.7 % NeoCart and 54.3% microfracture). Both groups had minimal hypertrophy, displacement, subchondral edema and similar scoring of bony overgrowth.

While T2 mapping has been thought to help determine the progression to mature hyaline cartilage there were no statistically significant differences in the relaxation time means for the superficial or deep layers or their standard deviations. The relaxation times at 6 months for the superficial cartilage was similar between treatment arms (62.5077msec NeoCart, 62.9531msec microfracture p=0.8191) The deep layer showed longer times for NeoCart while not statistically significant (57.4107 NeoCart vs 54.9544 msec microfracture p=0.2212). The one-year data showed a slight decrease in the relaxation times as would be expected for organizing tissue but there was no difference between the groups, and both were about 10msec longer than the normal tissue. Similar results were seen in the smaller Phase II study. Thus, for the NeoCart phase III clinical study MRI including T2 mapping provided minimal additional insight into the healing process.

11.3. Exploratory Endpoints

11.3.1. KOOS/IKDC:

KOOS and IKDC scoring systems are validated methods of evaluating the success of treatment for knee pain. The FDA guidance document on clinical trial for treatment of knee cartilage recognizes the need to evaluate improvement in pain and physical function and the requirement to use at least two validated measures as no one instrument is sufficient. The raw data to determine the dual threshold responder analysis comes directly from the prospectively collected data individual subscales of KOOS and IKDC, as change from baseline for NeoCart versus microfracture treatment. These de-coupled scores provide a more a holistic view of a variety of patient outcomes. On most of these measures, a statistically significant improvement versus microfracture at one year was observed. The results for Years 1 for all KOOS subscales and IKDC subscale are shown in Figure 1. The individual results show statistically significant improvement in the NeoCart arm across all KOOS measures to the $p \leq 0.050$ level.

Commented [LK45]: Needs further details with selected tables possible or put tables in the appendix

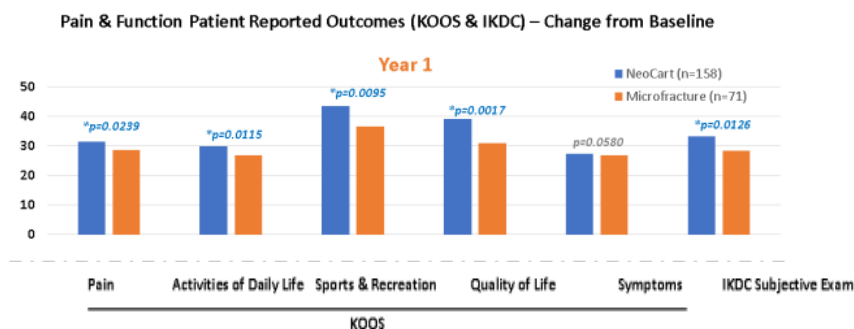
Commented [AG46R45]: This should be a prominent part of this write-up, just like in Briefing Document. IKDC and KOOS – all subscores

THEN, we need summary of other analyses by section or organized a bit more prominently

Commented [AG47R45]: This whole section is still weak. We still need to first talk IKDC and KOOS – take briefing document narrative. This is all solid verbiage.

Then talk about all of the other VAS, SF, ETC – that can be summarized in another paragraph or two

Figure 1: Pain & Function Patient Reported Outcomes (KOOS & IKDC) – Change from Baseline



From a functionality point of view the IKDC showed statistically significant improvement at the 1 year mark $P=0.0126$.

The KOOS pain scale is used to demonstrate the change in value between baseline and then at various pre-specified periods after treatment to assess improvement. Clinically meaningful improvement is established as an 8-10 point improvement on KOOS Pain. Table 12 shows KOOS pain scale improvement over time with statistically significant improvement in the NeoCart treatment arm at the 1-year time point and with further improvement at the 2-year mark with only about one half of the patients contributing data. There is also a trend for the overall population visible at the 6-month time period consistent with early performance of the implant. The results are the same for the ITT and MITT Populations as values are not entered for subjects without datapoints.

Commented [AG48]: Not just a trend in some cases?

11.3.2. Other exploratory endpoints

In addition to the previously provided subjective pain and function measurements in the IKDC and KOOS subscales, the Company also collected data including VAS pain, Marx Activity Rating Scale, SF-12 Health Survey, EQ-5D-5L Health Questionnaire, WPAI Questionnaire, and other medications/drug therapies. The majority of the analyses of these patient reported outcome measures did not demonstrate a statistically significant difference between the two treatment groups.

SF-12 Health Survey: This is a 12 question validated survey that provides insight into the mental and physical functioning and overall health related quality of life. Patients in both arms of the study had improved scores in the physical functioning score with a mean improvement from baseline for the NeoCart arm 15.7 points and 11.3 for microfracture meeting statistical significance $p=0.0057$. (table 14.2.4.3.2) For the physical component the mean change from baseline for the NeoCart group was 14.4 points and microfracture 11.8 with a trend towards significance ($p=0.097$). (table 14.2.4.1.2). Statistical significance was also reached for the Vitality scores at 1 year $P=0.0332$ Interestingly there was statistically significant improvement at the 6 month visit for bodily pain ($p=0.0096$), Social functioning ($p=0.0049$) and Emotional scores ($p=0.0326$). There was no difference in the general health or the mental component summary scores between the two treatment arms (table 14.2.4.5.2 and 14.2.4.10.2)

Marx Activity Rating Scale: a patient reported outcome measure that focuses on four activity points: running, deceleration, cutting and pivoting. There was no statistical difference between the two study arms and scores actually decreased because the test asks about activity for the preceding 12 months. Thus the period from baseline through rehabilitation and recovery would be expected to demonstrate a decrease of activity.

EQ-5D-5L Health Questionnaire: There was a highly statistically significant improvement for NeoCart compared to microfracture at all time points for pain and discomfort scores over time ($p=0.0066$ at 3 months, 0.0053 at 6 months and $p=0.0201$ at 1 year). For usual activities statistically significant improvement for NeoCart was seen at 1 year ($p=0.0482$). There was no difference for the anxiety-depression scores at 1 year.

The Work Productivity and Activity Impairment (WPAI) Questionnaire is an instrument used to measure how impaired an individual may be due to a specific condition. It is also used to compare work impairments between treatment groups in clinical studies. For the NeoCart Phase III trial both treatment groups had similar baseline and 3-month scores. At 6 months and 1 year more patients reported less impairment in the NeoCart treatment arm (<25% impairment reported in 79.9% of the NeoCart group and 67.7% of the microfracture group).

The study also measured the use concomitant medications and separated out those medications that are classified as used for pain. Patients were asked to record pain medication within the last seven days prior to their clinic visit. There was a 19% decrease in the pain medication use at 6 months between the NeoCart and the microfracture treated patients and a 10% decrease at the one-year primary endpoint visit. These data reinforce the survey data from the VAS pain scale noted above.

Commented [AB49]: Team – There was no discreet text related to each of the “Other Exploratory Endpoints.” Therefore, I’ve created subsection headings without text.

11.3.3. VAS pain scale

The VAS pain scale demonstrated a meaningful trend towards decreased pain in the NeoCart arm, with the magnitude of the trend increased at each clinic visit while not reaching statistical significance. (P=0.9703 at 6 months and P= 0.0590 at 1 year). Despite the difference in pain there were no statistically significant differences in mobility, self-care, anxiety and depression.

11.3.4. SF Quality of Life

For the various component scores for the SF-12 questionnaires there were statistical improvements of NeoCart over microfracture at 1 year in the ITT population for physical functioning (p=0.0057) and vitality (p= 0.0332) with either no difference or slight trend towards improvement in the other scores but nothing statistically significant.

11.3.5. Work Productivity Activity Impairment (WPAI)

There was no statistical difference between the NeoCart and microfracture groups when looking at the percentage of work time missed, percent impairment while working or percent activity impairment at 1 year however there was trend of improvement with NeoCart at the percentage of impairment < 25% of the responders of the questionnaire (80.0% NeoCart and 67.2% microfracture).

11.4. Efficacy Conclusions

NeoCart did not demonstrate statistically superior performance with the dual threshold responder analysis missing by approximately 2 patients. However, when the data is reviewed a number of very important conclusions can be drawn. First that for the independent scores of KOOS pain and IKDC NeoCart does demonstrate a clinically and statistically significance at both the 6m and 1 year time points. In addition, when the data is stratified by lesion size the performance separation is apparent at low lesion sizes and continues as lesion sizes increase. The primary endpoint for the Phase 3 clinical trial was a dual-threshold responder analysis measuring the improvement in KOOS pain and IKDC function scores for each patient treated with NeoCart compared to those treated with microfracture one year after the time of treatment. In clinical trials a threshold response is a way to take a continuous primary efficacy measure and dichotomize it into responders and non-responders. The threshold is set based upon expectations for what a clinically meaningful result would be. In a dual threshold endpoint, the subjects must meet the value set as a responder both of two different measurements to be considered a responder (often termed a super superiority trial). At the time of the NeoCart phase 3 clinical trial design and discussion with the agency for the special protocol assessment (SPA) there was limited data available regarding what was a clinically meaningful change from baseline for either KOOS or IKDC and there had not been any completed prospective randomized trials. As the field has evolved, clinicians have worked hard to identify the scores at which meaningful benefit can be determined. The current scores for IKDC starts at a 6-point improvement as early as 6 months and increases to around 15 or 16 points at later timeframes. For KOOS pain the meaningful clinically important difference is seen in the 8 to 10-point range. Both thresholds in

Commented [LK50]: Aaron: Is this referring to results of this study? If so, need to reference a table.

Commented [LK51]: Aaron: Does this mean 8-10 points is clinically important and this is what was observed in the study? If so, need to refer to the appropriate data table.

the NeoCart trial were higher than what was and is now seen as a minimum clinically meaningful or important response.

The NeoCart efficacy results from the pivotal study are clinically meaningful and relevant and should be put into the context of how the field of treatment for knee cartilage defects has evolved over the past 10 years since the NeoCart trial was designed. The NeoCart treatment group demonstrates a statistically significant improvement using the dual-threshold responder criteria at 6 months and continues to improve numerically to the one-year primary endpoint ([Table 24](#)), although not meeting the prespecified statistical superiority p-value. NeoCart treatment also demonstrated statistically significant improvements over the microfracture group for substantially all of the KOOS and IKDC subgroup analyses at 6 months, 1 years and 2 years (with roughly half the patients completing that visit) when measured independently as reflected in current FDA Guidance.

12. SAFETY EVALUATION

12.1. Extent of exposure

The NeoCart implant is (Cultured Autologous Neo-Cartilage) tissue implant is a 3-dimensional, white, opalescent disc (6cm² area, 1.5to 2.0mm thick) designed to treat symptomatic defects of the articular cartilage of the knee. Patients in the clinical trial undergo standard arthroscopy and if randomized to the NeoCart arm a biopsy of normal cartilage is obtained. This biopsy is then used to create the autologous cartilage graft that is placed into the defect at a second procedure approximately 8 weeks after the biopsy. Patients randomized to the control arm have the microfracture procedure performed at the time of arthroscopy and are exposed to a single operative procedure. NeoCart patients are required to undergo a mini-arthrotomy second procedure for preparation and placement of the graft into the defect. Patients in the NeoCart arm are exposed to the implant and an adhesive used to affix the implant. Safety data is collected for all patients in both arms of the study for the duration of three years post baseline entry into the study with the follow-up for patients in the NeoCart arm continued through three years post implantation.

Commented [LK52]: Aaron: This is more for the Investigational Plan section.

Commented [LK53]: Aaron: Important: So, no safety data are collected between arthroscopy and NeoCart implant procedure? I don't think that's correct.

Commented [LK54R53]: Andrea: AE data are collected from Baseline on

12.2. Adverse events

12.2.1. Brief summary of adverse events

There were no unexpected safety events reported in either arm of the study. NeoCart and microfracture procedures were tolerated equally well with a low number of adverse events related to the study procedure (Table of TAES-15 in BB). Treatment emergent adverse events (TEAEs) are all adverse events that occur during the study period independent of relatedness to procedure or treatment. There were 12 serious TEAEs in the NeoCart arm (6.8%) and 6 in the microfracture group (7.5%; Table 27). TEAEs were similarly equal in both arms and observed in 151 patients (445 events), (622% of the NeoCart population and 58.8% of the microfracture p=0.4808.

Commented [LK55]: Check definition

Commented [AB56R55]: From SAP of August 20, 2018: TEAE: defined as having onset on or after the study treatment (defined as the date of first treatment exposure, e.g. arthroscopic evaluation and subsequent microfracture procedure or NeoCart biopsy).

Commented [AB57]: Not according to Table 28.

Commented [LK58R57]: Andrea: There were 12 NeoCart patient with SAEs and 6 MFX pateints with SAEs. Table 28 needs updating

Commented [LK59]:

Commented [LK60]: Aaron: Where are the data to support this statement?

Commented [LK61]: Need to add timing as related to procedure to ensure causality

Commented [LK62R61]: Andrea: There are only 2 events of DVT.

- 1 occurring at 80 days deemed probably related
- 1 occurring at 53 days deemed unlikely related.

1 event of PE reported occurring at 11 days deemed possibly related

Commented [LK63]: Need to add more color around DVT and risk with any arthroscopy/arthrotomy

The two most common adverse events were arthralgia (NeoCart 24.5% and microfracture (23.8%) and joint swelling (NeoCart 13.6%, microfracture 7.5%). The only other adverse event that occurred above a 5% rate was headache (NeoCart 11.1% and microfracture (12.5%). The vast majority of adverse events were of short duration and self limited. There were 4 deep vein thromboses in the NeoCart group and none in the microfracture arm (p=0.3052) and one pulmonary embolism in a patient randomized to the NeoCart arm.

Table 27: Overall Summary of Treatment Emergent Adverse Events (TEAE)

Parameter	NeoCart (N = 162)	Microfracture (N = 80)	Fisher's Exact Test P-Value
Patients with Any Treatment-Emergent AE, n (%)	105 (64.8%)	47 (58.8%)	0.3973
Number of TEAEs	311	144	
Patients with Any Serious TEAE, n (%)	12 (7.4%)	6 (7.5%)	1.0000
Number of Serious TEAEs	17	7	
Patients with Any Procedure-Related TEAE, n (%)	39 (24.1%)	15 (18.8%)	0.4132
Number of Procedure-Related TEAEs	72	24	
Patients with Any NeoCart-Related TEAE, n (%)	30 (18.5%)	N/A	N/A
Number of NeoCart-Related TEAEs	44		

Source: [Table 14.3.1.1](#)

12.2.2. All adverse events

[Table 28](#) summarizes all treatment emergent adverse events.

Commented [LK64]: Aaron do you need table numbers for both of these?

Commented [AB65R64]: Lynne – I just need the number of the original source table (eg, Table 14.3.1.1. as n this Table 27). You can refer to the original table number in the text and I'll create the sequential number f the table caption.

All tables added to the CSR should include the original (eg, Table 14.3.1.1) table number as a source at the bottom of the table. See Table 28 as an example.

Table 28: Treatment Emergent Adverse Events, Incidence by System Organ Class and Preferred Term, by Patient (AT population)

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)	Fisher's Exact Test P- Value
Musculoskeletal and connective tissue disorders	73 (45.1%)	30 (37.5%)	0.2728
Arthralgia	29 (17.9%)	15 (18.8%)	0.8612
Joint swelling	16 (9.9%)	7 (8.8%)	1.0000
Back pain	5 (3.1%)	6 (7.5%)	0.1858
Pain in extremity	5 (3.1%)	3 (3.8%)	0.7217
Joint lock	6 (3.7%)	1 (1.3%)	0.4305
Joint effusion	6 (3.7%)	0 (0.0%)	0.1819
Joint crepitation	1 (0.6%)	3 (3.8%)	0.1068
Musculoskeletal pain	3 (1.9%)	1 (1.3%)	1.0000
Tendonitis	4 (2.5%)	0 (0.0%)	0.3052
Arthrofibrosis	3 (1.9%)	0 (0.0%)	0.5528
Intervertebral disc protrusion	2 (1.2%)	1 (1.3%)	1.0000
Neck pain	3 (1.9%)	0 (0.0%)	0.5528
Joint instability	1 (0.6%)	1 (1.3%)	0.5528
Joint range of motion decreased	2 (1.2%)	0 (0.0%)	1.0000
Muscle spasms	2 (1.2%)	0 (0.0%)	1.0000
Osteoarthritis	2 (1.2%)	0 (0.0%)	1.0000
Patellofemoral pain syndrome	2 (1.2%)	0 (0.0%)	1.0000
Arthritis	1 (0.6%)	0 (0.0%)	1.0000
Bone formation increased	0 (0.0%)	1 (1.3%)	0.3306
Bunion	0 (0.0%)	1 (1.3%)	0.3306
Cartilage atrophy	1 (0.6%)	0 (0.0%)	1.0000
Cervical spinal stenosis	0 (0.0%)	1 (1.3%)	0.3306
Coccydynia	1 (0.6%)	0 (0.0%)	1.0000
Groin pain	1 (0.6%)	0 (0.0%)	1.0000
Haemarthrosis	1 (0.6%)	0 (0.0%)	1.0000
Joint hyperextension	0 (0.0%)	1 (1.3%)	0.3306
Joint stiffness	1 (0.6%)	0 (0.0%)	1.0000
Medial tibial stress syndrome	1 (0.6%)	0 (0.0%)	1.0000
Muscle atrophy	1 (0.6%)	0 (0.0%)	1.0000
Muscular weakness	1 (0.6%)	0 (0.0%)	1.0000
Osteochondrosis	1 (0.6%)	0 (0.0%)	1.0000
Osteolysis	1 (0.6%)	0 (0.0%)	1.0000
Rotator cuff syndrome	1 (0.6%)	0 (0.0%)	1.0000
Spinal osteoarthritis	0 (0.0%)	1 (1.3%)	0.3306
Synovitis	1 (0.6%)	0 (0.0%)	1.0000
Tendon discomfort	1 (0.6%)	0 (0.0%)	1.0000
Infections and infestations	26 (16.0%)	16 (20.0%)	0.4731
Nasopharyngitis	8 (4.9%)	4 (5.0%)	1.0000
Sinusitis	3 (1.9%)	2 (2.5%)	0.6667
Upper respiratory tract infection	4 (2.5%)	0 (0.0%)	0.3052
Urinary tract infection	2 (1.2%)	2 (2.5%)	0.6008
Gastroenteritis viral	2 (1.2%)	1 (1.3%)	1.0000
Incision site infection	3 (1.9%)	0 (0.0%)	0.5528

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)	Fisher's Exact Test P- Value
Bronchitis	0 (0.0%)	2 (2.5%)	0.1084
Cellulitis	0 (0.0%)	2 (2.5%)	0.1084
Tinea pedis	1 (0.6%)	1 (1.3%)	0.5528
Acute sinusitis	1 (0.6%)	0 (0.0%)	1.0000
Appendicitis	0 (0.0%)	1 (1.3%)	0.3306
Diarrhoea infectious	0 (0.0%)	1 (1.3%)	0.3306
Epididymitis	1 (0.6%)	0 (0.0%)	1.0000
Fungal infection	1 (0.6%)	0 (0.0%)	1.0000
Gastroenteritis	0 (0.0%)	1 (1.3%)	0.3306
Herpes zoster	1 (0.6%)	0 (0.0%)	1.0000
Labyrinthitis	1 (0.6%)	0 (0.0%)	1.0000
Oral candidiasis	1 (0.6%)	0 (0.0%)	1.0000
Pneumonia	0 (0.0%)	1 (1.3%)	0.3306
Post procedural cellulitis	1 (0.6%)	0 (0.0%)	1.0000
Post procedural infection	0 (0.0%)	1 (1.3%)	0.3306
Postoperative abscess	1 (0.6%)	0 (0.0%)	1.0000
Skin bacterial infection	1 (0.6%)	0 (0.0%)	1.0000
Injury, poisoning and procedural complications	30 (18.5%)	10 (12.5%)	0.2731
Fall	5 (3.1%)	3 (3.8%)	0.7217
Meniscus injury	6 (3.7%)	0 (0.0%)	0.1819
Joint injury	3 (1.9%)	1 (1.3%)	1.0000
Contusion	3 (1.9%)	0 (0.0%)	0.5528
Cartilage injury	1 (0.6%)	1 (1.3%)	0.5528
Foot fracture	1 (0.6%)	1 (1.3%)	0.5528
Joint dislocation	1 (0.6%)	1 (1.3%)	0.5528
Muscle strain	1 (0.6%)	1 (1.3%)	0.5528
Post procedural constipation	2 (1.2%)	0 (0.0%)	1.0000
Tibia fracture	1 (0.6%)	1 (1.3%)	0.5528
Ankle fracture	1 (0.6%)	0 (0.0%)	1.0000
Arthropod bite	0 (0.0%)	1 (1.3%)	0.3306
Gun shot wound	1 (0.6%)	0 (0.0%)	1.0000
Hand fracture	0 (0.0%)	1 (1.3%)	0.3306
Incision site erythema	1 (0.6%)	0 (0.0%)	1.0000
Incision site rash	1 (0.6%)	0 (0.0%)	1.0000
Ligament sprain	1 (0.6%)	0 (0.0%)	1.0000
Limb injury	1 (0.6%)	0 (0.0%)	1.0000
Lumbar vertebral fracture	1 (0.6%)	0 (0.0%)	1.0000
Procedural pain	1 (0.6%)	0 (0.0%)	1.0000
Scar	1 (0.6%)	0 (0.0%)	1.0000
Spinal fracture	0 (0.0%)	1 (1.3%)	0.3306
Suture related complication	1 (0.6%)	0 (0.0%)	1.0000
Suture rupture	1 (0.6%)	0 (0.0%)	1.0000
Transplant failure	1 (0.6%)	0 (0.0%)	1.0000
Wound	0 (0.0%)	1 (1.3%)	0.3306
Wrist fracture	0 (0.0%)	1 (1.3%)	0.3306
Nervous system disorders	23 (14.2%)	15 (18.8%)	0.3552
Headache	16 (9.9%)	10 (12.5%)	0.5175

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)	Fisher's Exact Test P- Value
Migraine	1 (0.6%)	2 (2.5%)	0.2547
Paraesthesia	0 (0.0%)	2 (2.5%)	0.1084
Aphonia	1 (0.6%)	0 (0.0%)	1.0000
Cerebral cyst	1 (0.6%)	0 (0.0%)	1.0000
Cubital tunnel syndrome	0 (0.0%)	1 (1.3%)	0.3306
Neuralgia	1 (0.6%)	0 (0.0%)	1.0000
Restless legs syndrome	1 (0.6%)	0 (0.0%)	1.0000
Sciatica	1 (0.6%)	0 (0.0%)	1.0000
Syncope	1 (0.6%)	0 (0.0%)	1.0000
General disorders and administration site conditions	11 (6.8%)	7 (8.8%)	0.6078
Influenza like illness	2 (1.2%)	2 (2.5%)	0.6008
Local swelling	1 (0.6%)	1 (1.3%)	0.5528
Oedema peripheral	2 (1.2%)	0 (0.0%)	1.0000
Pain	1 (0.6%)	1 (1.3%)	0.5528
Pyrexia	2 (1.2%)	0 (0.0%)	1.0000
Surgical failure	0 (0.0%)	2 (2.5%)	0.1084
Chest pain	1 (0.6%)	0 (0.0%)	1.0000
Implant site reaction	1 (0.6%)	0 (0.0%)	1.0000
Inflammation	0 (0.0%)	1 (1.3%)	0.3306
Malaise	1 (0.6%)	0 (0.0%)	1.0000
Post procedural inflammation	1 (0.6%)	0 (0.0%)	1.0000
Skin discolouration	1 (0.6%)	0 (0.0%)	1.0000
Gastrointestinal disorders	9 (5.6%)	7 (8.8%)	0.4112
Diarrhoea	1 (0.6%)	2 (2.5%)	0.2547
Gastroesophageal reflux disease	1 (0.6%)	2 (2.5%)	0.2547
Abdominal pain	1 (0.6%)	1 (1.3%)	0.5528
Nausea	2 (1.2%)	0 (0.0%)	1.0000
Abdominal discomfort	1 (0.6%)	0 (0.0%)	1.0000
Abdominal mass	1 (0.6%)	0 (0.0%)	1.0000
Abdominal pain upper	0 (0.0%)	1 (1.3%)	0.3306
Constipation	1 (0.6%)	0 (0.0%)	1.0000
Dental caries	0 (0.0%)	1 (1.3%)	0.3306
Dysphagia	1 (0.6%)	0 (0.0%)	1.0000
Inguinal hernia	1 (0.6%)	0 (0.0%)	1.0000
Mallory-Weiss syndrome	0 (0.0%)	1 (1.3%)	0.3306
Peptic ulcer haemorrhage	1 (0.6%)	0 (0.0%)	1.0000
Umbilical hernia	1 (0.6%)	0 (0.0%)	1.0000
Psychiatric disorders	3 (1.9%)	6 (7.5%)	0.0629
Anxiety	1 (0.6%)	4 (5.0%)	0.0422
Depression	2 (1.2%)	0 (0.0%)	1.0000
Insomnia	1 (0.6%)	1 (1.3%)	0.5528
Anxiety disorder	0 (0.0%)	1 (1.3%)	0.3306
Sleep disorder	1 (0.6%)	0 (0.0%)	1.0000
Immune system disorders	4 (2.5%)	4 (5.0%)	0.4451
Drug hypersensitivity	2 (1.2%)	1 (1.3%)	1.0000

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)	Fisher's Exact Test P- Value
Seasonal allergy	0 (0.0%)	2 (2.5%)	0.1084
Allergy to arthropod bite	0 (0.0%)	1 (1.3%)	0.3306
Food allergy	1 (0.6%)	0 (0.0%)	1.0000
Hypersensitivity	1 (0.6%)	0 (0.0%)	1.0000
Respiratory, thoracic and mediastinal disorders	8 (4.9%)	0 (0.0%)	0.0552
Oropharyngeal pain	3 (1.9%)	0 (0.0%)	0.5528
Cough	2 (1.2%)	0 (0.0%)	1.0000
Allergic respiratory symptom	1 (0.6%)	0 (0.0%)	1.0000
Atelectasis	1 (0.6%)	0 (0.0%)	1.0000
Dyspnoea	1 (0.6%)	0 (0.0%)	1.0000
Pleural effusion	1 (0.6%)	0 (0.0%)	1.0000
Pulmonary embolism	1 (0.6%)	0 (0.0%)	1.0000
Skin and subcutaneous tissue disorders	3 (1.9%)	3 (3.8%)	0.4000
Alopecia	0 (0.0%)	1 (1.3%)	0.3306
Campbell de Morgan spots	1 (0.6%)	0 (0.0%)	1.0000
Dermatitis contact	0 (0.0%)	1 (1.3%)	0.3306
Erythema	1 (0.6%)	0 (0.0%)	1.0000
Precancerous skin lesion	0 (0.0%)	1 (1.3%)	0.3306
Rash	0 (0.0%)	1 (1.3%)	0.3306
Urticaria	1 (0.6%)	0 (0.0%)	1.0000
Vascular disorders	3 (1.9%)	3 (3.8%)	0.4000
Deep vein thrombosis	3 (1.9%)	0 (0.0%)	0.5528
Haematoma	0 (0.0%)	1 (1.3%)	0.3306
Hypertension	0 (0.0%)	1 (1.3%)	0.3306
Popliteal artery entrapment syndrome	0 (0.0%)	1 (1.3%)	0.3306
Ear and labyrinth disorders	4 (2.5%)	1 (1.3%)	1.0000
Tympanic membrane perforation	2 (1.2%)	0 (0.0%)	1.0000
Cerumen impaction	0 (0.0%)	1 (1.3%)	0.3306
Motion sickness	1 (0.6%)	0 (0.0%)	1.0000
Vertigo positional	1 (0.6%)	0 (0.0%)	1.0000
Investigations	5 (3.1%)	0 (0.0%)	0.1739
Alanine aminotransferase increased	1 (0.6%)	0 (0.0%)	1.0000
Aspartate aminotransferase increased	1 (0.6%)	0 (0.0%)	1.0000
Blood testosterone decreased	1 (0.6%)	0 (0.0%)	1.0000
Monocyte count increased	1 (0.6%)	0 (0.0%)	1.0000
Neutrophil count increased	1 (0.6%)	0 (0.0%)	1.0000
Prothrombin time prolonged	1 (0.6%)	0 (0.0%)	1.0000
White blood cell count decreased	1 (0.6%)	0 (0.0%)	1.0000
White blood cell count increased	1 (0.6%)	0 (0.0%)	1.0000
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	3 (1.9%)	2 (2.5%)	0.6667
Basal cell carcinoma	1 (0.6%)	0 (0.0%)	1.0000
Intraocular melanoma	1 (0.6%)	0 (0.0%)	1.0000

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)	Fisher's Exact Test P- Value
Lipoma of breast	0 (0.0%)	1 (1.3%)	0.3306
Malignant melanoma	0 (0.0%)	1 (1.3%)	0.3306
Thyroid neoplasm	1 (0.6%)	0 (0.0%)	1.0000
Reproductive system and breast disorders	3 (1.9%)	2 (2.5%)	0.6667
Breast mass	1 (0.6%)	0 (0.0%)	1.0000
Haemorrhagic ovarian cyst	1 (0.6%)	0 (0.0%)	1.0000
Ovarian cyst	0 (0.0%)	1 (1.3%)	0.3306
Polycystic ovaries	1 (0.6%)	0 (0.0%)	1.0000
Prostatic cyst	0 (0.0%)	1 (1.3%)	0.3306
Cardiac disorders	3 (1.9%)	1 (1.3%)	1.0000
Tachycardia	2 (1.2%)	0 (0.0%)	1.0000
Arrhythmia	0 (0.0%)	1 (1.3%)	0.3306
Sinus tachycardia	1 (0.6%)	0 (0.0%)	1.0000
Blood and lymphatic system disorders	2 (1.2%)	1 (1.3%)	1.0000
Bone marrow oedema	1 (0.6%)	1 (1.3%)	0.5528
Splenomegaly	1 (0.6%)	0 (0.0%)	1.0000
Pregnancy, puerperium and perinatal conditions	1 (0.6%)	2 (2.5%)	0.2547
Pregnancy	1 (0.6%)	2 (2.5%)	0.2547
Surgical and medical procedures	3 (1.9%)	0 (0.0%)	0.5528
Oral surgery	1 (0.6%)	0 (0.0%)	1.0000
Scar excision	1 (0.6%)	0 (0.0%)	1.0000
Vasectomy	1 (0.6%)	0 (0.0%)	1.0000
Eye disorders	1 (0.6%)	1 (1.3%)	0.5528
Giant papillary conjunctivitis	0 (0.0%)	1 (1.3%)	0.3306
Ocular rosacea	1 (0.6%)	0 (0.0%)	1.0000
Hepatobiliary disorders	2 (1.2%)	0 (0.0%)	1.0000
Cholecystitis	1 (0.6%)	0 (0.0%)	1.0000
Cholelithiasis	1 (0.6%)	0 (0.0%)	1.0000
Gallbladder disorder	1 (0.6%)	0 (0.0%)	1.0000
Renal and urinary disorders	1 (0.6%)	1 (1.3%)	0.5528
Nephrolithiasis	0 (0.0%)	1 (1.3%)	0.3306
Pollakiuria	1 (0.6%)	0 (0.0%)	1.0000
Endocrine disorders	1 (0.6%)	0 (0.0%)	1.0000
Hypothyroidism	1 (0.6%)	0 (0.0%)	1.0000

Source: [Table 14.3.1.2](#)

12.3. Deaths, other serious adverse events, and other significant adverse events

12.3.1. Deaths

There were no deaths in either treatment arm during the study.

12.3.2. Serious adverse events

There were no treatment or procedure related adverse events that led to prolonged or permanent disability. A summary of serious TEAEs is provided in [Table 29](#) and listed in [Table 30](#). Narratives for all patients with a serious TEAE may be found in Section [14.3.3](#).

Table 29: Number (%) of Patients with Serious TEAEs by System Organ Class and Preferred Term

System Organ Class/ Preferred Term	NeoCart (N = 162) n (%)	Microfracture (N = 80) n (%)
Injury, poisoning and procedural complications	5 (3.1%)	0 (0.0%)
Meniscus injury	2 (1.2%)	0 (0.0%)
Gun shot wound	1 (0.6%)	0 (0.0%)
Lumbar vertebral fracture	1 (0.6%)	0 (0.0%)
Transplant failure	1 (0.6%)	0 (0.0%)
Musculoskeletal and connective tissue disorders	3 (1.9%)	2 (2.5%)
Arthrofibrosis	2 (1.2%)	0 (0.0%)
Arthralgia	0 (0.0%)	1 (1.3%)
Cartilage atrophy	1 (0.6%)	0 (0.0%)
Cervical spinal stenosis	0 (0.0%)	1 (1.3%)
Vascular disorders	2 (1.2%)	2 (2.5%)
Deep vein thrombosis	2 (1.2%)	0 (0.0%)
Haematoma	0 (0.0%)	1 (1.3%)
Popliteal artery entrapment syndrome	0 (0.0%)	1 (1.3%)
Gastrointestinal disorders	2 (1.2%)	0 (0.0%)
Abdominal mass	1 (0.6%)	0 (0.0%)
Peptic ulcer haemorrhage	1 (0.6%)	0 (0.0%)
General disorders and administration site conditions	0 (0.0%)	2 (2.5%)
Surgical failure	0 (0.0%)	2 (2.5%)
Hepatobiliary disorders	2 (1.2%)	0 (0.0%)
Cholecystitis	1 (0.6%)	0 (0.0%)
Cholelithiasis	1 (0.6%)	0 (0.0%)
Infections and infestations	0 (0.0%)	1 (1.3%)
Appendicitis	0 (0.0%)	1 (1.3%)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	1 (0.6%)	0 (0.0%)
Intraocular melanoma	1 (0.6%)	0 (0.0%)
Respiratory, thoracic and mediastinal disorders	1 (0.6%)	0 (0.0%)
Pulmonary embolism	1 (0.6%)	0 (0.0%)

Source: [Table 14.3.1.3](#)

Table 30: Listing of all Serious TEAEs

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
03-013/ Microfracture	31/ M/ 11MAY2012	Musculoskeletal and connective tissue disorders/ Arthralgia/ INCREASED LEFT KNEE PAIN; FAILURE OF MICROFRACTURE SURGERY	19JUN2014/ 13MAY2015	329/ Moderate	Definitely Related/ N/A	Medically significant/ Medication; Surgery; ICE THERAPY	Recovered/Resolved
03-027/ Microfracture	35/ F/ 12DEC2014	General disorders and administration site conditions/ Surgical failure/ Re-operation on index knee due to failed microfracture surgery 2 years post op	02MAY2016/ 04AUG2016	95/ Severe	Definitely Related/ N/A	Medically significant/ Surgery	Recovered/Resolved
13-009/ Microfracture	37/ F/ 09JUN2016	Infections and infestations/ Appendicitis/ Appendicitis	23OCT2016/ 26OCT2016	4/ Moderate	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Surgery; appendectomy on 25Oct2016	Recovered/Resolved

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
23-003/ Microfracture	27/ F/ 17DEC2014	General disorders and administration site conditions/ Surgical failure/ FAILED MICROFRACTURE TO THE MEDIAL FEMORAL CONDYLE	26MAY2015/ 21DEC2016	576/ Moderate	Definitely Related/ Not Related	Medically significant/ Surgery	Recovered/Resolved
34-013/ Microfracture	44/ M/ 02JUL2015	Musculoskeletal and connective tissue disorders/ Cervical spinal stenosis/ C4-C5 Spinal Stenosis	JUL2016/ AUG2017	N/A/ Moderate	Not Related/ N/A	Hospitalized- prolonged hospitalization/ Medication; Surgery	Recovered/Resolved
34-013/ Microfracture	44/ M/ 02JUL2015	Vascular disorders/ Haematoma/ Hematoma Secondary to Cervical Surgery	27JUL2017/ 28JUL2017	2/ Moderate	Not Related/ N/A	Hospitalized- prolonged hospitalization/ Medication	Recovered/Resolved
41-003/ Microfracture	38/ M/ 30MAR2016	Vascular disorders/ Popliteal artery entrapment syndrome/ Left popliteal artery entrapment syndrome	16SEP2016/ 13FEB2017	151/ Moderate	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Surgery	Recovered/Resolved
03-002/ NeoCart	35/ M/ 13AUG2010	Musculoskeletal and connective tissue disorders/ Cartilage atrophy/ Chondral Degeneration	01JAN2011/ 22OCT2012	661/ Severe	Possibly Related/ Possibly Related	Hospitalized- prolonged hospitalization/ Surgery	Recovered/Resolved

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
03-002/ NeoCart	35/ M/ 13AUG2010	Injury, poisoning and procedural complications/ Gun shot wound/ GUN SHOT WOUND TO LEFT LEG	20DEC2012/ 16APR2013	118/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Medication; Surgery	Recovered/Resolved
03-005/ NeoCart	38/ M/ 10SEP2010	Injury, poisoning and procedural complications/ Meniscus injury/ LEFT MEDIAL MENISCUS TEAR	08AUG2010/ 10SEP2010	34/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Medication; Surgery; MRI	Recovered/Resolved
03-005/ NeoCart	38/ M/ 10SEP2010	Respiratory, thoracic and mediastinal disorders/ Pulmonary embolism/ PULMONARY EMBOLISM	20SEP2010/ 30SEP2010	11/ Severe	Probably Related/ Possibly Related	Life threatening/ Medication	Recovered/Resolved
03-005/ NeoCart	38/ M/ 10SEP2010	Injury, poisoning and procedural complications/ Meniscus injury/ LEFT KNEE, MEDIAL MENISCUS TEAR	06AUG2012/ Ongoing	N/A/ Severe	Not Related/ Not Related	Medically significant/ Unknown	Not Recovered/Not Resolved
05-001/ NeoCart	34/ F/ 26SEP2012	Injury, poisoning and procedural complications/ Lumbar vertebral fracture/ VERTEBRAL FRACTURE (L2 COMPRESSION FX)	03JUL2015/ 20AUG2015	49/ Moderate	Not Related/ Not Related	Medically significant/ Medication; Surgery	Recovered/Resolved

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
09-014/ NeoCart	40/ M/ 03MAY2016	Vascular disorders/ Deep vein thrombosis/ Deep Vein Thrombosis	09MAY2016/ 27JUL2016	80/ Moderate	Probably Related/ Probably Related	Medically significant/ Medication	Recovered/Resolved
10-003/ NeoCart	40/ F/ 22JUL2014	Musculoskeletal and connective tissue disorders/ Arthrofibrosis/ ARTHROFIBROSIS	22OCT2014/ 11NOV2014	21/ Moderate	Definitely Related/ Unlikely Related	Hospitalized- prolonged hospitalization; Medically significant/ Medication; Surgery; Physical therapy; MRI was ordered to diagnose further info in source documents	Recovered/Resolved
13-003/ NeoCart	43/ M/ 22JUL2014	Hepatobiliary disorders/ Cholelithiasis/ Cholelithiasis	08OCT2014/ 09OCT2014	2/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization; Life threatening/ Surgery; Gall bladder removed	Recovered/Resolved

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
17-003/ NeoCart	29/ M/ 16JUL2015	Hepatobiliary disorders/ Cholecystitis/ Inefficient evacuation of bile Gallbladder Removal/Cholecystectomy Cholecystitis	2016/ Unknown	N/A/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization; Medically significant/ Surgery	Unknown
21-001/ NeoCart	34/ F/ 14MAY2014	Musculoskeletal and connective tissue disorders/ Arthrofibrosis/ ARTHROFIBROSIS (LEFT KNEE)	16JUL2014/ 30JUL2014	15/ Moderate	Definitely Related/ Not Related	Medically significant/ Surgery	Recovered/Resolved
21-001/ NeoCart	34/ F/ 14MAY2014	Injury, poisoning and procedural complications/ Meniscus injury/ LATERAL MENISCUS TEAR (LEFT KNEE)	22SEP2014/ 22DEC2014	92/ Moderate	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Medication; Surgery; Physical therapy	Recovered/Resolved
23-005/ NeoCart	34/ F/ 26JAN2016	Injury, poisoning and procedural complications/ Transplant failure/ Right knee Medial femoral condyle graft failure	26JUL2016/ Ongoing	N/A/ Severe	Possibly Related/ Definitely Related	Medically significant/ Surgery	Ongoing, active study
23-005/ NeoCart	34/ F/ 26JAN2016	Gastrointestinal disorders/ Abdominal mass/ complex abdominopelvic mass	11OCT2017/ 11OCT2017	1/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization/ Surgery	Recovered/Resolved

Patient ID/ Actual Treatment	Age/ Sex/ Treatment Date	System Organ Class/ Preferred Term/ Reported Term	Start Date/ End Date	Duration (days)/ Severity	Relationship to Procedure/Relationship to NeoCart	Description of Serious/ Actions Taken	Event Outcome
33-003/ NeoCart	49/ M/ 03MAY2016	Gastrointestinal disorders/ Peptic ulcer haemorrhage/ Bleeding Peptic Ulcer (hematemesis & hematochezia)	15MAY2016/ 19MAY2016	5/ Severe	Possibly Related/ Unlikely Related	Hospitalized- prolonged hospitalization; Medically significant/ Medication; Surgery; 4 units of transfusion	Recovered/Resolved
36-001/ NeoCart	53/ M/ 29DEC2015	Vascular disorders/ Deep vein thrombosis/ DVT	11JAN2016/ 03MAR2016	53/ Severe	Definitely Related/ Unlikely Related	Hospitalized- prolonged hospitalization/ Medication	Recovered/Resolved
36-005/ NeoCart	38/ M/ 31JAN2017	Neoplasms benign, malignant and unspecified (incl cysts and polyps)/ Intraocular melanoma/ Ocular melanoma	02MAR2017/ Ongoing	N/A/ Severe	Not Related/ Not Related	Hospitalized- prolonged hospitalization; Medically significant/ treatment with eye implant	Ongoing, active study

Source: [Table 14.3.2.1](#)

12.3.3. Other significant adverse events

~~12.3.4.~~ There were three deep venous thromboses (1.9%) and one pulmonary embolism (0.6%) in the NeoCart arm while none in the microfracture arm. All were treated and resolved. One DVT occurred remote from the procedures. Due to the small numbers there is no statistically significant difference. Literature review reveals that DVT after arthroscopy for any reason has been reported to range from 1.5% to 17.9% (reference Chen et al Medicine (2017) 96:22) reducing any concern of causality related to the test device.

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12.4. Laboratory safety assessments

12.4.1. Laboratory values over time

There were no clinically relevant differences between treatment groups for any laboratory values.

12.4.2. Individual patient changes in laboratory values

Not applicable

12.4.3. Individual clinically significant abnormalities

Not applicable

12.5. Other safety assessments

Not applicable

13. DISCUSSION AND OVERALL CONCLUSIONS

13.1. Discussion

NeoCart's clinical effectiveness was demonstrated through clinically meaningful and statistically significant treatment related improvements in pain and function at the 1-year timepoint versus the microfracture control arm.

These data were generated from the largest adequate and well-controlled Phase III clinical study to date in a U.S representative patient population with stringent inclusion/exclusion standards based on FDA's input. The efficacy of NeoCart was measured using validated and FDAaccepted instruments such as KOOS pain and IKDC function that that can be directly correlated to clinically relevant benefits for patients.

NeoCart's safety was also established, with TEAEs and TSEAEs occurring in comparable rates in both arms of the study with no unexpected TEAEs or TSEAEs. NeoCart has a positive benefit/risk profile in the knee cartilage defect space, where treatment options are limited to treat pain and loss of function

The Phase 3 Clinical Trial of NeoCart did not meet the primary endpoint of a statistically significant superiority in pain and function in a dual threshold responder analysis 1 year after treatment as compared to microfracture. (Show table 14.2.11 Primary efficacy endpoint ITT population). However this failure to reach statistical significance in no way diminishes the facts that NeoCart provides a better alternative for the treatment of focal cartilage defects in the knee than current standard procedures. In the Phase III clinical trial the NeoCart treatment arm performed better than anticipated based upon what was predicted from the results of the Phase II clinical trial and in the statistical plan (74% success compared to 70% anticipated). Microfracture performed worse than NeoCart but better than was anticipated based upon the Phase II clinical results, what has been observed in clinical practice and what was in the statistical plan (62% vs the expected 50%). Thus, the pre-specified difference of approximately 15-20% was not attained and a statistically significant result for the total population was just missed ($p=0.074$ Pearson's chi-square for the ITT population and $p=0.0257$ for the mITT population).

The early benefit of the NeoCart arm is across all treatment groups and is sustained through the 1-year primary endpoint further reinforcing the durability of replacing a cartilage defect with a predictable cartilage tissue construct as opposed to chondrocytes. The results with respect to the primary endpoint (dual-threshold responder analysis 1 year after treatment) are provided for the ITT population, mITT population, and the PP population in Table 24, Table 25, and **Error! Reference source not found.**, respectively.

The NeoCart efficacy results from the pivotal study are clinically meaningful and relevant and should be put into the context of how the field of treatment for knee cartilage defects has evolved over the past 10 years since the NeoCart trial was designed. The NeoCart treatment group demonstrates a statistically significant improvement using the dual-threshold responder criteria at 6 months and continues to improve numerically to the one-year primary endpoint mark (Table 4), although not meeting the prespecified statistical superiority p-value. NeoCart treatment also demonstrated improvement with statistical significance over the microfracture group for the majority of both the KOOS and IKDC subgroup analyses when measured independently as reflected in current FDA Guidance. The early benefit of the NeoCart arm is across all treatment

Commented [AG66]: This is good, however we'll need to refresh/update once rest is built out.

Commented [LK67]: Table 14.2.1.1 Primary Efficacy endpoint ITT population

Commented [LK68R67]: Aaron: Check paragraph as repetitive information

Commented [LK69]: Aaron: What treatment groups are being referred to? Are you referring to Analysis Populations?

Commented [LK70R69]: Aaron: Need to include the original Table numbers for proper identification.

groups and is sustained through the 1-year primary endpoint further reinforcing the durability of replacing a cartilage defect with a predictable cartilage tissue construct as opposed to chondrocytes. In addition, from a “do no harm” perspective it is important to evaluate the safety performance of those patients who not only had positive results even if they did not meet the dual threshold responder criteria but those who appeared to respond poorly after treatment.

There are two previously FDA cleared or approved products for knee cartilage injuries (Carticel and MACI). Both products harvest healthy cartilage and then amplify the cells and return them to the area of the cartilage defect, Carticel as a cell suspension and MACI as a cell suspension seeded on a porcine collagen membrane. Carticel is no longer on the US market. There are fundamental differences between NeoCart and MACI product characteristics that may show distinct clinical advantages over time. NeoCart is a more mature cartilage tissue graft when it is implanted into the patient, rather than cells laid upon a collagen membrane. This is thought to account for the faster and potentially greater durability in result for patients. The demographics of the NeoCart phase III clinical study were different from those studied in the MACI Phase III SUMMIT clinical study that limit the ability for a direct comparison between these products. The lower limit of lesion sizes for inclusion in the Summit trial was 3.0cm² and the mean lesion size was 4.8cm² whereas in the NeoCart study the mean lesion size was 2.2cm² with lesions treated from 0.5cm² to 6cm². This is important as microfracture treatment effect is known to perform better in smaller lesion sizes.

The SUMMIT trial also reported on the KOOS subscales with the co-primary using the pain and function (sports and recreation subscale) at the two-year timepoint, versus the 1-year timepoint for NeoCart. Finally, the trial also allowed concomitant procedures (>30%) making it difficult to assign benefit to the primary cartilage defect repair in those patients and included previous microfracture treatments (>30%), which is known to result in poor subsequent outcomes and durability. The SUMMIT trial was conducted entirely in Europe with a 100% Caucasian demographic. The NeoCart Phase III study was conducted in the United States and Canada with almost 9% African American and almost 9% Hispanic patients which is more reflective of the U.S. population.

Since 2014, at least four Phase III or Pivotal clinical trials for knee cartilage have been initiated by other third parties to our knowledge based on publicly available information. One trial (NOVOCART) is being conducted under an IND as a biologic or combination product, and the other three products (Hyalofast, GelrinC, Agili-C) are medical devices in clinical studies being performed under IDEs. All of these trials appear to have similar pain and function primary endpoints using measures of KOOS and/or IKDC either individually or as co-primary endpoints, designed as superiority trials with microfracture as the active control, or more recently a historical control arm, with 2-year endpoints, and a three year follow up period.

Unlike the MACI trial, these trials also suggest that previous microfracture or other invasive surgeries should be excluded along with concomitant procedures and be limited to smaller lesions such as in the NeoCart trials. We also understand from publicly available information that each of these trials have had substantial difficulty in enrolling patients, similar to the sponsor's lengthy experience. The narrow inclusion/exclusion criteria as seen in the NeoCart study, as well as public perceptions regarding the utility and efficacy of microfracture, have continued to make these trials both financially and logistically challenging.

There were a few patients that did not respond favorably in both treatment arms. From a “do no harm” perspective it is important to evaluate the safety performance of those patients who not only had positive results even if they did not meet the dual threshold responder criteria but those who appeared to respond poorly after treatment. There was a very large difference between the treatment arms who reported worse outcomes after the procedure with 12.7% of microfracture compared to 2.5% in the NeoCart arm on the IKDC function scales. Getting patients back to good functional performance is key both from a quality of life point of view and an ability to perform daily activities including driving a car, climbing stairs and being able to work. When evaluating the risk/benefit of the procedures there are several elements that demonstrate that the NeoCart procedure is unique to other knee cartilage repair options and are supported by the data. These are elective procedures with the overall goal of decreasing pain and improving the functional status of patients. The potential to impact long term function and decrease the progression to osteoarthritis and reduce the need for joint replacement surgery remains to be seen but has never been demonstrated with the alternative therapy of microfracture due to the eventual failure rate over time (Solheim, 2018). The process of taking a patient’s own cartilage and amplifying it within a scaffold under the conditions present in the knee joint prior to implanting the tissue results in the differentiation of the cartilage cells to the optimum articular cartilage. This is further evidenced by the evaluation of articular hyaline cartilage biomarkers prior to release of the implant. Patients treated with NeoCart demonstrated a statistically significant faster recovery at 6 months post index procedure that is durable and continues to improve through up to the two years for which data is available.

In summary, based on our review of publicly available information regarding current standards, we believe the NeoCart Phase III study meets the FDA’s current guidance in terms of conduct, patient demographics and inclusion and exclusion criteria, and the pain and function endpoints as chosen are appropriate to demonstrate efficacy and safety of an investigational treatment for knee cartilage injury.

13.2. Overall conclusion

In conclusion, NeoCart’s clinical effectiveness was demonstrated through clinically meaningful and statistically significant treatment related improvements in pain and function at the 1-year timepoint versus the microfracture control arm. The data become even more powerful as lesion size increases above the 2.0cm² size. While the limited application of Microfracture in larger lesions has been recognized this is the first study to clearly define the lesion size above which the risk of even intermediate term (1-year) may be too great for physicians and patients to accept. NeoCart offers a durable and safe option. The NeoCart phase 3 clinical study results demonstrate clinical benefit with a very acceptable risk profile. The ability to get patients back on their feet in half the time has demonstrable advantages from a health economics perspective, particularly in a broad range of lesions where microfracture is most often used, not in limited populations where it is no longer used.

These data were generated from the largest adequate and well-controlled Phase III clinical study to date in a U.S representative patient population with stringent inclusion/exclusion standards based on FDA’s input. The efficacy of NeoCart was measured using validated and FDA accepted instruments such as KOOS pain and IKDC function that that can be directly correlated to clinically relevant benefits for patients.

Commented [AG71]: We need to further clarify how this is measured (decline in IKDC or other)

Commented [LK72]: Aaron: Important: How does this conclusion align with out time to weight bearing endpoint?

NeoCart’s safety was also established, with TEAEs and TSEAEs occurring in comparable rates in both arms of the study with no unexpected TEAEs or TSEAEs. NeoCart has a positive benefit/risk profile in the knee cartilage defect space, where treatment options are limited to treatpain and loss of function.

<add text>

Commented [LK73]: Aaron: need to address DVTs

Commented [AG74]: This is good, may need to build out a bit more (AG can do this)

14. TABLES AND FIGURES

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Table 14.2.20.2.1 Primary efficacy endpoint sensitivity analysis (ITT Population)

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Table 14.2.20.3.1 Primary efficacy endpoint sensitivity analysis (ITT Population)

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Table 14.2.22.3.2 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.22.4.1 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.22.4.2 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.22.5.1 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.22.5.2 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.22.6.1 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.22.6.2 Secondary endpoint responder rates, by treated lesion size - GE or LT 2.6cm² (MITT Population)

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Table 14.2.23.1.2 KOOS pain score, by treated lesion size - GE or LT 1.6cm² (MITT Population)

Table 14.2.23.2.1 KOOS pain score, by treated lesion size - GE or LT 1.8cm² (ITT Population)

Table 14.2.23.2.2 KOOS pain score, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.23.3.1 KOOS pain score, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.23.3.2 KOOS pain score, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.23.4.1 KOOS pain score, by treated lesion size - GE or LT 2.2cm² (ITT Population)

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Table 14.2.24.1.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 1.6cm² (MITT Population)

Table 14.2.24.2.1 IKDC subjective knee exam score, by treated lesion size - GE or LT 1.8cm² (ITT Population)

Table 14.2.24.2.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.24.3.1 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.24.3.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.24.4.1 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.24.4.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.24.5.1 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.24.5.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.24.6.1 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.24.6.2 IKDC subjective knee exam score, by treated lesion size - GE or LT 2.6cm² (MITT Population)

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Table 14.2.25.1.2 KOOS symptoms score, by treated lesion size - GE or LT 1.6cm² (MITT Population)

Table 14.2.25.2.1 KOOS symptoms score, by treated lesion size - GE or LT 1.8cm² (ITT Population)

Table 14.2.25.2.2 KOOS symptoms score, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.25.3.1 KOOS symptoms score, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.25.3.2 KOOS symptoms score, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.25.4.1 KOOS symptoms score, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.25.4.2 KOOS symptoms score, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.25.5.1 KOOS symptoms score, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.25.5.2 KOOS symptoms score, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.25.6.1 KOOS symptoms score, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.25.6.2 KOOS symptoms score, by treated lesion size - GE or LT 2.6cm² (MITT Population)

Table 14.2.26.1.1 KOOS activities of daily living score, by treated lesion size - GE or LT 1.6cm² (ITT Population)

Table 14.2.26.1.2 KOOS activities of daily living score, by treated lesion size - GE or LT 1.6cm² (MITT Population)

Table 14.2.26.2.1 KOOS activities of daily living score, by treated lesion size - GE or LT 1.8cm² (ITT Population)

Table 14.2.26.2.2 KOOS activities of daily living score, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.26.3.1 KOOS activities of daily living score, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.26.3.2 KOOS activities of daily living score, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.26.4.1 KOOS activities of daily living score, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.26.4.2 KOOS activities of daily living score, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.26.5.1 KOOS activities of daily living score, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.26.5.2 KOOS activities of daily living score, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.26.6.1 KOOS activities of daily living score, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.26.6.2 KOOS activities of daily living score, by treated lesion size - GE or LT 2.6cm² (MITT Population)

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Table 14.2.27.1.2 KOOS sports and recreation function score, by treated lesion size - GE or LT 1.6cm² (MITT Population)

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Table 14.2.27.5.1 KOOS sport and recreation function score, by treated lesion size - GE or LT 2.4cm2 (ITT Population)

Table 14.2.27.5.2 KOOS sports and recreation function score, by treated lesion size - GE or LT 2.4cm2 (MITT Population)

Table 14.2.27.6.1 KOOS sport and recreation function score, by treated lesion size - GE or LT 2.6cm2 (ITT Population)

Table 14.2.27.6.2 KOOS sports and recreation function score, by treated lesion size - GE or LT 2.6cm2 (MITT Population)

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Table 14.2.28.1.2 KOOS quality of life score, by treated lesion size - GE or LT 1.6cm2 (MITT Population)

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Table 14.2.28.5.2 KOOS quality of life score, by treated lesion size - GE or LT 2.4cm2 (MITT Population)

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Table 14.2.29.1.2 Comparative efficacy endpoint, 1 year, by treated lesion size - GE or LT 1.6cm² (MITT Population)

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Table 14.2.29.2.2 Comparative efficacy endpoint, 1 year, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.29.3.1 Comparative efficacy endpoint, 1 year, by treated lesion size - GE or LT 2.0cm² (ITT Population)

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Table 14.2.29.4.2 Comparative efficacy endpoint, 1 year, by treated lesion size - GE or LT 2.2cm² (MITT Population)

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Table 14.2.30.2.2 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.30.3.1 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.30.3.2 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.30.4.1 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.30.4.2 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.30.5.1 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.30.5.2 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.30.6.1 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.30.6.2 Comparative efficacy endpoint, 2 years, by treated lesion size - GE or LT 2.6cm² (MITT Population)

Table 14.2.31.1.1 Comparative responder endpoint, by treated lesion size - GE or LT 1.6cm² (ITT Population)

Table 14.2.31.1.2 Comparative responder endpoint, by treated lesion size - GE or LT 1.6cm² (MITT Population)

Table 14.2.31.2.1 Comparative responder endpoint, by treated lesion size - GE or LT 1.8m² (ITT Population)

Table 14.2.31.2.2 Comparative responder endpoint, by treated lesion size - GE or LT 1.8cm² (MITT Population)

Table 14.2.31.3.1 Comparative responder endpoint, by treated lesion size - GE or LT 2.0cm² (ITT Population)

Table 14.2.31.3.2 Comparative responder endpoint, by treated lesion size - GE or LT 2.0cm² (MITT Population)

Table 14.2.31.4.1 Comparative responder endpoint, by treated lesion size - GE or LT 2.2cm² (ITT Population)

Table 14.2.31.4.2 Comparative responder endpoint, by treated lesion size - GE or LT 2.2cm² (MITT Population)

Table 14.2.31.5.1 Comparative responder endpoint, by treated lesion size - GE or LT 2.4cm² (ITT Population)

Table 14.2.31.5.2 Comparative responder endpoint, by treated lesion size - GE or LT 2.4cm² (MITT Population)

Table 14.2.31.6.1 Comparative responder endpoint, by treated lesion size - GE or LT 2.6cm² (ITT Population)

Table 14.2.31.6.2 Comparative responder endpoint, by treated lesion size - GE or LT 2.6cm² (MITT Population)

14.3. Safety data

14.3.1. Extent of exposure

Table 14.3.1.1 Treatment emergent adverse events overview (AT Population)

Table 14.3.1.2 Treatment emergent adverse events, incidence by system organ class and preferred term, by patient (AT Population)

Table 14.3.1.3 Serious treatment emergent adverse events, incidence by system organ class and preferred term, by patient (AT Population)

Table 14.3.1.4 TEAEs, incidence by SOC and PT, by patient and event severity (AT Population)

Table 14.3.1.5 TEAEs, incidence by SOC and PT, by patient and relationship to procedure (AT Population)

Table 14.3.1.6 TEAEs, incidence by SOC and PT, by patient and relationship to NeoCart (AT Population)

14.3.2. Adverse events

Table 14.3.2.1 Listing of serious adverse events (AT Population)

Table 14.3.2.2 Listing of adverse events with action taken (AT Population)

14.3.3. Deaths, other serious adverse events, and other significant adverse events

Patient 03-002 (NeoCart):

Event 1: Patient experienced Arthralgia that was first reported January 1, 2011, approximately 5 months after implantation on August 13, 2010. The patient was diagnosed with Chondral Degeneration (MedDra Cartilage Atrophy) at arthroscopy on October 12, 2012, at which time chondroplasty of a flap tear adjacent to the NeoCart implant, as well as approximately 50% of the NeoCart implant, was performed. Following the procedure, the patient reported immediate relief of pain. The Investigator assessed the event of Arthralgia as severe in severity and possibly related to both NeoCart and the NeoCart procedure.

Event 2: On December 20, 2012, the patient was hospitalized for a self-inflicted gunshot wound to the non-study leg. The patient underwent an operation to fixate a fractured femur and irrigate and debride the wound. The patient was discharged on December 23, 2012 in stable condition. The Investigator assessed the event of Gun Shot Wound as severe in severity and unrelated to both NeoCart and the NeoCart procedure.

The patient completed the study at the 3-year follow-up visit on September 18, 2013.

Patient 03-005 (NeoCart):

Event 1: Patient suffered a Meniscus Injury (left medial meniscus tear) in the target knee on August 8, 2010, after randomization/biopsy, but prior to the implantation procedure. The Investigator assessed the event of Meniscus Injury as severe in severity and unrelated to both NeoCart and the NeoCart procedure. During the index implantation procedure on September 10, 2010, the medial meniscus tear was repaired.

Event 2: On September 20, 2010, the patient was hospitalized for shortness of breath and hemoptysis and was found to have a pulmonary embolism. The Investigator assessed the event of Pulmonary Embolism as life-threatening, severe in severity, possibly related to NeoCart and probably related to the NeoCart procedure. The patient was discharged on September 30, 2010 with no shortness of breath, prescriptions for coumadin and Lovenox, and instructions to follow-up with the primary care provider.

Event 3: On December 31, 2011, it was discovered that patient 03-005 suffered a Meniscus Injury (left medial meniscus tear) in the target knee. Meniscectomy was recommended, but to date no known intervention has been performed. The Investigator assessed the event of Meniscus Injury as medically significant, severe in severity, and unrelated to both NeoCart and the NeoCart procedure.

The patient has not been seen since September 26, 2012, following the 2-year follow-up visit and is considered lost to 3-year follow-up.

Patient 05-001 (NeoCart):

Patient suffered a Vertebral Fracture due to an accidental fall on July 3, 2015. The patient was treated with narcotic analgesics, and underwent L2 kyphoplasty on August 20, 2015, resulting in full recovery. The Investigator assessed the event as moderate in severity and unrelated to both NeoCart and the NeoCart procedure.

The patient completed the study at the 3-year follow-up visit on September 22, 2015.

Patient 09-014 (NeoCart)

Patient implanted with NeoCart on May 3, 2016, was diagnosed with deep vein thrombosis on May 9, 2016. Patient was treated with anti-coagulant therapy. Follow-up by ultrasound and laboratory tests on August 1, 2016 confirmed resolution of the deep vein thrombosis. The Investigator assessed the event as moderate in severity, probably related to both the NeoCart implant and the NeoCart procedure.

The patient completed the 1 year follow up visit on June 8, 2017 and remains on study.

Patient 10-003 (NeoCart):

Patient was diagnosed with Arthrofibrosis on October 22, 2014 after the patient failed to regain full range of motion of the target knee early after the implantation procedure. On November 11, 2014, the patient underwent adhesiolysis of the target knee and the event resolved. The Investigator assessed the event of Arthrofibrosis as moderate in severity, unlikely related to NeoCart, and definitely related to the NeoCart procedure.

The patient completed the study at the 3-year follow-up visit on September 6, 2017.

Patient 13-003 (NeoCart):

Patient was diagnosed with Cholelithiasis on October 8, 2014. Cholecystectomy was performed on October 9, 2014. The Investigator assessed the event as life threatening, severe in severity, and unrelated to both NeoCart and the NeoCart procedure.

The patient completed the study at the 3-year follow-up visit on August 30, 2017.

Patient 17-003 (NeoCart):

Inefficient evacuation of bile Gallbladder, Removal/Cholecystectomy/Cholecystitis

The patient had the NeoCart procedure.

Patient 21-001 (NeoCart):

Event 1: Patient was diagnosed with Arthrofibrosis on July 16, 2014 after the patient failed to regain full range of motion of the target knee early after the implantation procedure. On July 17, 2014, the patient underwent adhesiolysis of the target knee. The Investigator assessed the event of Arthrofibrosis as moderate in severity, unrelated to NeoCart, and definitely related to the NeoCart procedure.

Event 2: On September 22, 2014, the patient experienced a Lateral Meniscus Tear of the target knee. Partial meniscectomy was performed on December 22, 2014, resulting in resolution of associated symptoms. The Investigator assessed the event as unrelated to both NeoCart and the NeoCart procedure.

The patient completed the study at the 3-year follow-up visit on June 27, 2017.

Patient 23-005 (NeoCart):

Event 1: Patient experienced Femoral Condyle Graft Failure (MedDRA Transplant Failure) approximately 6 months after implantation performed on January 26, 2016, and underwent re-operation for chondroplasty on July 26, 2016. The event was assessed by the Investigator to be severe in severity, definitely related to NeoCart, and possibly related to the NeoCart procedure.

Event 2: October 2, 2017, Patient 23-005 was evaluated for an abdominopelvic mass. She admitted with increased abdominal pain, girth and bloating. CT scan noted an abdominopelvic cystic mass on the left ovary measuring 17.1x28.4x24.3cm. The patient underwent left salpingo-oophorectomy on October 11, 2017. She was discharged on October 15, 2017 and instructed to follow-up with primary physician.

Patient 33-003 (NeoCart)

Patient implanted with NeoCart on May 3, 2016, was diagnosed with Peptic Ulcer Hemorrhage on May 16, 2016. Patient underwent endoscopy and cauterization of the ulcer on May 18, 2016. Patient was found to have low hemoglobin levels and received 4 units of whole blood. Cause of ulcer is believed to be patient's concomitant use of aspirin with ibuprofen. Patient was discharged on May 19, 2016 and was asymptomatic as of follow up visit on May 27, 2016. The Investigator assessed the event as severe in severity, possibly related to the surgical procedure, and unlikely to be related to NeoCart implant.

The patient completed the 2 year follow up visit on May 2, 2018 and remains on study.

Patient 36-001 (NeoCart):

Patient suffered Deep Vein Thrombosis on January 11, 2016. The event was resolved March 3, 2016. The investigator assessed the event as severe in severity, unlikely related to NeoCart, and definitely related to the NeoCart procedure.

The patient completed the 2 year follow up visit on January 15, 2018 and remains on study.

Patient 36-005 (NeoCart)

Patient was diagnosed with ocular melanoma on March 2, 2017 and underwent treatment with an implantation of a radiation disc into his eye on March 13, 2017. The disc was explanted on March 17, 2017. The Investigator assessed the event as moderate in severity and unrelated to both NeoCart and NeoCart procedure.

The patient completed the 1 year follow up visit on February 22, 2018 and remains on study.

Patient 03-013 (Microfracture):

Patient was diagnosed with Increased Knee Pain and Failure of Microfracture Surgery on June 19, 2014 following the 2-year time point follow-up visit. On February 6, 2015, the patient underwent re-operation to remove damaged cartilage and perform osteochondral allografting.

The patient completed the study at the 3-year follow-up visit on October 14, 2015.

Patient 03-027 (Microfracture):

Patient was diagnosed with attritional failure of microfracture (MedDRA PT Surgical Failure) following microfracture treatment performed on December 12, 2014 with persistent and increasing pain requiring re-operation. After nearly 2 years of conservative treatment, patient underwent reoperation of the index knee on August 4, 2016, at which time an osteochondral allograft procedure was performed. The Investigator assessed the event as severe in severity, and possibly related to the microfracture procedure.

The patient has not been seen since December 9, 2015, following the 1-year follow-up visit and is considered lost to follow-up.

Patient 13-009 (Microfracture)

Patient was diagnosed with appendicitis on October 23, 2016 and underwent appendectomy on October 25, 2016 with full resolution on October 26, 2016. The Investigator assessed the event as moderate in severity and unrelated to the microfracture procedure.

The patient completed the 2 year follow up visit on June 4, 2018 and remains on study.

Patient 23-003 (Microfracture):

Patient experienced Failure of Microfracture Surgery on May 26, 2015 (MedDRA PT Surgical Failure) after initial surgery on December 17, 2014. Patient underwent re-operation in April 2017 at which time osteochondral allografting was performed. The Investigator assessed the event as moderate in severity and definitely related to the microfracture procedure.

The patient completed the study at the 3-year follow-up visit on October 18, 2017.

Patient 34-013 (Microfracture)

Patient developed C4-C5 cervical stenosis in July 2016. He was admitted and underwent spinal fusion surgery on July 27, 2017. The patient's post-operative course was complicated by a hematoma which prolonged the hospitalization. He underwent a re-operation on July 28, 2017 to remove the hematoma. These were reported as 2 separate serious events.

The patient completed the study at the 3-year follow-up visit on August 6, 2018.

Patient 41-003 (Microfracture):

Patient was diagnosed with Popliteal Artery Entrapment Syndrome on October 13, 2016. Popliteal artery release was performed on October 14, 2016, and the event was considered resolved at surgical follow-up on February 13, 2017. The Investigator assessed the event as moderate in severity and unrelated to the microfracture procedure.

The patient completed the 2 year follow up visit on May 1, 2018 and remains on study.

14.3.4. Laboratory data

14.3.5. Other safety data (if applicable)

Table 14.3.5.1 Frequency of re-operation or intervention to the target knee (AT Population)

Table 14.3.5.2 Pain medication use (AT Population)

Table 14.3.5.3 Changes in smoking status at follow-up visits (AT Population)

14.4. Original statistical output

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16. APPENDICES

16.1. Study Information

16.1.1. Protocol and Protocol Amendments

16.1.2. Sample Case Report Form and Annotated Case Report Form

16.1.3. List of IRBs and Representative Written Information for Subject and Sample Informed Consent Forms

16.1.3.1. List of IRBs

16.1.3.2. Sample Informed Consent Form

16.1.4. List and Description of Investigator and Other Important Participants in the Study

16.1.4.1. List of Principal Investigator

Site	Site Location	Site PI
Tucson Orthopaedic Institute	Tucson, AZ	Nebojsa Skrepnik, MD, PhD
Grossmont Orthopaedic Medical Group	La Mesa, CA	Scott Hacker, MD
Orthopaedic Specialty Institute/Newport Orthopaedic Institute (OSI/NOI)	Orange, CA	Benjamin Rubin, MD
UCSF Mission Bay Orthopaedic Institute	San Francisco, CA	C. Benjamin Ma, MD
Southern California Orthopedic Institute	Van Nuys, CA	Mark Getelman, MD
Colorado University Boulder	Boulder, CO	Eric McCarty, MD
Steadman Hawkins Clinic	Greenwood Village, CO	Thomas Noonan, MD
Shrock Clinical Research	Fort Lauderdale, FL	Kevin Shrock, MD
Jewett Orthopaedic Clinic	Orlando, FL	Sean McFadden, DO
Optim Healthcare	Savannah, GA	David Palmer, MD
St. Luke's Medical Center	Boise, ID	Kevin Shea, MD
Rush University Medical Center	Chicago, IL	Brian Cole, MD, MBA
OrthoIndy	Greenwood, IN	Jack Farr, II, MD
Ochsner Clinic Sports Medicine	New Orleans, LA	Deryk Jones, MD
Orthopaedic Associates	Portland, ME	Thomas Murray, Jr., MD
Harvard Vanguard Medical Associates	Boston, MA	Kai Mithoefer, MD
William Beaumont Hospital	Royal Oak, MI	Joseph Guettler, MD
TRIA Orthopaedic Center	Bloomington, MN	Brad Nelson, MD
Mayo Clinic	Rochester, MN	Aaron Krych, MD
Alpine Orthopaedics & Sports Medicine	Bozeman, MT	Timothy O'Brien, MD
University of New Mexico	Albuquerque, NM	Daniel Wascher, MD
Hospital for Special Surgery	New York, NY	Riley Williams, III, MD
Keller Army Community Hospital	West Point, NY	Steven Svoboda, MD
Duke Sports Medicine	Durham, NC	Dean Taylor, MD

Site	Site Location	Site PI
Ohio State University – Department of Sports Medicine	Columbus, OH	David Flanigan, MD
Oregon Health & Science Univ.	Portland, OR	Dennis Crawford, MD, PhD
University of Pennsylvania	Philadelphia, PA	Brian Sennett, MD
The San Antonio Orthopaedic Group (TSAOG)	San Antonio, TX	Brad Tolin, MD
The Orthopedic Specialty Hospital	East Murray, UT	Tyler Barker, MD
University of Washington	Seattle, WA	Albert Gee, MD
Simon Fraser Orthopaedic Fund	New Westminster, British Columbia	Robert McCormack, MD
Fowler Kennedy Sports Medicine Clinic	London, Ontario	Alan Getgood, MD
QEII Health Sciences Center	Halifax, Nova Scotia	Ivan Wong, MD
These Sites Consented but Did Not Enroll Patients due to Screen Failures		
Advanced Orthopaedic Surgery Center	Temecula, CA	R. Scott Herron, MD
JDP Medical Research	Aurora, CO	John Papilion, MD
Sunrise Medical Research	Lauderdale Lakes, FL	Richard Linn, MD
Florida Orthopaedics Associates	Deland, FL	Mark Hollmann, MD
Rhode Island Hospital	Providence, RI	Brett Owens, MD

16.1.4.2. Curricula Vitae of Principal Investigator

16.1.5. Sponsor CSR Approval Form

16.1.6. Certificate of Analysis for Drug Product

- 16.1.7. Randomization Scheme and Codes**
 - 16.1.8. Audit Certificates**
 - 16.1.9. Documentation of Statistical Methods**
 - 16.1.9.1. Final statistical analysis plan**
 - 16.1.9.2. Minutes of data review meeting(s)**
 - 16.1.9.3. Interim analyses**
 - 16.1.9.3.1. Statistical analysis plan**
 - 16.1.9.3.2. Interim analysis results**
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 - 16.3. Case Report Forms**
 - 16.4. Individual Patient Data Listings**
 - 16.5. Bioanalytical data report(s)**
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